Investigations of irrationality constructions for $\beta(4)$

Technical supplement: obstructions, reductions, and unresolved arithmetic

Research note prepared with Codex

1 November 2026

Abstract

An accurate rational approximation to a constant need not help prove its irrationality: multiplying the error by the denominator may make it large. This supplement examines that difficulty for $\beta(4) = 1 - 3^{-4} + 5^{-4} - \cdots$ through several natural constructions. It proves failure results for specified polynomial integral sources, asymptotic corrections to partial sums, differentiated continued fractions, and algebraic approximations built from roots of unity. Further arguments identify limitations imposed by positivity constraints and by analytic continuation. The conclusions concern the stated families and operations. Two results explain the limits of a broader negative conclusion: unrestricted positive polynomial sources can represent every positive rational linear form in $\beta(4)$, and a parameter-dependent moment construction gives positive errors with a sharp decay rate while its denominator growth remains unresolved. Self-contained case studies also develop interleaved and signed determinants, a common-base polylogarithm ladder, endpoint interpolation, modular transformations, coupled period reductions and paired hypergeometric identities. Each gives explicit constructions, established arguments and the remaining mathematical question. Exact algebraic certificates accompany the larger identities; shorter exploratory variants are catalogued separately. The companion paper develops two determinant obstructions in detail. Neither document settles the rationality or irrationality of $\beta(4)$.

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1 Introduction and guide to the results

Put $\theta = \beta(4) = \sum_{k\geq 0}(-1)^k/(2k+1)^4$. Suppose a construction gives a rational approximation a_n/q_n in lowest terms, with error $e_n = \theta - a_n/q_n$. The associated integer linear form is $q_n\theta - a_n = q_n e_n$. To prove irrationality by this route, one needs these forms to be nonzero and tend to zero. Indeed, if $\theta = a/b$ were rational with $b > 0$, every such nonzero form would have absolute value at least $1/b$. Thus the central comparison is between the decay of e_n and the growth of the approximant's reduced denominator q_n.

This supplement asks how that comparison works in several explicit constructions. Some start from integrals, some improve the alternating series by tail corrections or continued fractions, and some use algebraic approximations or differential equations. The companion paper [1] treats two determinant families in detail. Here we give further obstruction proofs and reproducible case studies of the substantial unfinished constructions. Each case states what is known and the exact additional estimate needed to continue it.

The results are organized around three questions: what makes the constructed value nonzero, how small is it as a real number, and what arithmetic normalization is unavoidable? To prove failure by comparing sizes, one must show that the last cost defeats the real decay. A large convenient denominator, or a finite run of unfavorable examples, is insufficient for such a conclusion.

The motivating determinant work of OpenAI [11, Section 1.2] separates nonvanishing, real size and arithmetic normalization. Zudilin [13] gives the broader positive-moment determinant framework. The distinction between real decay and primitive arithmetic remained decisive in this investigation. Carreira's work [4] provides directly related $\beta(4)$ constructions and tests; the general positive-weight and rationally modified moment frameworks also appear in Prévost [12] and Krattenthaler [7].

For a nonzero $Q \in \mathbb{Q}[X]$, write $\mathrm{Prim}(Q) = \kappa Q$, where $\kappa > 0$ and the resulting integer coefficients have gcd one. Then

$$v_p(\kappa) = -\min_j v_p([X^j]Q). \tag{1.1}$$

Here v_p counts powers of the prime p in the numerator minus those in the denominator of a nonzero rational number; $v_p(0) = +\infty$. For example, $v_p(p^{-3}) = -3$. A rational number is p-integral when its valuation is nonnegative and a p-unit when its valuation is zero. Reduction modulo p means inverting a denominator not divisible by p. The notation $[X^j]Q$ denotes a polynomial coefficient. The identity follows because the minimum valuation of a primitive integer coefficient list is zero at every prime. Multiplying Q by a nonzero rational scalar preserves its primitive representative up to sign. A lower bound on a convenient clearing integer is therefore not automatically a lower bound on κ. For example, 5 clears $(6X + 12)/5$, whose primitive multiplier is $5/6$.

If $\theta = a/b$ and $Z \in \mathbb{Z}[X]$ has degree at most d, then $b^d Z(\theta) \in \mathbb{Z}$. Thus nonzero values satisfying $|Z_n(\theta)|^{1/d_n} \to 0$ would prove irrationality. Merely having $Z_n(\theta) \to 0$ with unbounded degrees is insufficient: X^n at $1/2$ is a counterexample. For linear forms this reduces to the usual nonzero integer-form criterion.

A guide to the proved results. The table summarizes the selected statements proved in Sections 2–9. Each section introduces its construction and states its hypotheses. In particular, the continued-fraction conclusion fixes the endpoint and uses even depth, and the Mellin inequality concerns constrained nonnegative sources.

Construction	What is proved here	Section
Positive polynomial integrals	Every positive form $A\theta + B$, $A, B \in \mathbb{Q}$, is representable; an obstruction requires restrictions on the sources.	2
Repeated quadratics	Unavoidable prime denominators defeat the best real decay within the stated family.	3
Euler–Boole corrections	Reduced integer forms grow for every escaping endpoint and truncation schedule.	4
Differentiated continued fractions	A fixed-endpoint, even-depth family has growing reduced forms.	5
Roots of unity	Large conjugates defeat the accuracy of one algebraic approximation.	6
Completed Mellin source	Positivity and two moment constraints force a lower bound after normalization.	7
Linear regularization	Removing a prescribed branch also removes the constant to be tested.	8
Moment differentiation	The error is positive with a sharp decay rate; the denominator question remains open.	9

A guide to the unfinished constructions. The following case studies are designed as starting points for further work. Each defines its coefficients without numerical fitting and gives the arguments supporting its established properties. The last column identifies what is still missing, rather than suggesting that an unfavorable convenient denominator proves failure.

Construction	Established part	Remaining issue
Interleaved frames (10)	Positivity, exact analytic rate and local prime layers.	Common numerator content after rational primitive normalization.
Quarter poles (11)	Explicit pencils; positivity in one version and conditional prime-based nonvanishing in the other.	Useful all-index primitive size estimates.
Common-base ladder (12)	Infinitely many nonzero values; decay beats the cost at 2.	A bound for the aggregate cost at odd primes.
Endpoint interpolation and prime deletion (13)	Exact endpoint parametrization and selected-prime savings.	Nonvanishing and decay compatible with the full normalization.
Modular source (14)	Exact period cancellation, a signed cusp error and coefficient arithmetic.	An integer-form construction retaining the analytic gain.
Coupled periods (15)	A five-period reduction and an explicit elimination procedure.	Nonvanishing and arithmetic after the signed elimination.
Paired identities (16)	Two exact coefficient presentations, character projection and a scoped limitation on transferred estimates.	Stronger primitive arithmetic or projected decay, and nonvanishing.

Construction	Established part	Remaining issue
Compatible multipliers (17)	A common target coefficient and eventual nonzero exponential decay.	A sufficiently small primitive multiplier.

For a first reading, Sections 3–4 give direct examples of the analytic-size versus denominator comparison. Sections 9 and 12 show what remains when the analytic estimates succeed and the arithmetic is unresolved. The signed quarter-pole case illustrates nonvanishing without positivity. The other sections can be read by method; notation and dependencies are introduced locally. The paired-identity case includes machine-readable certificates because its exact polynomial identities are too large to print usefully. Its replay program requires no original research workspace.

Four conclusions must be kept separate. A *primitive obstruction* bounds actual primitive values. An *estimate limitation* says only that a specified bound or normalization is inadequate. A *structural obstruction* disproves a particular proposed implication or transformation. *Finite evidence* concerns only the enumerated instances. The guide to the wider investigation in Section 18 preserves these distinctions, including where a later correction weakened an earlier interpretation. Stable labels such as S03 or P07 point to the source register in Appendix A and the 57-entry repository inventory. They identify records, not additional theorems proved here. The organization is by construction, rather than by the chronology of eighteen research rounds.

Standard background includes the prime number theorem $\vartheta(x) = \sum_{p\le x} \log p = x + o(x)$, elementary complex integration, Fourier convergence and the stated elementary polynomial identities. Throughout, $\chi_4(m)$ is zero for even m and $(-1)^{(m-1)/2}$ for odd m. The completed Mellin application has an additional, explicitly identified polylogarithmic input in Section 7. The modular case imports a precisely normalized classical transformation formula; the paired case uses a stated asymptotic recurrence theorem. The interleaved case states the kernel comparison imported from the companion paper. These external inputs are identified where used. The source register distinguishes the expanded cases from exploratory variants retained only as archived claims. The papers and accompanying code have passed an independent model review.

2 Positive polynomial sources: a scope boundary

Multiplying an integral representation of θ by a nonnegative polynomial is a natural way to preserve the sign while reducing the integral's size. Before restricting to a particular sequence of polynomials, we ask how large a class of rational linear forms this procedure can represent. The answer is every positive one. Thus positivity alone cannot explain failure of this approach; the restrictions on a chosen polynomial family matter.

Define

$$\mathrm{d}\nu(t) = \frac{(-\log t)^3}{96\sqrt{t}}\,\mathrm{d}t, \qquad I(P) = \int_0^1 \frac{P(t)}{1+t}\,\mathrm{d}\nu(t). \tag{2.1}$$

The density is chosen to reproduce the beta denominators:

$$\int_0^1 t^{j-1/2}(-\log t)^3\,\mathrm{d}t = \frac{6}{(j+1/2)^4} = \frac{96}{(2j+1)^4}.$$

Hence $\nu_j = \int t^j\,\mathrm{d}\nu = (2j+1)^{-4}$, and expansion of $1/(1+t)$ gives $I(1) = \theta$. Polynomial division by $1+t$ now separates any integral into a multiple of θ and a rational term. For a rational polynomial

define its *formal pair* by

$$A(P) = P(-1), \qquad B(P) = \int_0^1 \frac{P(t) - P(-1)}{1+t}\,\mathrm{d}\nu(t). \tag{2.2}$$

The quotient is a polynomial, so $B(P) \in \mathbb{Q}$ and $I(P) = A(P)\theta + B(P)$. This defines the coefficients without assuming that 1 and θ are linearly independent. For example, $P = 1$ gives $(A, B) = (1, 0)$ and $P = 1 + t$ gives $(0, 1)$. The word "formal" records the coefficients produced by this division, rather than assuming uniqueness of a numerical expression for $I(P)$.

Theorem 2.1 (The unrestricted positive cone; S01)**.** *The formal-pair image of $P \in \mathbb{Q}[t]$ nonnegative on $[0, 1]$ is*

$$\{(0, 0)\} \ \cup \ \{(A, B) \in \mathbb{Q}^2 : A\theta + B > 0\}.$$

The same positive primitive integer pairs are obtained if the source polynomials are required to have integer coefficients.

Proof. Nonzero nonnegative polynomials have strictly positive integrals. Conversely let $L = A\theta+B > 0$. If $A = 0$, use $P = B(1 + t)$, since $\nu_0 = 1$. For $A \neq 0$, first choose a nonnegative polynomial with the required value at -1 but with integral smaller than L. Then add a positive multiple of $1 + t$: this increases only the rational coefficient. For $A > 0$ use $P_n = A((1 - t)/2)^{2n}$; for $A < 0$ use $P_n = (-A)t((1 - t)/2)^{2n}$. Both have $A(P_n) = A$, are nonnegative, and satisfy $I(P_n) \leq |A|\theta 4^{-n}$. Choose n making this less than L. Then

$$P = P_n + (B - B(P_n))(1 + t)$$

is nonnegative and has exactly the desired pair; the added coefficient is rational and equals $L - I(P_n) > 0$. Clearing the rational coefficients of P multiplies its pair by a positive integer. Primitive normalization returns the original primitive pair. □

This construction encodes an already specified approximation; it does not produce a new one. In particular, shrinking positive primitive integer forms in this unrestricted cone exist if and only if θ is irrational. Necessity follows from the rational lower bound $1/b$; sufficiency follows from rational approximation to an irrational real number and Theorem 2.1. A negative theorem for all such sources would therefore be a rationality theorem, not a consequence of failures of a few prescribed weights.

Proposition 2.2 (Cost in degree; S01)**.** *Let $\rho = 3 + 2\sqrt{2}$. For $A \neq 0$ and $L = A\theta + B > 0$, the least degree $d(A, B)$ of a nonnegative rational witness satisfies*

$$d(A, B) = \frac{\log(|A|/L)}{\log \rho} + O\big(\log\log(|A|/L)\big) \qquad (|A|/L \longrightarrow \infty).$$

Proof. The point $t = -1$, where the coefficient A is fixed, lies outside the integration interval; $t \mapsto 2t - 1$ sends it to -3. The exterior growth of Chebyshev polynomials lets us fix this value while keeping the polynomial small on $[0, 1]$. Let $q_n(t) = T_n(2t - 1)/T_n(-3)$, where T_n is the Chebyshev polynomial. Then $q_n(-1) = 1$ and the identity $|T_n(-3)| = (\rho^n + \rho^{-n})/2$ gives $\|q_n\|_{[0,1]} \leq 2\rho^{-n}$. Replace P_n in the preceding proof by Aq_n^2 or $(-A)tq_n^2$. Its integral is at most $4|A|\rho^{-2n}$ and its degree at most $2n + 1$. This gives the upper bound, with an absolute additive constant.

For the converse, a polynomial cannot have a large maximum on an arbitrarily narrow region. We use a derivative bound to turn that maximum into an integral lower bound. For a nonnegative polynomial P of degree $D \geq 1$, set $M = \max_{[0,1]} P$. Its Chebyshev expansion has constant coefficient

at most M in absolute value and other coefficients at most $2M$, by the cosine coefficient integrals. The identity $T_k(\cos u) = \cos(ku)$ gives $\|T_k'\|_{[-1,1]} \le k^2$, hence $\|P'\|_{[0,1]} \le 4MD^3$. On an interval of length $\ell = 1/(8D^3)$ adjoining a maximizer, $P \ge M/2$. On its central half, $1 - t \ge \ell/4$, and $\mathrm{d}\nu/(1+t) \ge (1-t)^3\,\mathrm{d}t/192$. Thus

$$I(P) \ge \frac{M}{3 \cdot 2^{26} D^{12}}.$$

Interpolation at the $D+1$ Chebyshev extrema gives $|P(-1)| \le MT_D(3) \le M\rho^D$: the exterior Lagrange coefficients alternate in sign, and their absolute sum is $T_D(3)$. Therefore

$$I(P) \ge \frac{|P(-1)|}{3 \cdot 2^{26} D^{12} \rho^D}. \tag{2.3}$$

Taking logarithms gives $\log(|A|/L) \le D\log\rho + 12\log D + O(1)$. The upper construction already gives $D = O(\log(|A|/L))$ for a least-degree witness, so the logarithmic correction is $O(\log\log(|A|/L))$. This proves the assertion. □

The archived bounded-degree extremizers and a degree-16 witness for $5719097 - 5783031\theta > 0$ are finite evidence [record S02]. Their role is to check scope: unfavorable boundary examples do not exclude favorable interior forms. The complete bounded-degree boundary formulas are retained in the source record; they are not needed for either proof above.

3 Rational Euler sources and repeated quadratics

One way to build a small linear form is to cancel many initial terms of a generating series, leaving a rapidly decaying tail. A rational correction is attractive because integration by parts can turn its contribution into a rational boundary value. The Euler operator below multiplies t^k by k; applying it four times matches the fourth powers in the beta series. We first reduce such a construction to a polynomial integral, then test one natural repeated-factor family. The obstruction comes from fourth powers of primes that survive in the reduced form.

Let $E = t\,\mathrm{d}/\mathrm{d}t$, $g(t) = t/(1+t^2)$, and

$$J(R) = \frac{1}{6}\int_0^1 (-\log t)^3 R(t)\frac{\mathrm{d}t}{t}.$$

If $H \in \mathbb{Q}(t)$ is regular on $[0,1]$ and $H(t) = O(t)$ at zero, put $h(z) = H(e^{-z})$. Then $E = -\mathrm{d}/\mathrm{d}z$, and $J(E^4H) = \frac{1}{6}\int_0^\infty z^3 h^{(4)}(z)\,\mathrm{d}z$. Four integrations by parts leave $h(0) = H(1)$; the other boundary terms at zero contain z^3, z^2 or z, and those at infinity vanish by exponential decay. Since $J(g) = \theta$, this gives

$$J(E^4H) = H(1), \qquad J(Ag + E^4H) = A\theta + H(1). \tag{3.1}$$

This isolates the target exactly, but imposes no favorable arithmetic bound on the coefficient A.

Put $F(y) = \sum_{k\ge0}(-1)^k y^k/(2k+1)^4$. Let $Q(y) = \sum_{j=0}^d q_j y^j \in \mathbb{Q}[y]$ have no zero on $[0,1]$, and let C_m be the Taylor polynomial of QF through degree $m-1$. The choice $H(t) = -AtC_m(t^2)/Q(t^2)$ cancels the first m odd coefficients of $Ag + E^4H$: indeed $E^4(tF(t^2)) = g(t)$, and $t(F(t^2) - C_m(t^2)/Q(t^2))$ vanishes to order at least $2m+1$. To see what remains at $t = 1$, define

$$W_{Q,m}(y) = \sum_{j=m}^{d} q_j + \sum_{j=0}^{\min(m-1,d)} q_j(-y)^{m-j}.$$

For a term $q_j y^j$ with $j < m$, the factor F contributes the omitted tail $I((-t)^{m-j})$ at $y = 1$; for $j \geq m$, the whole factor F contributes $I(1)$. Weighting by q_j and summing gives $Q(1)\theta - C_m(1) = I(W_{Q,m})$, hence the exact formal-pair identity

$$\theta - \frac{C_m(1)}{Q(1)} = \frac{I(W_{Q,m})}{Q(1)}. \tag{3.2}$$

This reduction includes $m < d$ and introduces no period-independence assumption. For $Q = (v+uy)^n$ and $m = 2n$, $W_{Q,2n} = (-y)^n(u - vy)^n$. The rational Euler source therefore has the same primitive pair, up to sign, as the following scalar source.

Theorem 3.1 (Repeated-quadratic sources; S03)**.** *For even n, let u_n, v_n be positive integers with $\log(u_n + v_n) = o(n)$. Set $P_n(y) = y^n(u_n - v_n y)^n$ and let $L_n > 0$ be the value of its primitive formal pair under (2.2). Then*

$$\liminf_{\substack{n\to\infty \\ n \text{ even}}} \frac{\log L_n}{n} \geq \frac{8}{3} - \log(5 + 4\sqrt{2}) > 0.$$

Proof. The proof has two parts: primes in $(4n/3, 2n)$ force a denominator even after gcd removal, and a minimax estimate limits how small the normalized positive integral can be. Write $A = P_n(-1) = (u+v)^n$ and $I(P_n) = A\theta + N/D$ in lowest terms. The first n coefficients of $(P_n - A)/(1+y)$ are $(-1)^{j+1}A$. For a prime

$$\frac{4n}{3} < p < 2n, \qquad p \nmid (u + v),$$

only the denominator p among $1, 3, \ldots, 4n-1$ is divisible by p. Its index $(p-1)/2 < n$ has a p-unit coefficient. Consequently $v_p(N/D) = -4$, exactly. If $g = \gcd(DA, N)$, then $g \mid A$, so

$$L_n = \frac{D}{g} I(P_n) \geq \prod_{\substack{4n/3<p<2n \\ p\nmid(u+v)}} p^4 \, \frac{I(P_n)}{A}. \tag{3.3}$$

This controls the actual primitive value, including all common content. The gcd step uses $\gcd(D, N) = 1$, which implies $g \mid A$ and $D/g \geq D/A$. The selected prime factors are unavoidable factors of D, whereas A accounts for the most that gcd removal could save.

Let $r = v/(u+v)$ and $f_r(y) = y(1 - r - ry)$. With $\mathrm{d}\mu = \mathrm{d}\nu/(1+y)$, $I(P_n)/A = \int |f_r(y)|^n \, \mathrm{d}\mu(y)$. An elementary minimax calculation gives

$$M_* := \min_{0\leq r\leq 1} \max_{0\leq y\leq 1} |f_r(y)| = \frac{1}{5 + 4\sqrt{2}}.$$

Indeed for $r \geq 1/3$ the interior positive maximum is $(1-r)^2/(4r)$; for $r \geq 1/2$ the negative endpoint has magnitude $2r - 1$. They are equal at $r = (1 + 2\sqrt{2})/7$. For $r < 1/3$ the maximum is at least $1/3$.

Uniformity for moving r follows from $|f_r'| \leq 2$. In an interval of length $\ell = M_*/(4n)$ adjacent to a maximizer, $|f_r| \geq M(r)(1 - 1/(2n))$. On its central half, $1 - y \geq \ell/4$ and $\mathrm{d}\mu \geq (1-y)^3 \, \mathrm{d}y/192$. Since $(1 - 1/(2n))^n \geq 1/2$,

$$\frac{I(P_n)}{A} \geq \frac{M_*^{n+4}}{3 \cdot 2^{22} n^4}.$$

The prime number theorem in (3.3) now gives

$$\log L_n \geq \left(\frac{8}{3} - \log(5 + 4\sqrt{2})\right) n - 4 \sum_{\substack{4n/3<p<2n \\ p\mid u+v}} \log p + o(n).$$

The exceptional sum is at most $\log(u + v) = o(n)$. □

It suffices more generally that the displayed exceptional prime sum be $o(n)$. The theorem does not cover odd n, different numbers of cancelled Taylor coefficients, general denominator polynomials, or sums of these forms. The source-level statement is weaker: if $H = P/Q$ with $P, Q \in \mathbb{Z}[t]$ and $p \nmid Q(0)$, cancellation of the first m odd coefficients forces $v_p(A) \geq 4 \max_{k<m} v_p(2k+1)$. This follows coefficientwise from $A(-1)^k + (2k+1)^4[t^{2k+1}]H = 0$. It does not by itself control primitive endpoint content; that was the purpose of the prime argument.

4 Euler–Boole truncations at every endpoint

The alternating partial sums can be improved by adding an asymptotic expansion of their tails. There are two choices: where to stop the original series and how many correction terms to retain. The best real error in this family is exponentially small, on the scale $e^{-\pi N}$. However, the reduced denominators grow at rate at least $e^{8N+o(N)}$, uniformly in the number of corrections. The theorem below compares these quantities and also handles a fixed endpoint with an increasing number of correction terms.

The Euler–Boole method is classical; the point here is the uniform arithmetic conclusion for this particular family. Define integers E_{2j} by $\operatorname{sech} t = \sum_{j\geq 0} E_{2j} t^{2j}/(2j)!$ near zero. Multiplication by $\cosh t$ gives their integral recurrence. The exact tail is an integral of $t^3 e^{-2Nt} \operatorname{sech} t/12$, as shown in (4.3) below. Inserting the first m Taylor terms of $\operatorname{sech} t$ into that integral gives the proposed correction; $\int_0^\infty t^{2j+3} e^{-2Nt}\,\mathrm{d}t = (2j+3)!/(2N)^{2j+4}$ explains its coefficients. For $N \geq 1$ and $m \geq 0$ put

$$\begin{aligned} S_N &= \sum_{k<N} \frac{(-1)^k}{(2k+1)^4}, \\ T_{N,m} &= \frac{1}{12} \sum_{j<m} \frac{E_{2j}(2j+1)(2j+2)(2j+3)}{(2N)^{2j+4}}, \\ A_{N,m} &= S_N + (-1)^N T_{N,m} = a_{N,m}/q_{N,m}, \end{aligned} \tag{4.1}$$

in lowest terms with $q_{N,m} > 0$. The Euler-number convention and classical summation formula are recorded in DLMF [5, Sections 24.2, 24.17]; the needed remainder is derived below.

Theorem 4.1 (Uniform primitive obstruction; S04)**.** *Every $q_{N,m}\theta - a_{N,m}$ is nonzero. Uniformly in $m \geq 1$,*

$$|\theta - A_{N,m}| \geq \frac{\pi^2 e^{-\pi N}}{144N}. \tag{4.2}$$

Moreover

$$\liminf_{N\to\infty} \inf_{m\geq 0} \frac{\log q_{N,m}}{N} \geq 8, \qquad \liminf_{N\to\infty} \inf_{m\geq 0} \frac{\log |q_{N,m}\theta - a_{N,m}|}{N} \geq 8 - \pi > 0.$$

The primitive absolute values tend to infinity whenever $N + m \to \infty$.

Proof. Real error and its sign. Laplace integration of the alternating tail gives

$$\theta - S_N = \frac{(-1)^N}{12} \int_0^\infty t^3 e^{-2Nt} \operatorname{sech} t\,\mathrm{d}t. \tag{4.3}$$

Let $a = \pi/2$ and $a_k = (2k+1)a$. The partial fractions formula

$$\operatorname{sech} t = 2 \sum_{k\geq 0} \frac{(-1)^k a_k}{a_k^2 + t^2}$$

follows by expanding $\cosh(tx)/\cosh t$ on $[0,1]$ in the mixed cosine basis $\cos(a_k x)$: twice integrating its differential equation gives coefficient $2a_k(-1)^k/(t^2+a_k^2)$, and Fourier convergence at $x=0$ gives the formula. Applying the finite geometric identity to each term gives, for $m \geq 1$,

$$\operatorname{sech} t - \sum_{j<m} \frac{E_{2j} t^{2j}}{(2j)!} = (-1)^m t^{2m} C_m(t),$$

$$C_m(t) = 2 \sum_{k \geq 0} \frac{(-1)^k a_k^{1-2m}}{t^2 + a_k^2}. \tag{4.4}$$

This remainder series is absolutely convergent. Its positive terms decrease, and the ratio of the second to the first is at most $3^{1-2m} \leq 1/3$. Hence

$$(1 - 3^{1-2m}) \frac{2a^{1-2m}}{t^2+a^2} \leq C_m(t) \leq \frac{2a^{1-2m}}{t^2+a^2}, \qquad C_m(t) > 0. \tag{4.5}$$

Thus $\theta - A_{N,m} = (-1)^{N+m} \epsilon_{N,m}$ with $\epsilon_{N,m} = \int_0^\infty t^{2m+3} e^{-2Nt} C_m(t)\, \mathrm{d}t/12 > 0$. On $t \geq a$, use $(t/a)^{2m} \geq 1$ and $t^3/(t^2+a^2) \geq a/2$. The lower bound becomes $\epsilon_{N,m} \geq (\pi/18)(a/2) e^{-2Na}/(2N)$, proving (4.2). For $m = 0$, $\operatorname{sech} t \geq e^{-t}$ gives $|\theta - S_N| \geq 1/[2(2N+1)^4]$.

Unavoidable reduced denominators. The correction terms have denominators supported only on primes dividing $6N$. Primes elsewhere that survive in the partial sum therefore cannot be cancelled by taking more correction terms. Let p be odd, $p^2 > 2N$ and $p < 2N$, and let J count the positive odd r with $pr < 2N$. Reduction yields

$$p^4 S_N \equiv \chi_4(p) S_J \pmod{p}, \qquad \chi_4(p) = (-1)^{(p-1)/2}. \tag{4.6}$$

If $p \nmid 6N$, the correction $T_{N,m}$ is p-integral for every m. Whenever S_J is a nonzero residue, therefore, $v_p(q_{N,m}) = 4$ exactly. Fix $K \geq 1$. For all sufficiently large N, primes in $(2N/(2K+1), 2N)$ have $1 \leq J \leq K$, satisfy $p^2 > 2N$, and avoid the finitely many numerator and denominator prime divisors of the positive rationals $S_1, \ldots, S_K$. Uniformly in m,

$$\log q_{N,m} \geq 4\{\vartheta(2N) - \vartheta(2N/(2K+1)) - \log(6N)\} - O_K(\log N).$$

First take the lower limit in N, then let $K \to \infty$; no uniformity in K is required. This gives the rate 8, and combining with the error estimates gives $8 - \pi$.

The remaining parameter regime. The estimates above handle every sequence with $N \to \infty$, uniformly in m. For bounded N and $m \to \infty$, restrict the positive remainder integral to $[b, b+1]$, where $b > a$. Its lower bound is $c_{N,b}(b/a)^{2m} \to \infty$, uniformly over a finite set of N. Since $q \geq 1$, this also proves divergence on every escaping parameter sequence. □

The best number of corrections balances the factor t^{2m} against the Laplace decay near the first pole scale $a = \pi/2$. This suggests $m \sim aN$. The raw exponential constant is indeed sharp: for $m = \pi N/2 + O(1)$,

$$|\theta - A_{N,m}| \sim \frac{\pi^3}{48\sqrt{2N}} e^{-\pi N}. \tag{4.7}$$

Indeed (4.5) reduces the integral to its first pole with relative error $O(3^{1-2m})$. The exponent is $N(2a\log(t/a) - 2t)$, with unique maximum at $t = a$, second derivative $-2/a$, and remaining factor $(\pi/12)t^3/(t^2+a^2)$ there equal to $\pi a/24$. Gaussian localization gives (4.7), uniformly for the bounded rounding error in m.

There are finitely many small primitive values, for example those of $X-1$, $81X-80$ and $32X-31$. They do not contradict divergence. Nonvanishing and divergence off finite parameter sets give $c := \inf_{N,m} |q_{N,m}\theta - a_{N,m}| > 0$. By Gauss's lemma a product of d of these primitive linear polynomials is primitive, and its dth-root absolute value is at least c. Products alone therefore cannot supply the degree-normalized smallness criterion. Sums, resultants and general determinant combinations are outside this result.

5 Differentiated continued fractions at fixed endpoints

Continued fractions give rational approximations to a tail integral without truncating its asymptotic series. Here three derivatives with respect to the Laplace parameter introduce the factor t^3 needed for the fourth-power beta tail. The resulting approximation has an explicit error asymptotic, but differentiating a quotient also raises powers of its denominator. For a fixed tail endpoint and increasing even depth, the error decays only polynomially while the reduced denominator grows exponentially. The proof checks this cost after all cancellation.

For $\Re a > 0$ let $S(a) = \int_0^\infty e^{-at} \operatorname{sech} t \, \mathrm{d}t$. The numerator and denominator polynomials of the approximants, called continuants, are defined by

$$q_0 = 1,\ q_1 = a, \quad r_0 = 0,\ r_1 = 1, \qquad (n+1)y_{n+1} = ay_n + ny_{n-1} \quad (n \geq 1).$$

For example, $r_1/q_1 = 1/a$ and $r_2/q_2 = a/(a^2+1)$. We use q_n, r_n for analytic generating functions and the integer polynomials $Q_n = n!q_n$, $R_n = n!r_n$ for prime divisibility. They satisfy $Y_{n+1} = aY_n + n^2 Y_{n-1}$. Their quotient is the finite continued fraction with successive numerators $1^2, 2^2, \ldots$. Its convergence will follow from an explicit error, rather than an imported convergence theorem. Since three derivatives of $S(a)$ insert $-t^3$, the exact tail identity (4.3) reads $\theta = S_N + (-1)^{N+1} S'''(2N)/12$. Replacing S by its rational approximant therefore gives

$$\alpha_{n,N} = S_N + \frac{(-1)^{N+1}}{12} \frac{\mathrm{d}^3}{\mathrm{d}a^3} \frac{r_n(a)}{q_n(a)}\bigg|_{a=2N} = \frac{b_{n,N}}{d_{n,N}}, \tag{5.1}$$

in lowest terms, $d_{n,N} > 0$. Here S_N is the finite sum in (4.1). The entire quotient is differentiated.

Theorem 5.1 (Fixed-endpoint obstruction; S05)**.** *For each fixed integer $N \geq 1$,*

$$\theta - \alpha_{n,N} \sim (-1)^{N+n} \frac{2^{-2N}\Gamma(N+\frac{1}{2})^2}{12} n^{-2N} (\log n)^3. \tag{5.2}$$

Along even depth $n = 2m$,

$$\liminf_{m\to\infty} \frac{\log |d_{2m,N}\theta - b_{2m,N}|}{2m} \geq 4.$$

No uniformity for growing N, or arithmetic conclusion at odd depth, is part of this theorem.

5.1 The exact error and its derivative

Define the ordinary generating functions $\mathcal{Q}(a,z) = \sum_{n\geq 0} q_n(a) z^n$ and $\mathcal{R}(a,z) = \sum_{n\geq 0} r_n(a) z^n$. Summing the recurrence gives

$$(1-z^2)\partial_z \mathcal{Q} = (a+z)\mathcal{Q}, \qquad (1-z^2)\partial_z \mathcal{R} = (a+z)\mathcal{R} + 1.$$

Their initial values are 1 and 0. Solving these equations gives

$$\mathcal{Q}(a,z) = (1+z)^{(a-1)/2}(1-z)^{-(a+1)/2},$$
$$\mathcal{R}(a,z) = \mathcal{Q}(a,z)\int_0^z \frac{\mathrm{d}t}{(1-t^2)\mathcal{Q}(a,t)}.$$

Substitution $z = \tanh u$ and translation of the tail integral give

$$\mathcal{Q}(a,z)S(a) - \mathcal{R}(a,z) = \int_0^\infty \frac{e^{-av}}{\cosh v + z\sinh v}\,\mathrm{d}v.$$

Expanding on compact subdisks $|z| < 1$ proves

$$q_n(a)S(a) - r_n(a) = (-1)^n I_n(a), \qquad I_n(a) = \int_0^\infty e^{-at}\operatorname{sech} t\tanh^n t\,\mathrm{d}t. \tag{5.3}$$

Since $\theta = S_N + (-1)^{N+1}S'''(2N)/12$,

$$\theta - \alpha_{n,N} = \frac{(-1)^{N+n+1}}{12}\,\frac{\mathrm{d}^3}{\mathrm{d}a^3}\frac{I_n(a)}{q_n(a)}\bigg|_{a=2N}.$$

Uniformly on a small fixed complex disk about $2N$ with $\Re a > 1$,

$$q_n(a) = \frac{2^{(a-1)/2}n^{(a-1)/2}}{\Gamma((a+1)/2)}(1 + O_N(n^{-1})),$$
$$I_n(a) = 2^{-(a+1)/2}\Gamma((a+1)/2)n^{-(a+1)/2}(1 + O_N(n^{-1})).$$

For the first formula, expand the displayed generating function at $z = 1$ and use the coefficient formula for $(1-z)^{-b}$; the second singularity $z = -1$ has relative size $O(n^{-\Re a})$. For the second, put $y = \tanh t$ and expand $(1+y)^{-(a+1)/2}$ at $y = 1$ in

$$I_n(a) = \int_0^1 y^n(1-y)^{(a-1)/2}(1+y)^{-(a+1)/2}\,\mathrm{d}y.$$

The beta integral $\int_0^1 y^n(1-y)^{b-1}\,\mathrm{d}y = \Gamma(n+1)\Gamma(b)/\Gamma(n+b+1)$ and the gamma-ratio estimate give the stated constants. The remainders are holomorphic and uniform on a slightly larger disk; the leading factor of q_n is nonzero there for large n. Cauchy's derivative formula therefore justifies three derivatives of $2^{-a}\Gamma((a+1)/2)^2 n^{-a}(1 + O_N(n^{-1}))$. Its leading third derivative is the negative of its leading term times $(\log n)^3$, proving (5.2). The standard gamma-ratio expansion is also recorded in DLMF [5, Section 5.11(iii)]. Pointwise real asymptotics without this complex uniformity would not justify the differentiation.

5.2 An exact reduced-denominator law

For even depth $2m$, primes in $(m, 2m)$ contribute one factor to $(2m)!$ and one to the central binomial factor below. A third derivative of a quotient can raise the denominator to its fourth power, suggesting an exponent of at least eight. To prove that those factors survive reduction, we must also show that the relevant numerator and first derivative are p-units.

The following polynomials separate the dependence on the fixed endpoint N from the growing depth m. The polynomial $C_N(m)$ will identify the exceptional primes; its fixed degree makes

their total contribution negligible in the prime sum. Write $X^{\underline{j}} = X(X-1)\cdots(X-j+1)$ and $h_N = (2N-1)!!$. Define integer polynomials

$$\begin{aligned}
A_N(X) &= \sum_{j=0}^{N} \binom{2N}{2j} 2^j X^{\underline{j}} \prod_{\ell=0}^{N-j-1} (2X+2\ell+1),\\
B_N(X) &= \sum_{j=0}^{N-1} \binom{2N}{2j+1} 2^j X^{\underline{j}} \prod_{\ell=0}^{N-j-1} (2X+2\ell+1),\\
C_N(m) &= B_N(m-\tfrac{1}{2})\\
&= \sum_{j=0}^{N-1} \binom{2N}{2j+1} \prod_{s=0}^{j-1}(2m-1-2s) \prod_{\ell=0}^{N-j-1} (2m+2\ell).
\end{aligned}$$

The last expression is integral for integer m, positive for $m \geq N$, and of fixed degree N. With $c_m = \binom{2m}{m}/4^m$,

$$q_{2m}(2N) = c_m A_N(m)/h_N, \qquad q_{2r+1}(2N) = c_r B_N(r)/h_N. \tag{5.4}$$

To check these, write $\mathcal{Q}(2N,z) = (1+z)^{2N}(1-z^2)^{-N-1/2}$ and separate even and odd coefficients. With $(u)_j = u(u+1)\cdots(u+j-1)$ denoting the rising factorial, the coefficient ratio used is

$$\frac{(N+\frac{1}{2})_{m-j}/(m-j)!}{c_m} = \frac{m^{\underline{j}}(m+\frac{1}{2})_{N-j}}{(\frac{1}{2})_N}.$$

Lemma 5.2. *If $m < p < 2m$, $2N < p$, p is prime and $p \nmid C_N(m)$, then*

$$v_p(d_{2m,N}) = 8 + 4v_p(A_N(m)).$$

Proof. Unit factors. We first establish $Q_p(a) \equiv a^p - a \pmod p$. To make this congruence transparent, realize Q_p as a characteristic polynomial. Use $D = (1-z^2)\mathrm{d}/\mathrm{d}z + (p-1)z$ on polynomials of degree at most $p-1$ over $\mathbb{F}_p$. Since $D(z^j) = jz^{j-1} + (p-1-j)z^{j+1}$, its matrix is tridiagonal. Adjacent off-diagonal products are $j(p-j)$; expanding characteristic determinants gives the continuant recurrence modulo p. Its eigenvectors $(1+z)^j(1-z)^{p-1-j}$ have distinct eigenvalues $2j-(p-1)$, which run through $\mathbb{F}_p$. Thus its characteristic polynomial is $\prod_{c\in\mathbb{F}_p}(a-c) = a^p - a$. Starting again at index p, the recurrence reduces to the same recurrence as at zero, because $(p+j)^2 \equiv j^2$. It gives

$$Q_{p+j}(a) \equiv (a^p - a)Q_j(a) \pmod p \quad (0 \leq j < p).$$

The continuant Wronskian is $Q_k R_{k+1} - Q_{k+1}R_k = (-1)^k(k!)^2$. At $k = p-1$, $a = 2N$, it shows that $R_p(2N)$ is a unit. Recurrence after p and differentiation of the last congruence give

$$R_{p+j}(2N) \equiv R_p(2N)Q_j(2N), \qquad Q'_{p+j}(2N) \equiv -Q_j(2N) \pmod p.$$

Here $j = 2m-p$ is odd, and (5.4) shows that $Q_j(2N)$ is a unit precisely when $B_N(m-(p+1)/2) \equiv C_N(m)$ is. Thus $R_{2m}(2N)$ and $Q'_{2m}(2N)$ are units.

Order of the denominator. Meanwhile (5.4) gives

$$e := v_p(Q_{2m}(2N)) = 2 + v_p(A_N(m)),$$

because both $(2m)!$ and c_m have valuation one.

Survival under differentiation. Write $Q = Q_{2m}$ and $R = R_{2m}$, and expand the integer polynomials as $Q(2N + x) = \sum u_i x^i$ and $R(2N + x) = \sum v_i x^i$. The coefficient of x^3 in R/Q is

$$\frac{v_3u_0^3 - v_2u_1u_0^2 + v_1(u_1^2u_0 - u_2u_0^2) + v_0(-u_1^3 + 2u_1u_2u_0 - u_3u_0^2)}{u_0^4}.$$

Its numerator has the unique unit term $-v_0u_1^3$, so its valuation is $-4e$. Formula (5.1) multiplies this Taylor coefficient by $\pm 1/2$; S_N is p-integral. It follows that the reduced denominator has valuation $4e$, as asserted. □

For fixed N, $C_N(m)$ is a positive integer of polynomial size for all sufficiently large m, so $\log C_N(m) = O_N(\log m)$. The primes in $(m, 2m)$ dividing it have at most this total logarithm. The lemma and the prime number theorem imply $\log d_{2m,N} \geq 8m + o(m)$. Equation (5.2) supplies an eventually nonzero error with logarithm $O_N(\log m)$, completing Theorem 5.1. The exception condition is material: four archived cases excluded by $C_N(m)$ have denominator exponent four, not eight. None of this extends automatically to N growing with m.

A shift-and-binomial repair has an exact reduction: combining the error integrals at $a = 2(L + j)$ uses

$$e^{-2Lt} \tanh^n t(1 + e^{-2t})^n = e^{-2Lt}(1 - e^{-2t})^n.$$

After $x = e^{-t}$ this is the scalar Mellin weight $x^{2L}(1 - x^2)^n$. The proposed repair is therefore an old scalar family in a different presentation, not an additional approximation.

6 A sine-normalized cyclotomic family

Algebraic approximations offer another route: choose an algebraic number α close to θ and evaluate its integer minimal polynomial at θ. This evaluation contains a factor $\theta - \alpha$, but also a factor for every other conjugate of α. A good approximation at one real embedding can therefore be defeated by large conjugates. The roots-of-unity family below exhibits this explicitly, including the removal of common factors from the polynomial coefficients.

The topological-signature motivation came from the cubic cotangent formula recorded by Marché and Masbaum [10, Corollary 3.2], who credit its simplification to Murakami. That signature formula is not an even-power integrality theorem and is not used below. The following algebraic and analytic claims are derived directly for one specified normalization.

For an odd prime p, set $h = \pi/(2p)$ and $w = e^{2\pi i/(4p)}$. The ratio $\sin^4 h / \sin^4(nh)$ tends to n^{-4} for fixed n, while $\cos^3(nh)$ tends to one. Thus the following trigonometric sum is a natural candidate approximation to the beta series:

$$\alpha_p = \sin^4 h \sum_{\substack{1 \leq n < p \\ n \text{ odd}}} \chi_4(n) \frac{\cos^3(nh)}{\sin^4(nh)}. \tag{6.1}$$

Let $F_p = \mathbb{Q}(w + w^{-1})$, of degree $d = p - 1$, and $M_p(X) = \mathrm{Norm}_{F_p/\mathbb{Q}}(X - \alpha_p)$. Here the norm polynomial is the product of $X - \sigma(\alpha_p)$ over all embeddings $\sigma : F_p \hookrightarrow \mathbb{R}$.

Theorem 6.1 (One cyclotomic norm obstruction; S06)**.** *The number $8\alpha_p$ is integral, α_p has degree $p - 1$, and its primitive minimal polynomial P_p satisfies*

$$\log |P_p(\theta)| = 4(p - 1) \log p + O(p).$$

In particular these primitive evaluations grow along the odd primes.

Proof. We first control denominators and the principal approximation, then bound every conjugate and form their product. The latter step is essential: a small factor at one embedding does not make a small norm.

Integrality. The identity

$$8\alpha_p = \sum_{\substack{n<p \\ n \text{ odd}}} \chi_4(n) \left(\frac{w - w^{-1}}{w^n - w^{-n}} \right)^4 (w^n + w^{-n})^3$$

proves integrality. Indeed $z = w^2$ is a primitive $2p$th root and $1 - z^n$ is a cyclotomic unit for odd $0 < n < p$, since $\Phi_{2p}(1) = 1$. Here a cyclotomic unit means an algebraic integer whose inverse is also integral. Polynomial division of $\Phi_{2p}(x) - \Phi_{2p}(1)$ by $x - 1$ proves the unit assertion. The displayed expression is real and belongs to F_p.

The principal approximation. The signed expansion is

$$\alpha_p = \theta - \frac{\pi^2(4\theta + 5\beta(2))}{24p^2} + O(p^{-4}). \tag{6.2}$$

To prove it, write $\cos^3 x / \sin^4 x = x^{-4} - (5/6)x^{-2} + R(x)$, where R is smooth on $[0, \pi/2]$. Alternating samples of R have uniformly bounded sum by its total variation, and the alternating tails of n^{-4} and n^{-2} are bounded by p^{-4} and p^{-2}. Multiply by $\sin^4 h$ and use $(\sin h/h)^4 = 1 - (2/3)h^2 + O(h^4)$. The coefficient of p^{-2} is strictly positive.

The other embeddings. An embedding sends $w + w^{-1}$ to $w^k + w^{-k}$; k and $-k$ give the same real embedding. Represent them by odd $1 \le k < 2p$, $k \ne p$. The conjugate $\alpha_{p,k}$ is given by (6.1) with each angle multiplied by k. Let

$$f(x) = \frac{\cos^3 x}{\sin^4 x}, \qquad C = \sum_{n \ge 1,\ n \text{ odd}} n^{-4} = \frac{\pi^4}{96}, \qquad \delta = 2 - C > 0.$$

The function $x^4 f(x)$ strictly decreases on $(0, \pi/2)$. The sign of its logarithmic derivative reduces to positivity of $H(x) = x(3 + \cos^2 x) - 4 \sin x \cos x$; indeed $H(0) = 0$ and $H'(x) = 7 \sin^2 x - 2x \sin x \cos x \ge 5 \sin^2 x$. Thus $f(mh) \le f(h)/m^4$ for odd $1 \le m < p$.

Multiplication by k permutes the pairs of odd nonzero residues $\{n, 2p - n\}$ modulo $2p$. The inner conjugate sum is therefore $\sum_m \eta_m f(mh)$, where each $\eta_m = \pm 1$ and each odd $0 < m < p$ occurs once. Its first term exceeds the sum of the absolute values of the rest. Uniformly in k,

$$\delta \cos^3 h \left(\frac{\sin(kh)}{\sin h} \right)^4 \le |\alpha_{p,k}| \le C \cos^3 h \left(\frac{\sin(kh)}{\sin h} \right)^4. \tag{6.3}$$

For $p \ge 5$ and $k \notin \{1, 2p - 1\}$, the sine ratio is at least $\sin(3h)/\sin h \ge 2$, so these conjugates have absolute value greater than 10 (use $\delta > 0.98$, $\cos^3 h \ge 3\sqrt{3}/8$). The principal conjugate is in $(0, C)$ and the $2p - 1$ conjugate is its negative. Hence it is the unique conjugate in $(0, 2)$ and its stabilizer is trivial. The degree is d. For $p = 3$, directly $\alpha_3 = 3\sqrt{3}/8$ has degree two.

The full norm and primitive normalization. Individual conjugates have very different sizes, but their sine factors have an exact product. The odd sine product is

$$\prod_{\substack{1 \le k < 2p \\ k \text{ odd},\ k \ne p}} \sin(kh) = 2^{1-p} = 2^{-d}.$$

It follows by evaluating $x^p + 1$ at $x = 1$ in its root factorization and removing the sine-one factor. Therefore (6.3) implies

$$\left(\frac{\delta \cos^3 h}{16 \sin^4 h}\right)^d \le \prod_k |\alpha_{p,k}| \le \left(\frac{C \cos^3 h}{16 \sin^4 h}\right)^d.$$

For $k \notin \{1, 2p-1\}$, replacing $|\alpha_{p,k}|$ by $|\theta - \alpha_{p,k}|$ changes the product logarithm by $O(d)$, since $0 < \theta < 1$ and their magnitudes exceed 10. For the two remaining conjugates, (6.2) gives a total change $-2\log p + O(1)$. Since $\sin h \sim \pi/(2p)$, $\log |M_p(\theta)| = 4d \log p + O(p)$.

Finally $J_p(X) = \mathrm{Norm}(8X - 8\alpha_p) = 8^d M_p(X)$ is integral, and its content g_p divides its leading coefficient 8^d. Thus $P_p = J_p/g_p$ is the actual primitive minimal polynomial and $|M_p(\theta)| \le |P_p(\theta)| \le 8^d |M_p(\theta)|$. This proves the theorem after all content is removed. □

The norm polynomial is evaluated at the real number θ; no assertion that $\theta - \alpha_p \in F_p$ is made without first assuming θ rational. Under $\theta = a/b$, the integer test is $b^d P_p(a/b) \in \mathbb{Z} \setminus \{0\}$. Its degree cost cannot be omitted. Neither other normalizations nor other nonlinear norms are excluded.

7 A constrained completed Mellin source

Functional identities can express θ through polylogarithms at arguments strictly inside the unit disk, where their series converge geometrically. This suggests a source of accurate approximations. Turning such identities into polynomially weighted integrals, however, introduces lower-order logarithmic terms. Cancelling those terms imposes two moment constraints. For the resulting kernel, these constraints force a positive lower bound whenever the multiplier polynomial is nonnegative; the hoped-for small positive forms cannot arise in that class.

This section separates an analytic inequality for an explicit rational kernel from its polylogarithmic identification with θ. The latter uses a published identity; it is not a newly discovered functional equation.

7.1 The external identity and the rational kernel

For $|z| < 1$, the polylogarithm is $\mathrm{Li}_s(z) = \sum_{k \ge 1} z^k/k^s$. Its relation to the integrals here is

$$\int_0^1 x^{j-1} \frac{(-\log x)^{s-1}}{(s-1)!}\, \mathrm{d}x = j^{-s} \qquad (j, s \text{ positive integers}),$$

obtained by $x = e^{-t}$ and integration by parts. A *completion* adds lower-order polylogarithms with logarithmic coefficients to obtain the cancellation required by the identity. In this case set

$$D_4(z) = \Im\left(\mathrm{Li}_4(z) - \log|z|\,\mathrm{Li}_3(z) + \frac{\log^2|z|}{3}\,\mathrm{Li}_2(z)\right).$$

The weight-two, three and four ladder identities of Broadhurst [2, equations (39)–(46), (54)–(55)] give

$$\theta = \frac{225}{136} D_4(i/2) - \frac{25}{34} D_4((1+i)/4) + \frac{9}{68} D_4((1-i)/8). \tag{7.1}$$

For clarity about completion, write P_s, Q_s, R_s for the raw ladders

$$2^{s-1}\Im\,\mathrm{Li}_s((1+i)/2), \quad (2/3)^{s-1}\Im\,\mathrm{Li}_s((1+i)/4) + \Im\,\mathrm{Li}_s(i/2), \quad (2/5)^{s-1}\Im\,\mathrm{Li}_s((1-i)/8) + 2\Im\,\mathrm{Li}_s(i/2).$$

With $\ell = \log 2$ and $\widehat{X} = X_4 + \ell X_3 + (\ell^2/3)X_2$, those source identities give $\widehat{Q} - (2/3)\widehat{P} = -(80/27)\theta$ and $\widehat{R} - (4/5)\widehat{P} = -(384/125)\theta$. Their linear combination eliminating $\widehat{P}$ yields (7.1). The $\beta(3)\ell$ and $\beta(1)\ell^3$ terms cancel only after this completion. Broadhurst derives the source identities by specializing Kummer's functional equations. That external derivation is not reproduced or independently certified here. The inequality below is a direct theorem about U and W and does not depend on accepting (7.1); its interpretation as a θ-form uses this explicitly declared input.

The three arguments in (7.1) have absolute values $2^{-m/2}$ with $m = 2, 3, 5$. Using x^m in their geometric series therefore puts all three completions under the same logarithmic weight: the factor m^4 compensates for the fourth-power moment denominator. Let $r = -\log x$, $L = \log 2$, and

$$\begin{aligned} W(r) &= r^3/6 + Lr^2/4 + L^2 r/12, \\ U(x) &= \frac{900x^2}{17(x^4+4)} - \frac{2025x^3}{17(x^6-4x^3+8)} - \frac{5625x^5}{17(x^{10}-8x^5+32)}. \end{aligned} \tag{7.2}$$

For $m = 2, 3, 5$, respectively, put

$$(z_m) = (i/2, (1+i)/4, (1-i)/8), \qquad (c_m) = (225/136, -25/34, 9/68).$$

Then $U = \sum c_m m^4 \Im[z_m x^m/(1 - z_m x^m)]$. For the kth geometric term, integration of the common weight gives

$$m^4 \int_0^1 x^{mk} W(-\log x)\frac{\mathrm{d}x}{x} = \frac{1}{k^4} + \frac{mL}{2k^3} + \frac{m^2L^2}{12k^2}.$$

Because $-\log|z_m| = mL/2$, these are exactly the three coefficients in $D_4(z_m)$. Thus geometric expansion shows that

$$\Theta := \int_0^1 U(x)W(-\log x)\frac{\mathrm{d}x}{x} = \sum_m c_m D_4(z_m) = \theta, \tag{7.3}$$

where only the final equality uses (7.1).

Write $p_2 = x^4 + 4$, $p_3 = x^6 - 4x^3 + 8$, $p_5 = x^{10} - 8x^5 + 32$ and

$$D = \frac{p_2 p_3 p_5}{(x^2+2x+2)^2}, \qquad N = DU.$$

The three denominators share the factor $x^2 + 2x + 2$; removing its two repeated copies gives the common denominator D of degree 16. Exact division shows that N is a polynomial and D is monic and positive on $[0, 1]$.

We seek polynomial weights whose contribution is a multiple of Θ plus a rational correction. Choosing $P = A + DV$ gives $UP = AU + NV$, so the correction is a polynomial integral. With $A \in \mathbb{Q}$, $V \in \mathbb{Q}[x]$, write $NV = \sum_{j\ge1} a_j x^j$ and impose

$$\sum_j \frac{a_j}{j^3} = 0, \qquad \sum_j \frac{a_j}{j^2} = 0. \tag{7.4}$$

Termwise integration proves

$$I_U(P) := \int_0^1 UPW\frac{\mathrm{d}x}{x} = A\Theta + \sum_j \frac{a_j}{j^4}. \tag{7.5}$$

Without both constraints, there are extra $L/2$ and $L^2/12$ times the two sums; congruence modulo D alone does not isolate θ.

7.2 A positive-source lower bound

The difficulty is that U is not everywhere positive. It has exactly one sign change. The two moment constraints allow us to subtract linear and quadratic terms from W until the remainder has the same sign as U. Their product is then positive, producing a lower bound that no choice of the nonnegative weight P can evade.

Theorem 7.1 (Constrained positive sources; S09)**.** *If $P = A + DV$ is nonzero, rational and nonnegative on $[0,1]$, and satisfies (7.4), then $A > 0$ and $I_U(P) > 1309A/2600$. Consequently every positive rational normalization making A a nonzero integer gives a value greater than $1/2$. This includes primitive normalization of (7.5).*

Proof. Locate the sign change. One has $U = 225x^2 f(x)/(17D(x))$, with

$$f(x) = 4x^{12} - 25x^{11} + 68x^{10} - 145x^9 + 220x^8 - 232x^7 + 260x^6 \\ - 464x^5 + 880x^4 - 1160x^3 + 1088x^2 - 800x + 256.$$

A rational Sturm sequence has degrees $12, 11, \dots, 0$ and variation counts $7, 7, 6, 6$ at $0, 3/5, 5/8, 1$. The rational certificate accompanying this supplement recomputes the sequence. Also $f(0) = 256$, $f(1) = -50$, $f(3/5) = 802361314/244140625 > 0$, and $f(5/8) = -31248983321/17179869184 < 0$. Thus U changes sign exactly once, at $3/5 < x_0 < 5/8$, and its sign is the sign of $r - r_0$ with $r_0 = -\log x_0$.

Use the constrained moments. Let $M_j = \int_0^1 U(x) r^j \, \mathrm{d}x/x$, $j = 1, 2$. The constraints give $\int UPr^j \, \mathrm{d}x/x = AM_j$. Hence

$$\int_0^1 UP\, r(r - r_0) \frac{\mathrm{d}x}{x} = A(M_2 - r_0 M_1) > 0.$$

The same integral with $P = 1$ proves $M_2 - r_0 M_1 > 0$, so $A > 0$. Now

$$W(r) - (L/4 + r_0/6)r^2 - (L^2/12)r = r^2(r - r_0)/6.$$

Multiplication by UP and integration gives the strict bound $I_U(P) > A[(L/4 + r_0/6)M_2 + (L^2/12)M_1]$.

Bound the numerical constants. The exact rational tail estimates described below prove $M_1 > 9/10$ and $M_2 > 193/100$. The elementary inequality $\log a > 2(a-1)/(a+1)$ for $a > 1$ gives $L > 2/3$ and $r_0 > \log(8/5) > 6/13$. Therefore the bracket exceeds

$$(1/6 + 1/13)(193/100) + (1/27)(9/10) = 1309/2600 > 1/2.$$

If κA is a nonzero integer and $\kappa > 0$, it is at least one. This proves the normalization conclusion. □

Here the finite certificate has a precise mathematical role. The moment formulas are

$$M_1 = \sum_m c_m m^2 \Im \operatorname{Li}_2(z_m), \qquad M_2 = 2 \sum_m c_m m \Im \operatorname{Li}_3(z_m).$$

Use $\rho_m = 1/2, 3/8, 3/16$, respectively. For $s = 2, 3$, the remainder after $K = 32$ Gaussian-rational terms is at most

$$\frac{\rho_m^{K+1}}{(K+1)^s(1 - \rho_m)}.$$

All endpoints are rational. Their resulting intervals are contained in

$$(0.9006560116125, 0.9006560116154), \qquad (1.9321066776805, 1.9321066776807),$$

respectively; the exact fractions, not those decimal displays, are used in the checks. Together with the Sturm sequence, this is a finite exact certificate for the theorem's constants.

For signed P, the proof no longer bounds the actual integral. It still gives a limitation on the absolute-integrand estimate:

$$\int_0^1 |UP|W\frac{\mathrm{d}x}{x} \geq \frac{L^2}{12}|A|M_1.$$

After κA is a nonzero integer, the right side exceeds $1/30$. Signed cancellation is not excluded. In fact the archived Rodrigues subspace [record L03] reduces the rational moment denominator to a single lcm, but also contains explicit nonzero sources with zero formal pair. Small source coefficients alone do not settle endpoint nonvanishing.

8 A linear regularization that erases its endpoint constant

The arithmetic-holonomy framework of Calegari, Dimitrov and Tang [3] motivates trying to turn rationality of a constant into a restriction on analytic continuation. Here we examine one preliminary step of such a proposal. Adding a constant makes all Taylor coefficients at one point rational, but leaves a logarithmic branch at another point. Could a linear differential operation remove that branch while retaining the constant we want to test?

The answer for this particular function is no: any such operation also annihilates constants. The proof uses *monodromy*, the change in a function after analytic continuation around a singularity. One circuit followed by a second produces a nonzero additive constant, forcing the conclusion. No general holonomy bound is applied.

Let $E = x\,\mathrm{d}/\mathrm{d}x$ and $F(x) = \sum_{m\geq 1}\chi_4(m)x^m/m^4$. Put $F_j = E^jF$ and choose the logarithmic coefficients below so that differentiating successive terms cancels all but F_4:

$$C(x) = F_0 - \log x\,F_1 + \frac{\log^2 x}{2}F_2 - \frac{\log^3 x}{6}F_3. \tag{8.1}$$

Since $E^4F = x/(1+x^2)$, differentiation telescopes:

$$C'(x) = -\frac{\log^3 x}{6(1+x^2)}, \quad C(x) = \int_0^x \frac{-\log^3 t}{6(1+t^2)}\,\mathrm{d}t, \quad C(1) = \theta.$$

The derivative C' has rational Taylor coefficients at $x = 1$, because both $\log x$ and $(1+x^2)^{-1}$ do. Thus the only potentially irrational coefficient of the Taylor expansion of C there is its constant term $C(1) = \theta$. For any rational q, $C_q = C + q - \theta$ has an entirely rational Taylor expansion (a rational germ) at 1. This change does not remove the logarithmic branch at zero. The question is whether a linear differential operation can remove that branch without also erasing $q - \theta$.

Proposition 8.1 (Scoped monodromy obstruction; E04)**.** *Let L be a linear differential operator with rational-function coefficients in x, even over $\mathbb{C}$. If $L(C_q)$ is meromorphic near zero, then $L(1) = 0$. In particular $L(C_q) = L(C)$ is independent of the additive constant $q - \theta$.*

Proof. Let M_0 denote continuation once positively about zero and $\Delta = (M_0 - 1)C_q$. All F_j are holomorphic at zero, and

$$\Delta'(x) = -\frac{(\log x + 2\pi i)^3 - \log^3 x}{6(1+x^2)}.$$

Because the rational-function coefficients of L are single-valued, L commutes with continuation. If $L(C_q)$ is meromorphic near zero, a loop about zero leaves it unchanged; hence $L(\Delta) = (M_0 - 1)L(C_q) = 0$. Continue to $x = i$ on the branch with $\log i = \pi i/2$. Then

$$\operatorname{Res}_{x=i} \Delta' = -\frac{(5\pi i/2)^3 - (\pi i/2)^3}{12i} = \frac{31\pi^3}{24} \neq 0.$$

A primitive of a function with residue r acquires the constant $2\pi i r$ after a loop. Thus a small circuit about i adds $31i\pi^4/12$ to Δ. The coefficients of L are single-valued. Continuing $L(\Delta) = 0$ and subtracting thus gives $(31i\pi^4/12)L(1) = 0$. Paths can avoid the finite set of poles of L. □

One might instead change coordinates to exploit symmetry. Near $x = 1$, $C(1/x) = C(x)$ follows by differentiation and equality at 1. The coordinate $x = (1 - z)/(1 + z)$ turns $x \mapsto 1/x$ into $z \mapsto -z$, so the resulting function is even in z and can be written using $y = z^2$. It yields

$$C(x) = \theta + J(y), \qquad J'(y) = -\frac{2}{3}\frac{yA(y)^3}{1+y}, \qquad A(y) = \frac{\operatorname{atanh}\sqrt{y}}{\sqrt{y}}, \quad J(0) = 0.$$

Writing $B = yA$ gives the rational system

$$(B^k)' = \frac{kB^k}{2y} + \frac{kB^{k-1}}{2(1-y)} \ (k = 1, 2, 3), \qquad J' = -\frac{2B^3}{3y^2(1+y)}.$$

The span over $\mathbb{C}(y)$ of $1, B, B^2, B^3, J$ is unchanged by replacing J with $q + J$. It has dimension five: B has infinitely many monodromy values about 1, so is transcendental over $\mathbb{C}(y)$, while J' has nonzero residue $\pi^3/96$ at $y = -1$, where the first four functions are locally meromorphic. That residue gives J a logarithmic branch there, so it cannot be their rational-function linear combination. Adding a rational constant changes neither this module nor its continuation domains. These observations explain this proposed shortcut's failure; they do not exclude nonlinear expressions, extra functions or different differential systems.

9 An unresolved model with complete analytic control

Differentiating a parameter-dependent approximation to $\beta(2)$ can produce one for $\beta(4)$: differentiation raises the denominator power in the defining series. The difficulty is to cancel the remaining $\beta(2)$ term and control the sign of the new error. A moment matrix does both here. Its block determinant isolates a rational approximation, and a variational formula shows that the differentiated error is positive. This section therefore reaches the final arithmetic question rather than proving another failure theorem: how quickly do the reduced denominators of these rational approximations grow?

The ordinary-moment first variation [record P07] is retained because its analytic conclusions are stronger than a negative summary of the project might suggest. It has all-index positivity and a sharp scalar error rate. The remaining issue is primitive arithmetic.

The elementary Laplace identity

$$\int_0^\infty e^{-at}\frac{\sinh(\sqrt{u}\,t)}{\sqrt{u}}\,\mathrm{d}t = \frac{1}{a^2 - u} \qquad (a \geq 1,\ |u| < 1)$$

suggests a parameter whose derivative raises the beta denominator from the second to the fourth power. For $|u| < 1$ let

$$F(u) = \sum_{k\geq 0} \frac{(-1)^k}{(2k+1)^2 - u}, \qquad q_j(u) = \frac{1}{(2j+1)^2 - u}, \qquad w_u(x) = \frac{\sinh(\sqrt{u}(-\log x))}{\sqrt{u}(1+x^2)}.$$

Use the analytic value of the quotient at $u = 0$. The density is positive for $0 \le u < 1$; it need not be positive for negative u. The moments $m_j(u) = \int_0^1 x^{2j} w_u(x)\,\mathrm{d}x$ obey $m_j + m_{j+1} = q_j$, with $m_0 = F$, $F(0) = \beta(2)$ and $F'(0) = \theta$. These identities follow by expansion and Laplace integration, locally uniformly in u.

Define $H_n(u) = \det[m_{i+j}(u)]_{0 \le i,j < n}$. The identity $m_j + m_{j+1} = q_j$ suggests using multiples of $1 + t$ as basis elements: their inner products with monomials are rational functions of u. For $d = n - 1$ set $c_j = q_j$ and $B_{ij} = q_{i+j} + q_{i+j+1}$, $0 \le i, j < d$. The same moment identities give $B_{ij} = \int_0^1 x^{2(i+j)}(1 + x^2)^2 w_u(x)\,\mathrm{d}x$. Thus $B(0)$ is positive definite and B^{-1} is analytic near zero. Put $a_n = \det B$ and $r_n = c^T B^{-1} c$; for $n = 1$ use $a_1 = 1, r_1 = 0$. In the basis $1, (1 + t), (1 + t)t, \ldots$, $t = x^2$, the moment matrix is

$$\begin{pmatrix} F & c^T \\ c & B \end{pmatrix}, \qquad H_n = a_n(F - r_n).$$

The change of basis has determinant one; the formula for H_n is the Schur-complement identity for the displayed block matrix. The functions $a_n(u)$ and $r_n(u)$ are rational functions with rational coefficients, regular at zero; their values and derivatives there are rational. The following antisymmetric combination of the determinant and its derivative eliminates $\beta(2)$ exactly:

$$W_n = a_n(0)H_n'(0) - a_n'(0)H_n(0) = a_n(0)^2(\theta - r_n'(0)). \tag{9.1}$$

This is a formal identity, with no independence assumption about periods. For example, at $n = 2$ one obtains $r_2(u) = q_0(u)^2/(q_0(u) + q_1(u))$ and $r_2'(0) = 49/50$. The resulting primitive form is $50\theta - 49$.

Proposition 9.1 (Positivity and the scalar rate; P07)**.** *Let $E_n = \theta - r_n'(0)$ and $\rho = 3 + \sqrt{8}$. Then $E_n > 0$ for every n, and*

$$\lim_{n \to \infty} \frac{\log E_n}{n} = -2 \log \rho.$$

Proof. Positivity of the differentiated error. For $0 \le u < 1$, completing the square in the block matrix gives the following minimum over real polynomials:

$$F(u) - r_n(u) = \min_{\substack{\deg P < n \\ P(-1) = 1}} \int_0^1 P(x^2)^2 w_u(x)\,\mathrm{d}x.$$

Every polynomial of degree less than n with $P(-1) = 1$ has the form $1 + (1 + t)Q(t)$, $\deg Q < n - 1$. Minimizing this quadratic form in the coefficients of Q gives the same $B^{-1}c$ as the block determinant. The unique minimizing polynomial is $P_{n,u}(t) = 1 - (1 + t)\sum_{j<d}(B^{-1}c)_j t^j$ and is analytic near zero. Differentiate from $u \ge 0$ at zero. The derivative of the minimizing polynomial satisfies $\dot{P}(-1) = 0$, and stationarity gives $\int P \dot{P} w_0\,\mathrm{d}x = 0$. Thus only the derivative of the weight contributes. Expansion of $\sinh(\sqrt{u}(-\log x))/\sqrt{u}$ shows that this derivative is $(-\log x)^3/6$ divided by $1 + x^2$. Thus

$$E_n = \int_0^1 P_{n,0}(x^2)^2 \frac{(-\log x)^3}{6(1 + x^2)}\,\mathrm{d}x > 0. \tag{9.2}$$

The exponential rate. The same polynomial minimizes the weight-two norm, but its weight-four norm is the error we need. A lower bound follows from the weight-four minimum. For an upper bound we must control its supremum norm from its weight-two norm; the ratio of the two densities is unbounded near zero, so a direct pointwise comparison would not suffice.

Let $\lambda_{n,s}$ be the analogous constrained minimum for the weight $(-\log x)^{s-1}/[(s-1)!(1+x^2)]$, $s = 2, 4$. Then $E_n \geq \lambda_{n,4}$, while $P_{n,0}$ minimizes the weight-two norm. In the variable $t = x^2$ the densities are

$$\eta_2(t) = \frac{-\log t}{4\sqrt{t}(1+t)}, \qquad \eta_4(t) = \frac{(-\log t)^3}{96\sqrt{t}(1+t)}.$$

The test polynomial $T_{n-1}(2t-1)/T_{n-1}(-3)$ gives $\limsup n^{-1} \log \lambda_{n,s} \leq -2 \log \rho$.

For the opposite bound, restrict to a fixed interior interval $I_\epsilon = [\epsilon, 1-\epsilon]$. Each density has a positive minimum there. An elementary Legendre evaluation estimate for an interval $I = [a, b]$ of length ℓ is

$$|P(t_*)|^2 \leq \frac{n^2}{\ell} R^{2(n-1)} \int_I P(t)^2 \, \mathrm{d}t \quad (\deg P < n), \tag{9.3}$$

where $R = 1$ if $t_* \in I$, and otherwise $R = |z_*| + \sqrt{z_*^2 - 1}$, $z_* = (2t_* - a - b)/\ell$. To see this, expand in the orthonormal basis $\sqrt{(2j+1)/\ell}\, L_j((2t-a-b)/\ell)$ and apply Cauchy–Schwarz. The bounds $|L_j(z)| \leq 1$ on $[-1, 1]$ and $|L_j(z)| \leq (|z| + \sqrt{z^2-1})^j$ outside follow from

$$L_j(z) = \frac{1}{\pi} \int_0^\pi (z + \sqrt{z^2-1} \cos v)^j \, \mathrm{d}v,$$

using the imaginary square root inside the interval and parity for negative z. The formula is a polynomial identity by expansion. At $t_* = -1$, the constraint $P(-1) = 1$ turns (9.3) into a lower bound on the integral. Here $z_* = -3/(1-2\epsilon)$. First letting $n \to \infty$ and then $\epsilon \downarrow 0$ gives the matching lower exponent for both $\lambda_{n,s}$.

Finally use $I = [0, 1-\epsilon]$, where $\eta_2 \geq \epsilon/8$. The same evaluation estimate, uniformly over $t_* \in [0, 1]$, gives

$$\|P_{n,0}\|_{[0,1]}^2 \leq \frac{8n^2 S_\epsilon^{2(n-1)}}{\epsilon(1-\epsilon)} \lambda_{n,2}, \quad S_\epsilon = s_\epsilon + \sqrt{s_\epsilon^2 - 1}, \quad s_\epsilon = \frac{1+\epsilon}{1-\epsilon}.$$

Integrating against η_4 bounds E_n by θ times this quantity. Now $S_\epsilon \to 1$ as $\epsilon \downarrow 0$; combined with $E_n \geq \lambda_{n,4}$ this proves the asserted rate. □

Write $r_n'(0) = p_n/A_n$ in lowest terms, $A_n > 0$. Its actual primitive form is

$$L_n = A_n\theta - p_n = A_n E_n, \qquad \log L_n = \log A_n - 2n \log \rho + o(n). \tag{9.4}$$

Along any proposed subsequence, $\limsup n^{-1} \log A_n < 2 \log \rho$ is sufficient for shrinking values. The weak inequality is necessary, but equality requires control of the lower-order terms. No useful all-index reduced-denominator estimate has been established. The quadratic logarithmic decay of the raw determinant cannot be used as the scalar error rate: its target coefficient carries the same scale.

At $n = 2$, the first-variation pair is $50\theta - 49$ and the direct weight-four pair is $82\theta - 81$. The archived calculations cover exactly the direct and first-variation forms at $n = 1, \ldots, 12$, with full gcd removal. All twenty forms at $n = 3, \ldots, 12$ exceed one; the four at $n = 1, 2$ are smaller than one. Some diagnostic logarithms are shown below, rounded from the archived certified evaluations.

n	$\log A_n$	$\log E_n$	$\log L_n/n^2$
3	21.369976	−8.523832	1.427349
6	141.949801	−19.363907	3.405164
9	354.356711	−30.034542	4.003977
12	660.303908	−40.661541	4.303072

These are finite failures, not a growth theorem for A_n. An earlier table had cleared denominators without removing the common numerator factor; at $n = 12$ that overstated $\log L_n/n^2$ as 6.271803. The corrected data above are the retained conclusion. The archived arcsine realization, by contrast, was identified exactly with an earlier family; no such coefficient-sequence identification was established for this ordinary-moment variation.

10 Interleaved determinants: changing the arithmetic without losing positivity

The principal determinant in the companion paper has the same row and column nodes. Its primitive values grow. Changing the basis of that matrix cannot help, but changing the node sets changes the determinant polynomial itself. Interleaved borders are a natural test: retain a common block carrying the target θ, and add distinct row and column nodes to create new cancellations at primes. This geometry was suggested by Litsey's trigamma construction [9]. The estimates below concern the beta kernel defined here, not the constants in that work. They develop record P06.

10.1 The matrix and its exact analytic benchmark

For positive integers a define

$$S_\ell(a) = \sum_{j=0}^{a-1} ((4j+1)^{-\ell} - (4j+3)^{-\ell}),$$

and the rational pencil

$$\Phi_{ab}(X) = \begin{cases} \dfrac{bS_3(b) - aS_3(a)}{6(b^2-a^2)}, & a \neq b, \\ X - S_4(a) + \dfrac{S_3(a)}{12a}, & a = b. \end{cases} \tag{10.1}$$

Fix $0 < c \leq 1$, put $m = \lfloor cr/2 \rfloor$, $N = r + m$ and $M = r + 2m$, and take the ordered node sets

$$\mathcal{A} = \{1, \ldots, r\} \cup \{r+1, r+3, \ldots, r+2m-1\}, \qquad \mathcal{B} = \{1, \ldots, r\} \cup \{r+2, r+4, \ldots, r+2m\}.$$

Write $P_{r,c}(X) = \det[\Phi_{ab}(X)]_{a\in\mathcal{A}, b\in\mathcal{B}}$. This formula alone specifies every rational coefficient and permits exact reproduction; no numerical value of θ is needed.

Here is the analytic input from the companion paper, stated with its normalization [1, Proposition 3.1 and Theorem 3.3]:

$$\Phi_{ab}(\theta) = \int_0^\infty \frac{\omega(s)}{(a^2+s^2)(b^2+s^2)}\,\mathrm{d}s, \qquad \omega(s) = \frac{\pi^2 s^2}{96}(1 + A(s) - 2A(s)^3), \tag{10.2}$$

where $A(s) = \operatorname{sech}(2\pi s)$, and

$$L^N \det\left[\frac{1}{a+b}\right] \leq P_{r,c}(\theta) \leq U^N \det\left[\frac{1}{a+b}\right], \tag{10.3}$$

with $L = (\pi^3/192)/(1 + 1/(\sqrt{2}\pi))^2$ and $U = (\pi^3/192)(1 + 2/(3\sqrt{6}))$. Both constants are positive and independent of all nodes. Positivity also follows directly from (10.2): expand the integral determinant as the integral of the product of its two Cauchy determinants. For ordered nodes the product is positive off coincidences. The coefficient of X^r is the disjoint-border minor, positive by the same argument (and equal to one if $m = 0$). Thus $P_{r,c}$ has degree exactly r, and $P_{r,c}(\theta) > 0$.

The Cauchy product evaluates its leading real decay. Put

$$I(a,b)=\frac{(a+b)^2\log(a+b)+(a-b)^2\log|a-b|-2a^2\log a-2b^2\log b}{2},\qquad 0^2\log 0=0,$$

and

$$E(c)=\frac{I(1,1)+2I(1,1+c)+I(1+c,1+c)}{4}.\qquad \log P_{r,c}(\theta)=-E(c)r^2+o(r^2).\tag{10.4}$$

To verify the limit, the row and column counting measures divided by r both converge to $\rho_c=(\mathbf{1}_{[0,1]}+\mathbf{1}_{[0,1+c]})/2$. In the Cauchy product, the two Vandermonde numerators supply the $\log|x-y|$ integral and the denominator supplies $\log(x+y)$. The leftover scale factor is $-N\log r=o(r^2)$. Truncate the logarithms at distance ϵ from their singularities. The bounded grid density bounds the omitted normalized sums by $O(\epsilon|\log\epsilon|)+o(1)$; then let ϵ decrease to zero. Finally two integrations give

$$I(a,b)=\int_0^a\int_0^b\log\frac{x+y}{|x-y|}\,\mathrm{d}x\,\mathrm{d}y.$$

The $O(N)$ error in (10.3) does not change the limit. Thus there is no unknown leading analytic gain in this specified family.

10.2 The arithmetic quantity that would suffice

Let $D_{r,c}$ be the least positive integer clearing the coefficients of $P_{r,c}$, let $g_{r,c}$ be the gcd of the coefficients of $D_{r,c}P_{r,c}$, and put $\kappa_{r,c}=D_{r,c}/g_{r,c}$. A sufficient irrationality criterion for one fixed c is

$$\limsup_{r\to\infty}\frac{\log\kappa_{r,c}}{r^2}<E(c).\tag{10.5}$$

Indeed, rationality $\theta=u/v$ would make $v^r\kappa_{r,c}P_{r,c}(\theta)$ a positive integer; its logarithm would tend to minus infinity. Replacing $\kappa_{r,c}$ by $D_{r,c}$ gives a stronger sufficient condition, not an equivalent one.

The following calculation explains both the available estimate and its limitation. For a prime p, the Gauss valuation of a polynomial is the minimum valuation of its coefficients. It extends to a discrete valuation on $\mathbb{Q}_p(X)$; reduction of integral rational functions gives the residue field $\mathbb{F}_p(X)$. We use elementary pivot elimination over this valuation ring. This is the setting for the Smith exponents below; it does not specialize X to θ.

10.3 Isolating the four negative prime layers

Suppose $p>3$ and $p^2>4M$. Separate the multiples of p in each tail:

$$H_\ell(a;p)=\sum_{\substack{t>0\ \text{odd}\\ pt<4a}}\frac{\chi_4(t)}{t^\ell},\qquad U_\ell(a;p)=\sum_{\substack{0<n<4a\ \text{odd}\\ p\nmid n}}\frac{\chi_4(n)}{n^\ell},$$
$$S_\ell(a)=\chi_4(p)p^{-\ell}H_\ell(a;p)+U_\ell(a;p).$$

All denominators in H_ℓ are p-units. In a block of $4p$ integers, odd positions differing by $2p$ have the same residue modulo p and opposite characters. Reflecting a complete block by $n\mapsto 4p-n$ gives, respectively,

$$U_3(a+p)\equiv U_3(a),\qquad U_3(p-a)\equiv -U_3(a),\qquad U_3(pk)\equiv 0\pmod p.$$

The reflection identity is stated for $0 \le a \le p$; periodicity extends it to the corresponding residue classes. Consequently the matrix obtained by replacing S_ℓ by U_ℓ in (10.1) is p-integral. When $a \equiv \pm b \neq 0$ the numerator cancels the factor of p in $b^2 - a^2$; when $a \equiv b \equiv 0$, both factors of p are cancelled by the node and U_3 factors. The inequality $2M < p^2/2$ excludes higher node-factor valuations. The diagonal follows in the same way.

We therefore have $\Phi = \Psi + R(X)$ with R integral and

$$\Psi_{ab} = \frac{\chi_4(p)(bH_3(b;p) - aH_3(a;p))}{6p^3(b^2-a^2)} \quad (a \neq b), \qquad \Psi_{aa} = \chi_4(p)\left(-\frac{H_4(a;p)}{p^4} + \frac{H_3(a;p)}{12ap^3}\right).$$

In particular $p^4\Phi \equiv p^4\Psi \pmod{p^4}$. Let μ_i be the Smith exponents of $p^4\Phi$, capped at four, and $\tau_p = \sum_i (4 - \mu_i)$. Successive unit-pivot elimination, dividing out the minimum power of p at each step, gives

$$P_{r,c} = p^{-\tau_p} u_p \det T_p, \qquad v_G(u_p) = 0, \quad T_p \text{ integral.} \tag{10.6}$$

The matrix T_p is what remains after the negative layers have been removed. Thus $v_G(P_{r,c}) = -\tau_p + t_p$, where $t_p = v_G(\det T_p) \ge 0$. Knowing all four negative layers does not bound t_p and does not determine primitive numerator content.

10.4 An explicit large-prime calculation

Proposition 10.1 (Four-layer estimate; P06)**.** *For $2M + 4 < p < 4r$, put $h = r - \lfloor p/4 \rfloor$. The capped Smith exponents of $p^4\Phi$ are*

$$0 \ (h \text{ times}), \quad 1 \ (2m \text{ times}), \quad 2 \ (h \text{ times}), \quad 4 \ (N - 2h - 2m \text{ times}).$$

For $\max(2M + 4, 4r) < p < 4M$, put $H = M - \lfloor p/4 \rfloor$. They are 1 repeated H times and 4 repeated $N - H$ times. Consequently the contribution of these primes to $\sum_p \tau_p \log p$ is $(3 + 3c^2)r^2 + o(r^2)$.

Proof. All node sums and differences are p-units in this range. The only possible odd multiple of p in a tail is p itself, so $H_3(a;p) = H_4(a;p) = \mathbf{1}_{4a>p}$. Call these nodes high. There are h high common nodes and all borders are high in the first range. Up to the unit $\chi_4(p)$ the singular matrix is $D + pF$, where D is minus the identity on those h diagonals and zero elsewhere. The low-low block of F is zero; its low-high entry is $b/[6(b^2 - a^2)]$, a Cauchy entry times a unit column factor. Every admissible maximal minor is a unit. There are $l = r - h \ge h + m$ low common nodes.

Eliminate the h unit pivots. On the remaining low and border nodes, the first layer has block form $\left(\begin{smallmatrix} 0 & C_1 \\ C_2 & C_3 \end{smallmatrix}\right)$, where C_1 has column rank m and C_2 has row rank m. Its rank is $2m$. Left and right null vectors can be supported on the low common nodes and chosen to annihilate the border Cauchy columns and rows exactly. On these kernels the second layer is, up to a sign,

$$(YF_{\text{low,high common}})(F_{\text{high common,low}}Z).$$

Adjoining the h high-common columns to the m border columns is a unit full-rank Cauchy matrix, and the analogous assertion holds for rows. The first factor is injective and the second surjective on an h-dimensional space; their product has rank h. Eliminating the first layer changes this block only in the third layer. The exact singular matrix $D + pF$ has rank $2(h + m)$: its low-low block is zero and its two low-high blocks have full rank $h + m$. The $h + 2m + h$ pivots found account for this rank, so there are no further pivots of valuation less than four.

In the second range no common node is high. The low-high Cauchy blocks now give rank H, all in the first layer; the same argument leaves every other capped exponent equal to four. The

respective values of τ_p are $6h+6m$ and $3H$. With $x=p/r$, the prime number theorem gives the two integrals

$$\int_{2+2c}^{4} 6(1+c/2-x/4)\,\mathrm{d}x = 3(1-c^2), \qquad \int_{4}^{4+4c} 3(1+c-x/4)\,\mathrm{d}x = 6c^2.$$

Bounded endpoint strips and the floors contribute $o(r^2)$. □

This estimate cannot by itself prove (10.5), for any fixed $0<c\le 1$. The energy integrand is nonnegative and $\rho_c \le 1$, so

$$E(c) \le 2(1+c)^2 \log 2 \le 4(1+c^2)\log 2 < 3(1+c^2). \tag{10.7}$$

Even omitting every other prime leaves an unfavorable comparison for the bound obtained by setting $t_p=0$ in (10.6). This proves failure of that bound, not growth of the actual primitive value. For completeness, using $4N$ at the remaining primes above $\sqrt{4M}$ gives

$$\log D_{r,c} \le (11+12c+7c^2)r^2 + o(r^2).$$

At smaller primes every entry costs $O(\log r/\log p)$ powers of p, so their total is $O(r^{3/2}\log r) = o(r^2)$. This includes 2 and 3.

10.5 A genuine obstruction for integer multipliers

For $2M+4<p<4r$ a specific coefficient has the exact valuation

$$v_p([X^{r-h}]P_{r,c}) = -4h-3m. \tag{10.8}$$

Indeed, expand this coefficient using minors containing h common nodes and all borders. A minor containing only $j<h$ high common nodes has valuation at least $-4j-3(h+m-j) > -4h-3m$. In the unique minor using all high common nodes, only terms selecting all h diagonals attain the minimum. The remaining border determinant, after scaling by p^{3m}, is a unit multiple of the Cauchy determinant $[1/(a+b)]$; its node differences and sums are units. Hence there is no cancellation of the term of minimum valuation. Summing (10.8) gives

$$\log D_{r,c} \ge (2-c-c^2)r^2 + o(r^2).$$

At $c=1/5$, this lower exponent is $44/25 = 1.76$, whereas the explicit formula (10.4) gives $1.6523 < E(1/5) < 1.6524$. The latter interval can be checked using $\log x = 2\sum_{j\ge 0} z^{2j+1}/(2j+1)$, $z=(x-1)/(x+1)$, with geometric tail bounds. Every positive integer coefficient-clearing multiplier therefore produces growing values at this width. Division by common numerator content remains possible, so this is still not a primitive-value obstruction.

10.6 What remains open, and how to reproduce the construction

The nine archived examples ($r=8,16,24$ and $c=1/5,1/2,1$) remain finite evidence. Each used the entire rational polynomial, its least clearing denominator and the complete coefficient gcd. They do not bound $g_{r,c}$ for general r. A reader can reproduce such a calculation from (10.1) using the block matrix on common and border nodes. If its blocks are XI_r+U, V, W, Z, then Z is invertible by positivity and

$$P_{r,c}(X) = \det Z \,\det(XI_r + U - VZ^{-1}W).$$

Compute the characteristic polynomial over $\mathbb{Q}$, clear all coefficient denominators, and divide the complete integer coefficient gcd. This also specifies a practical exact procedure independently of our scripts.

The unresolved quantity is the primitive multiplier $\kappa_{r,c}$. A new estimate must control the residual factors t_p in (10.6), or common content at other primes, strongly enough to reach (10.5). The four-layer estimate, the integer-clearing obstruction and the finite primitive failures have different scopes; none closes that question.

11 Quarter-pole determinants with and without a positivity guarantee

The positive constructions in the main paper remove unwanted constants before estimating a determinant. A different possibility is to use Hurwitz zeta values on the quarter grid. Their common $\zeta(4)$ term can be removed by interpolation, leaving only θ and rational numbers. There are two versions. Squared poles preserve a positive Gram representation but require an extra numerator. Unsquared poles avoid that numerator but require a separate nonvanishing argument. This section supplies both constructions and proves the conditional nonvanishing result of record P05; record P04 is the positive version.

11.1 A functional with explicit rational moments and pole values

Let $f(y) = (e^{2\pi y} - 1)^{-1}$ and

$$w(y) = -\frac{y^4}{3} f'''(y), \qquad \nu(R) = \int_0^\infty R(y^2) w(y)\, \mathrm{d}y.$$

The exponential expansion of f shows $w > 0$. It is bounded near zero and decays exponentially at infinity. Define Bernoulli numbers by $z/(e^z - 1) = \sum_{j \geq 0} B_j z^j / j!$, so $B_1 = -1/2$. The needed evaluations are

$$\nu(t^e) = \frac{(2e+4)!\,|B_{2e+2}|}{6(2e+2)!}, \quad e \geq 0, \tag{11.1}$$

$$\nu\left(\frac{1}{t+a^2}\right) = a^3 \zeta(4, a) - \frac{1}{2a} - \frac{1}{3}, \quad a > 0. \tag{11.2}$$

For the first formula, integrate by parts three times and then use $\int_0^\infty y^{s-1} f(y)\, \mathrm{d}y = \Gamma(s)\zeta(s)/(2\pi)^s$ and the classical even-zeta evaluation. For the second put $g(y) = y^4/(a^2 + y^2)$. All boundary terms vanish, and

$$\nu((t+a^2)^{-1}) = \frac{1}{3} \int_0^\infty g'''(y) f(y)\, \mathrm{d}y = 8a^4 \int_0^\infty \frac{y(a^2 - y^2) f(y)}{(a^2 + y^2)^4}\, \mathrm{d}y.$$

Hermite's integral for the Hurwitz zeta function at $s = 4$ [5, Eq. 25.11.29] identifies this with the right side of (11.2). Thus the normalization and the two rational correction terms are fixed; they cannot be dropped.

Set $a_j = (2j+1)/4$ and

$$T_j = 256 \sum_{k=0}^{\lfloor j/2 \rfloor - 1} (4k + 1 + 2(j \bmod 2))^{-4}.$$

Separating the two odd residue classes gives

$$\zeta(4, a_j) = 120\zeta(4) + 128(-1)^j\theta - T_j. \tag{11.3}$$

For $j = 0, 1$ the sum and difference are $240\zeta(4)$ and 256θ; integer shifts of a_j subtract the displayed finite tail. This proves (11.3) without any independence assumption about constants.

11.2 Squared poles: positivity and exact coefficient recovery

Here "squared" refers to the pole locations $-a_j^2$; the poles themselves are simple. The second construction uses locations $-a_j$. Let $D_K(t) = \prod_{j<K}(t + a_j^2)$, and let S_K be the polynomial of degree less than K interpolating $S_K(-a_j^2) = 1/a_j$. This is an explicit rational polynomial, obtainable by Lagrange interpolation. It is positive on $[0, \infty)$: every derivative of $(-t)^{-1/2}$ is positive on the negative axis, so its divided differences are positive; in the Newton interpolation expression all factors are positive when $t \geq 0$.

For $K \geq 2h + 1$ consider the h by h matrix

$$G_{ik} = \nu\left(\frac{S_K(t)t^{i+k}}{D_K(t)}\right), \qquad 0 \leq i, k < h.$$

To compute an entry, divide $S_K(t)t^\ell$ by $D_K(t)$ and write

$$\frac{S_K(t)t^\ell}{D_K(t)} = Q_\ell(t) + \sum_{j<K} \frac{c_{j,\ell}}{t + a_j^2}, \qquad c_{j,\ell} = \frac{(-a_j^2)^\ell}{a_j D_K'(-a_j^2)}. \tag{11.4}$$

Evaluate the polynomial part by (11.1), and use (11.2)–(11.3) on the poles. The $\zeta(4)$ coefficient is zero, since

$$\sum_{j<K} a_j^3 c_{j,\ell} = -\sum_{j<K} \frac{(-a_j^2)^{\ell+1}}{D_K'(-a_j^2)} = 0 \quad (\ell \leq K - 3).$$

The last equality is the coefficient of t^{K-1} in the Lagrange interpolation formula for $t^{\ell+1}$. We obtain rational matrices $G(X) = B + XA$. The moment representation proves $G(\theta)$ positive definite. Also

$$A = 128 \sum_{j<K} \frac{(-1)^j a_j^2}{D_K'(-a_j^2)} v_j v_j^T, \qquad v_j = (1, -a_j^2, \ldots, (-a_j^2)^{h-1})^T.$$

Each weight is positive, because $D_K'(-a_j^2)$ has sign $(-1)^j$. The vectors span $\mathbb{R}^h$, so A is positive definite and $\det G(X)$ has degree exactly h. Both nonvanishing and coefficient recovery are therefore settled for this family.

11.3 Why the additional numerator matters analytically

The interpolation numerator partly undoes the benefit of adding poles. The following comparison makes that statement quantitative without claiming an arithmetic obstruction. Directly multiplying the node differences in (11.4) for $\ell = 0$ gives

$$\frac{S_K(t)}{D_K(t)} = \sum_{j<K} \frac{(-1)^j 4^K}{(K - 1 - j)!(K + j)!} \frac{1}{t + a_j^2}.$$

Its coefficient at $1/t$ is $c_K = 4^K/((2K-1)(K-1)!^2)$, by Pascal telescoping of the alternating binomial sum. Put

$$b_j = \frac{(2K-1)(K-1)!^2}{(K-1-j)!(K+j)!}, \qquad b_0 = 2 - 1/K.$$

The positive terms $b_j/(t+a_j^2)$ decrease with j for $t \geq 0$, since $b_{j+1}/b_j = (K-1-j)/(K+j+1) < 1$. The alternating-sum bounds therefore give, for $K \geq 2$,

$$\frac{2b_0}{K+1}\frac{1}{t+1/16} \leq \frac{S_K(t)}{c_K D_K(t)} \leq \frac{b_0}{t+1/16}. \tag{11.5}$$

For any h independent real polynomial test functions, their Gram matrices for these weights satisfy the same matrix inequalities. Conjugating by the inverse square root of the fixed-pole Gram matrix and multiplying its eigenvalues gives determinant factors between $(2b_0/(K+1))^h$ and b_0^h. When $h = O(K)$, their logarithms differ by only $O(K \log K)$. This is an analytic comparison; it does not identify the two determinant polynomials or their primitive content. The three archived tests $(K,h) = (3,1), (5,2), (7,3)$ are finite failures after primitive normalization. An all-index primitive estimate is still missing.

11.4 Unsquared poles: an explicit signed pencil

Now use the same weight on the imaginary line:

$$\Lambda(R) = \frac{1}{2}\int_0^\infty (R(iy) + R(-iy))w(y)\,\mathrm{d}y.$$

For real rational functions with negative real poles this converges, and (11.1)–(11.2) give

$$\Lambda(x^j) = \frac{(j+4)(j+3)}{6}B_{j+2}, \qquad \Lambda((x+a)^{-1}) = a^4\zeta(4,a) - \frac{1}{2} - \frac{a}{3}. \tag{11.6}$$

Odd polynomial moments vanish. Unlike a real-line Gram functional, $\Lambda(P^2)$ need not be positive.

Take integers $M \geq 0$, $h \geq 1$, $K \geq 4M + 2h + 4$, and set

$$V_M(x) = \prod_{m=1}^{M}(x-m)^4, \qquad D_K^*(x) = \prod_{j<K}(x+a_j), \qquad r_{j,\ell} = \frac{V_M(-a_j)(-a_j)^\ell}{(D_K^*)'(-a_j)}.$$

All entries in the matrix below are now explicit rational linear forms:

$$H_{ik}(X) = \sum_{j<K} r_{j,i+k}a_j^4(128(-1)^j X - T_j), \qquad 0 \leq i,k < h. \tag{11.7}$$

Indeed, apply partial fractions and (11.6) to $\Lambda(V_M x^{i+k}/D_K^*)$. The numerator has degree at most $K-6$. Lagrange interpolation cancels the $\zeta(4)$ coefficient (multiply the numerator by x^4), and also the elementary corrections (multiply by 1 and x). Formula (11.7) retains all rational terms through T_j. Since

$$(D_K^*)'(-a_j) = \frac{(-1)^j j!(K-1-j)!}{2^{K-1}},$$

the coefficient matrix of X is positive definite: its weights are $128(-1)^j a_j^4 V_M(-a_j)/(D_K^*)'(-a_j) > 0$ at K distinct nodes. Thus the polynomial determinant has degree h. This does not prove that its value at θ is nonzero.

There is also an exact derivative-sum interpretation. With $F_\ell(x) = x^4 V_M(x) x^\ell / D_K^*(x) = O(x^{-2})$, partial fractions give

$$H_{ik}(\theta) = -\frac{1}{6} \sum_{n \geq 0} F_{i+k}'''(n). \tag{11.8}$$

The equality follows by differentiating each simple pole three times; the pole coefficients sum to zero. The series converges absolutely. The fourth-order zeros at $0, 1, \ldots, M$ remove the first $M+1$ terms. They do not make the remaining terms, or their determinant, positive.

11.5 Nonvanishing under hypothetical rationality

An auxiliary prime can replace positivity for the nonvanishing step. This is the elementary Vandermonde mechanism used in the related literature [16, Section 3]; the specialization is proved here.

Proposition 11.1 (Conditional nonvanishing; P05)**.** *Use $M = n$, $h = 2n$, $K = 8n + 4$ in (11.7). If $p = 8n + 5$ is prime and u/v is p-integral, then*

$$v_p(\det H(u/v)) = -8n.$$

Consequently, if θ were rational, these determinants would be nonzero for infinitely many n.

Proof. Here $K = p - 1$. All factors of $(D_K^*)'(-a_j)$ are p-units. After writing the tails as sums over odd integer denominators, their only possible denominator divisible by p is p itself. Put $j_0 = (p-1)/2 = 4n + 2$. The denominator p occurs in T_j precisely for $j = j_0 + 2s$, $1 \leq s \leq h$. At these nodes $a_j = p/4 + s$, so $a_j \equiv s$ and $p^4 T_j \equiv 256 \pmod{p}$. The tails at all other nodes are integral. Thus $p^4 H(u/v)$ reduces to

$$\sum_{s=1}^{h} w_s v_s v_s^T, \qquad v_s = (1, -s, \ldots, (-s)^{h-1})^T, \qquad w_s = \frac{-256 s^4 V_n(-s)}{(D_K^*)'(-a_{j_0+2s})} \pmod{p}.$$

Every weight is a unit: each $s + m$ in $V_n(-s)$ is between 2 and $3n < p$. The h nodes are distinct. The determinant is therefore $\prod_s w_s \prod_{s<t} (t-s)^2 \neq 0 \pmod{p}$, proving the valuation.

For completeness, infinitely many primes 5 mod 8 avoid any finite set of odd primes without needing a prime-progression theorem. Choose an odd integer A divisible by the forbidden primes and by 5. Every prime divisor of $A^2 + 4$ avoids that set and is 1 mod 4, since -1 is a square modulo it. As $A^2 + 4 \equiv 5 \pmod{8}$, at least one prime divisor is 5 mod 8. It is not 5. Repeat with the enlarged set, including the odd denominator primes of the assumed rational θ. This supplies infinitely many admissible n. □

11.6 The exact remaining task

For the last family write $Z_n = \mathrm{Prim}(\det H_n)$; its degree is $2n$. The proposition supplies the required nonvanishing subsequence under rationality. It would suffice to prove, along that subsequence,

$$\frac{1}{n} \log |Z_n(\theta)| \longrightarrow -\infty.$$

No such estimate, or all-index primitive obstruction, is established. The seven earlier tests used $(M, h, K) = (0,1,6), (0,2,8), (0,3,10), (1,1,10), (1,2,12), (1,3,14), (2,2,16)$; all primitive absolute values exceeded one. These are different finite fixtures, not a test of every index on the nonvanishing

ray. The first primitive polynomial was $118682030250X - 117369943777$. Formula (11.7), followed by an exact determinant and complete coefficient-gcd removal, specifies an independent reproduction procedure. For the positive squared version use polynomial division (11.4) first; omitting its polynomial quotient would change the matrix.

12 A common-base ladder determinant with an unresolved odd-prime cost

Approximating the terms of a functional identity separately can destroy its arithmetic advantage: eliminating their unwanted constants multiplies large coefficients. Here the polylogarithm arguments remain powers of one variable throughout the construction. This gives a determinant whose analytic decay beats its entire cost at the prime 2. The missing step is the other primes, not nonvanishing. We develop record L03 and then a related signed-integral alternative.

12.1 A rational-coefficient germ and its endpoint determinant

Put $q = (1-i)/2$, $z_m = -q^m$ for $m = 2, 3, 5$, and $(c_2, c_3, c_5) = (225/136, -25/34, 9/68)$. Write

$$\mathcal{I}_s(z_m t^m) = \frac{\mathrm{Li}_s(z_m t^m) - \mathrm{Li}_s(\overline{z}_m t^m)}{2i}, \qquad L(t) = -\log(1 - t^2/2).$$

This defines an analytic germ, not the imaginary part operation on complex t. All its Taylor coefficients are rational. Define

$$H(t) = \sum_{m=2,3,5} c_m \left(\mathcal{I}_4(z_m t^m) + \frac{m}{2} L(t) \mathcal{I}_3(z_m t^m) + \frac{m^2}{12} L(t)^2 \mathcal{I}_2(z_m t^m) \right) = \sum_{j \geq 1} h_j t^j. \tag{12.1}$$

The completed Broadhurst identity (7.1) gives $H(1) = \theta$. Its use here has the same external dependency as Section 7. Each h_j is obtained by finite convolution of the displayed series, so the following polynomial is reproducible over $\mathbb{Q}$:

$$b_i = h_i, \quad T_{ij} = h_{i+j} - h_{i+j-1} \quad (1 \leq i, j \leq n), \qquad J_n(X) = \det \begin{pmatrix} X & -b^T \\ b & T \end{pmatrix}. \tag{12.2}$$

For $n = 0$ put $J_0(X) = X$. Its degree is at most one, even though its matrix size grows. With $c_j(X) = X - \sum_{\ell=1}^{j} h_\ell$, adjacent row and column differences give the exact identity

$$\det[c_{i+j}(X)]_{0 \leq i,j \leq n} = (-1)^n J_n(X), \qquad \sum_{j \geq 0} c_j(X) t^j = \frac{X - H(t)}{1 - t}. \tag{12.3}$$

No invertibility of T or Padé normality is required.

12.2 Nonvanishing from a logarithmic branch

As t increases to $\sqrt{2}$, the polylogarithm arguments tend to $i, e^{i\pi/4}, e^{-i\pi/4}$, none equal to 1. The coefficient of $L(t)^2$ at that endpoint is

$$A_+ = \frac{75}{272}(2\beta(2) - 3\,\mathrm{Cl}_2(\pi/4)), \qquad \mathrm{Cl}_2(\phi) = \sum_{j \geq 1} \frac{\sin(j\phi)}{j^2}.$$

The alternating series gives $\beta(2) < 1$. The first three Clausen terms and the absolute tail bound $\sum_{j\ge 4} j^{-2} < 1/3$ give

$$\mathrm{Cl}_2(\pi/4) > \frac{5\sqrt{2}}{9} + \frac{1}{4} - \frac{1}{3} > \frac{25}{36}, \qquad A_+ < -\frac{25}{1088}.$$

Thus H has a genuine logarithm-squared singularity. Subtracting any constant X and dividing by $1 - t$ cannot remove it. Kronecker's classical Hankel criterion says that a power series is rational exactly when all its sufficiently large leading Hankel determinants vanish [18, Theorem 1]. Applied to (12.3), it proves that $J_n(X) \neq 0$ for infinitely many n for every fixed complex X, in particular $X = \theta$. No irrationality assumption or positivity argument enters this step.

12.3 A common analytic domain and a determinant bound

Let $E = t\,d/dt$. The rational function U in (7.2) satisfies

$$U(t) = \sum_m c_m m^4 \mathcal{I}_0(z_m t^m).$$

Define $Y_0 = U$, $EY_j = Y_{j-1}$ and $Y_j(0) = 0$ for $1 \le j \le 4$. Then

$$H = Y_4 + \tfrac{1}{2} L Y_3 + \tfrac{1}{12} L^2 Y_2, \qquad Y_j(t) = \frac{1}{(j-1)!} \int_0^1 U(tx)(-\log x)^{j-1} \frac{\mathrm{d}x}{x}. \tag{12.4}$$

These identities follow coefficientwise. They also specify analytic continuation by integration on radial segments.

For $p_2 = t^4 + 4$, $p_3 = t^6 - 4t^3 + 8$, $p_5 = t^{10} - 8t^5 + 32$, let $D = p_2 p_3 p_5/(t^2 + 2t + 2)^2$. Polynomial division shows

$$(2 - t^2) D(t) \mid t^{120} - 2^{60}.$$

All possible finite singularities lie at radius $\sqrt{2}$ on the 120th-root rays. Remove those rays outward from that radius. The remaining domain is star-shaped, and (12.4) gives single-valued holomorphic functions there. There is no singularity at zero because $U = O(t^2)$.

In the reciprocal variable put

$$G(w) = \frac{\theta - H(1/w)}{w - 1}, \qquad K = \{w : w^{120} \in [0, 2^{-60}]\}.$$

The apparent pole at $w = 1$ is removable precisely because $H(1) = \theta$. Thus G is holomorphic off the compact star K and vanishes at infinity. The logarithmic capacity of a segment of length a is $a/4$; the monic polynomial-preimage rule gives

$$\mathrm{cap}(K) = \left(2^{-60}/4\right)^{1/120} = 2^{-31/60}.$$

For this particular preimage the rule also follows by composing the segment's Green function with w^{120} and dividing by 120.

Pólya's determinant inequality [17, Theorem 2.1, one variable] now yields

$$\log |J_n(\theta)| \le -\frac{31}{60}(\log 2) n^2 + o(n^2), \tag{12.5}$$

with $\log 0 = -\infty$. To check the exponent normalization, the Laurent coefficients of G are $c_j(\theta)$. Their Hankel determinant is

$$\frac{1}{(n+1)!(2\pi i)^{n+1}} \int_{\Gamma^{n+1}} \prod_j G(w_j) \prod_{i<j} (w_j - w_i)^2 \prod_j \mathrm{d}w_j.$$

Here Γ surrounds K in its analytic complement. The maximum squared Vandermonde on successively smaller contour neighborhoods has leading logarithm $n(n+1)\log \operatorname{cap}(K)$. The remaining factors contribute only $O(n \log n)$ for a fixed contour. This explains why there is one capacity exponent on the n^2 scale, not twice that exponent.

12.4 The complete leading cost at 2

For $n \geq 1$ with $J_n \neq 0$, write $\operatorname{Prim}(J_n) = \kappa_n J_n$. We give a short bound sufficient for the leading rate; the stronger finite-index residue-rank correction is unnecessary here. In $\mathbb{Q}_2(i)$ take $\pi_2 = 1 - i$, with $v_{\pi_2}(2) = 2$. Substitute $x = qt$, so $v_{\pi_2}(q) = -1$. The polylogarithm blocks become

$$\mathcal{B}_{m,s}(x) = \frac{\operatorname{Li}_s(-x^m) - \operatorname{Li}_s(-i^m x^m)}{2i}, \qquad [x^{mj}]\mathcal{B}_{m,s} = \frac{(-1)^j(1 - i^{mj})}{2ij^s}.$$

For $m = 2$ these vanish at even j. For $m = 3, 5$ they vanish when $4 \mid j$; at odd j the valuation is -1, and at $j \equiv 2 \pmod 4$ it is $-2s$. Including $c_m, m/2, m^2/12$, multiplication by -64 makes each of the three blocks in (12.1) integral. The worst valuations before that multiplication, for each of the three blocks, are $-6, -10, -12$ at $m = 2, 3, 5$.

In the x variable $L = -\log(1 - ix^2)$. Through degree $2n$ its coefficients have 2-adic valuation at least $-r$, where $r = \lfloor \log_2 n \rfloor$; those of L^2 have valuation at least $-2r$. Hence $f_j = \lambda q^{-j} h_j$ is integral for $j \leq 2n$, where $\lambda = -64\, 2^{2r}$. Scale row and column i of the bordered matrix by q^{-i}, with index zero for the border, and scale the whole matrix by λ. Its entries are λX, $\pm f_i$ and $f_{i+j} - q^{-1} f_{i+j-1}$, all integral. The determinant multiplier has 2-adic valuation

$$B_n = \frac{n(n+1)}{2} + (6 + 2r)(n+1).$$

Since the original coefficients are rational, their valuations over $\mathbb{Q}_2(i)$ are twice their ordinary 2-adic valuations. We conclude

$$v_2(\kappa_n) \leq B_n = \tfrac{1}{2}n^2 + O(n \log n). \tag{12.6}$$

Combining (12.5) and (12.6) gives

$$\log |2^{B_n} J_n(\theta)| \leq -\frac{\log 2}{60} n^2 + o(n^2).$$

These are shrinking, infinitely often nonzero linear forms with 2-integral coefficients. They are not necessarily integer forms.

12.5 The precise missing inequality

Put $d_n = 2^{\max(0, v_2(\kappa_n))}$. It would suffice to prove

$$\limsup_{\substack{n \to \infty \\ J_n \neq 0}} \frac{\log(\kappa_n / d_n)}{n^2} < \frac{\log 2}{60}. \tag{12.7}$$

This includes all odd-prime costs and any compensating numerator content; it is a sufficient target, not a necessary condition for every use of the germ. No bound of this strength is established. Finite primitive failures at $n = 12, 16, 20$ do not settle the asymptotic question. To reproduce a form, expand (12.1) through degree $2n$, build (12.2) over $\mathbb{Q}[X]$, and clear its two coefficients followed by their full gcd. This procedure does not require solving a possibly singular Padé system.

12.6 A signed-integral alternative with only one LCM layer

The same rational kernel also gives a different, completely explicit starting point. Retain D above and put $N_0 = 17DU \in \mathbb{Z}[x]$; it has a zero of order two at zero. For arbitrary $S \in \mathbb{Z}[x]$ and $A \in \mathbb{Z}$ set

$$F = x(1-x)^2 N_0^4 S, \qquad V = 17\frac{E^3 F}{N_0}, \qquad P = A + DV.$$

At most three differentiations can remove at most three of the four N_0 factors, so $V \in \mathbb{Z}[x]$. With $N = DU = N_0/17$ one has $NV = E^3 F$. If $F = \sum f_j x^j$, the constraints of (7.4) become $F(1) = EF(1) = 0$, which hold by construction. Consequently (7.5) gives exactly

$$I_U(P) = A\theta + \sum_j \frac{f_j}{j}. \tag{12.8}$$

Thus $\operatorname{lcm}(1, \dots, \deg F)$ clears the rational remainder, rather than its fourth power. This is an arithmetic improvement, not a smallness or nonvanishing theorem.

There are infinitely many nonzero source polynomials giving the zero form. For any nonzero $Z \in \mathbb{Z}[x]$, let

$$R_0 = N_0^5 (1-x)^3 Z, \qquad F = ER_0, \qquad S = (5N_0'(1-x) - 3N_0)Z + N_0(1-x)Z'.$$

Then this F has the prescribed form and $\sum f_j/j = R_0(1) - R_0(0) = 0$. With $A = 0$ the source is nonzero, since E^4 is injective on polynomials with zero constant coefficient. A small-vector argument must therefore distinguish the kernel of the map to the final two coefficients from genuine small nonzero forms.

There is also a limitation on an absolute-integral estimate. In the notation of Section 7, the constraint preserves $\int PU(-\log x)\,\mathrm{d}x/x = AM_1$, where $M_1 > 9/10$ is proved there. As $W(\rho) \ge (\log 2)^2 \rho/12$,

$$\int_0^1 |PU| W(-\log x)\frac{\mathrm{d}x}{x} \ge \frac{(\log 2)^2}{12}|A| M_1.$$

After a positive normalization making the nonzero target coefficient an integer this is greater than $1/30$. Hence a successful signed choice must exploit cancellation in the actual integral and establish nonvanishing separately. Formula (12.8) supplies free parameters and exact arithmetic; it does not yet supply that choice or those two proofs.

13 Two-endpoint interpolation and exact prime deletion

These are two distinct ways to design an approximation before taking its primitive form. Endpoint interpolation forces zeros in an integral source. Prime deletion instead removes selected coefficient denominators from a generating function. Both operations are exact, but neither alone controls the normalized error. We give enough information to construct the endpoint pairs without recovering a large historical linear system. The records are H01–H03 and A04–A05.

13.1 A four-by-five system for all rational Hermite sources

Write $\mathcal{F}(x) = \sum_{j \ge 1} \chi_4(j) x^j / j^4$ and $E = x\,d/dx$, so $E^4 \mathcal{F} = x/(1+x^2)$ and $\mathcal{F}(1) = \theta$. For a source f put

$$\mathcal{I}(f) = \frac{1}{6}\int_0^1 f(x)(-\log x)^3\,\mathrm{d}x.$$

Fix integers $N \geq 1$, $a, b \geq 1$ with $a + b = 2N$, and a quadratic $D \in \mathbb{Z}[x]$ with $D(0) = 1$ and no zero on $[0, 1]$. Seek $A \in \mathbb{Q}$ and $R \in \mathbb{Q}[x]$, $\deg R \leq 2N$, $R(0) = 0$, such that, with $G = R/D^N$,

$$f = \frac{A}{1+x^2} + \frac{E^4 G}{x} = O(x^a) \text{ at } 0, \qquad f = O((x-1)^b) \text{ at } 1. \tag{13.1}$$

Four integrations by parts, or the monomial identity $\mathcal{I}(E^4 x^j/x) = 1$ for $j \geq 1$, give

$$\mathcal{I}(f) = A\theta + G(1). \tag{13.2}$$

For the integration argument the rational G is regular on the integration interval and vanishes at zero; all logarithmic boundary terms at zero vanish, and the surviving endpoint term is $G(1)$. Zeros of f may make its integral small, but its sign and normalization still require separate analysis.

Define

$$R_{\mathrm{low}}(x) = [-D(x)^N \mathcal{F}(x)]_{\leq a}, \qquad L(z) = \prod_{j=a+1}^{2N} (z - j), \qquad D(x)^N = \sum_{\ell} d_\ell x^\ell,$$

where brackets mean Taylor truncation. Let

$$T(y) = \frac{1}{2\cosh y}, \qquad V_b(y) = -\sum_{m=4}^{b+3} \frac{T^{(m-4)}(0) y^m}{m!}.$$

All these finite coefficients are rational. For $0 \leq r, k \leq 3$ set

$$K_{rk} = \sum_{\ell} d_\ell \, \frac{1}{k!} \frac{\mathrm{d}^k}{\mathrm{d}z^k} (z^r L(z)) \Big|_{z=\ell}, \tag{13.3}$$

$$v_r = \partial_y^r L(\partial_y)(R_{\mathrm{low}}(e^y) - D(e^y)^N V_b(y)) \Big|_{y=0}. \tag{13.4}$$

Proposition 13.1 (Exact endpoint parametrization; H01)**.** *The source space* (13.1) *is naturally identified with*

$$Kg = Av, \qquad g_k = \partial_y^k G(e^y) \Big|_{y=0}, \quad 0 \leq k \leq 3.$$

Its endpoint form is $A\theta + g_0$. This includes singular K and zero-target solutions; no determinant is divided out.

Proof. The condition at zero is exactly $R = AR_{\mathrm{low}} + \sum_{j=a+1}^{2N} r_j x^j$. Put $H(y) = G(e^y)$. Multiplying the source by $x = e^y$ changes it to $AT(y) + H''''(y)$, so the condition at one is

$$H(y) = \sum_{k=0}^{3} g_k y^k/k! + AV_b(y) + O(y^{b+4}).$$

The condition at zero is equivalently

$$L(\partial_y)(D(e^y)^N H(y) - AR_{\mathrm{low}}(e^y)) = 0.$$

Apply ∂_y^r, $r = 0, 1, 2, 3$, and evaluate at zero. The identity $P(\partial_y)(e^{\ell y} y^k/k!)|_0 = P^{(k)}(\ell)/k!$ gives exactly $Kg = Av$.

Conversely, use the prescribed Taylor polynomial for H as initial data for the displayed order-b differential equation. Its leading coefficient at zero is $D(1)^N \neq 0$. The four equations say that the residual of that polynomial vanishes to order four, so the differential recurrence makes the solution agree through degree $b + 3$. This also handles $b < 4$; the same recurrence forces the additional specified derivatives. The equation implies $D(e^y)^N H - AR_{\mathrm{low}}(e^y) = \sum_{j=a+1}^{2N} r_j e^{jy}$. Its distinct integer exponents and rational initial data give rational r_j by a Vandermonde system. Thus it reconstructs the required R. □

13.2 Primitive endpoint pairs and the actual error

Every entry of K is integral, since a divided derivative of an integer polynomial at an integer is integral. Let d_a be the lcm of the odd positive integers at most a and $C = 2d_a^4$. Then Cv is integral. For R_{low} this follows from its coefficient denominators; for V_b it follows from $T^{(j)}(0)$ being half-integers and from Leibniz's rule, whose evaluated factorial ratios are integers. Cramer's identity supplies the integer endpoint pair

$$(A, B) = \big(C \det K,\ \det K[\text{column } 0 \leftarrow Cv]\big). \tag{13.5}$$

If this pair is zero it is not a witness. If the augmented matrix has rank less than four, compute its full rational kernel instead of using vanishing maximal minors. For every nonzero pair, divide its complete gcd before any numerical comparison.

There is an exact formula for the signed error, not merely its source. Set $S_a(x) = R_{\mathrm{low}}(x) + D(x)^N \mathcal{F}(x)$ and $\ell_r = \partial_y^r L(\partial_y) S_a(e^y)|_0$. Since $\partial_y^4 \mathcal{F}(e^y) = T(y)$, the column relation is $v = \ell - K\eta$, where $\eta_k = \partial_y^k \mathcal{F}(e^y)|_0$ and $\eta_0 = \theta$. Multilinearity therefore gives

$$\det K\, \theta + \det K[0 \leftarrow v] = \det K[0 \leftarrow \ell]. \tag{13.6}$$

All lower-period endpoint derivatives disappear through repeated columns. The identity is valid even when K is singular.

A useful factorial normalization leaves the primitive endpoint pair unchanged. With $d_b = \mathrm{lcm}(1, \dots, b)$, $d_b^k K_{rk}/b!$ is integral: expand $L(z)/b! = \binom{z-a-1}{b}$ at an integer by Vandermonde's identity, and use $\binom{t}{j} = (-1)^{j-1}(t/j)\prod_{s=1}^{j-1}(1 - t/s)$. The coefficient of t^k has denominator dividing d_b^k. Replacing K by $K\,\mathrm{diag}(1, d_b, d_b^2, d_b^3)/b!$ and v by $v/b!$ multiplies both determinants by $d_b^6/(b!)^4$. It improves a representation, not the primitive pair.

For $D = 1 + x^2$, $a = b = N = n$, the scalar remainder S_n is positive on $x > 0$. To see this, use $\mathcal{F}(x) = x\int_0^1 (-\log u)^3/(1 + x^2u^2)\,\mathrm{d}u/6$. Put $m = \lceil n/2 \rceil$, $y = x^2$, $t = u^2$. The remainder of $(1 + y)^n/(1 + ty)$ after degree $m - 1$ in y is

$$\frac{y^m}{1 + ty}\left(c_m(t) + \sum_{k=m+1}^{n} \binom{n}{k} y^{k-m}\right), \qquad c_m(t) = \sum_{k=0}^{m} \binom{n}{k}(-t)^{m-k}.$$

In reverse order the alternating terms defining c_m decrease in magnitude on $0 \le t < 1$, giving strict positivity. Integrate the remainder to obtain $S_n > 0$. The determinant in (13.6) contains high derivatives of S_n; its positivity does not follow. The unresolved questions for a selected schedule are nonzero endpoint error and the comparison of that error with the reduced pair's size. Equations (13.3)–(13.5) give a fixed-size exact procedure for investigating them.

13.3 Prime deletion changes the generating function

Return to $\mathcal{F}(t) = \sum \chi_4(j) t^j/j^4$. For an odd prime p,

$$\mathcal{F}(t) - \frac{\chi_4(p)}{p^4}\mathcal{F}(t^p) = \sum_{p \nmid j} \frac{\chi_4(j) t^j}{j^4}.$$

For a finite set S of odd primes put $P = \prod_{p \in S} p$ and

$$G_S(t) = \sum_{(j,P)=1} \frac{\chi_4(j) t^j}{j^4}, \qquad \alpha_S = \prod_{p \in S}(1 - \chi_4(p)/p^4), \qquad G_S(1) = \alpha_S \theta. \tag{13.7}$$

Build the bordered pencil (12.2) from the coefficients of G_S, and evaluate its variable at $\alpha_S\theta$. For each $p \in S$ every coefficient entry is p-integral, whereas $v_p(\alpha_S) \geq -4$. Therefore the actual primitive multiplier of this endpoint pencil satisfies $v_p(\kappa) \leq 4$, independently of matrix size. The endpoint factor is essential to this conclusion.

For matrix order n, delete every odd prime $y < p \leq 2n$. The surviving prefix through degree $2n$ is cleared by

$$D_{y,2n} = \prod_{\substack{p \leq \min(y,2n) \\ p \text{ odd}}} p^{4\lfloor \log_p(2n) \rfloor}.$$

Scaling the bordered matrix gives a sufficient overall multiplier $P^4 D_{y,2n}^{n+1}$, whence

$$\log^+ \kappa \leq 4 \log P + 4(n+1)\pi(y)\log(2n) = o(n^2)$$

if $\pi(y_n)\log n = o(n)$; $y_n = \sqrt{n}$ suffices even using $\pi(y) \leq y$ and $\log P \leq 2n\log(2n)$. This is a genuine arithmetic result with no matching analytic theorem.

The analytic cost is concrete. In the divisor expansion $G_S = \sum_{d|P} \mu(d)\chi_4(d)d^{-4}\mathcal{F}(t^d)$, apply E^4. At a primitive $4P$th root of unity only $d = P$ can have a pole, since $E^4\mathcal{F}(t^d) = d^4 t^d/(1+t^{2d})$. Its coefficient is nonzero. Thus growing prime sets create genuine new singularities. Neither the analytic bound nor the nonvanishing subsequence for a fixed germ can simply be reused for this varying family.

13.4 A sparse family whose tiny determinant factor cancels

There is a useful explicit test of the last warning. For an integer $k \geq 1$, let $M = 3^k$, $n = M - 1$, and delete every odd prime at most $B \geq 2n$ except 3. The coefficient prefix is $r(t) + ct^M$, where

$$r(t) = \sum_{0 \leq a < k} \frac{(-1)^a t^{3^a}}{3^{4a}}, \qquad c = \frac{(-1)^k}{M^4}, \qquad \deg r = M/3 < M/2.$$

The polynomials $q(t) = r(t)/t - ct^{M-1}$ and $p(t) = r(t)^2/t$ obey $q(t)(r(t) + ct^M) - p(t) = -c^2 t^{2M-1}$, so they give a diagonal Padé pair for this prefix. More precisely its bordered determinant is

$$J_{n,S}(X) = (-1)^{n(n-1)/2+1} c^{M-2}((r(1) - c)X - r(1)^2). \tag{13.8}$$

Here is a verification of the scalar, which matters for normalization. Reverse the columns of T and multiply by $(I - \mathrm{shift}_{\mathrm{down}})^{-1}$. The result is $cI + N - \mathbf{1}v^T$, with $N_{ij} = r_{M+i-j}$, $v_j = r_{M-j}$, $N^2 = 0$ and $v^T N = 0$. The determinant lemma gives $\det T = \delta c^{n-1}(c - r(1))$, $\delta = (-1)^{n(n-1)/2}$. Also $\mathrm{adj}(T)_{11} = \delta c^{n-1}$ and $T\mathbf{1} = ce_1 - b$. Using these identities in $X \det T + b^T \mathrm{adj}(T) b$ proves (13.8) without assuming T invertible.

The entire factor c^{M-2} disappears after primitive normalization. The remaining form is $(r(1) - c)\alpha_S\theta - r(1)^2$. At the natural cutoff $B = 2n$ its useful real and primitive estimates remain unresolved. Tiny raw determinants alone therefore do not settle this prescribed sparse family. An additional large-cutoff obstruction in the catalog concerns a different scale; it does not close the natural cutoff.

14 Modular transformations and prime-filtered cusp errors

[record M01][record M02] A modular transformation can turn slow convergence near one boundary point into fast convergence near another. For this problem it also introduces an unwanted multiple of π^4. The construction below removes that term exactly. It then separates three questions that are easy to confuse: whether the transformed error is small and nonzero, whether the coefficients have manageable denominators, and whether these facts produce small integer forms.

14.1 The classical identity and its cancellation

Write $\chi = \chi_4$, $q = e^{2\pi i\tau}$, with $\Im\tau > 0$, and define

$$L_F(\tau) = \sum_{d\geq 1} \frac{\chi(d)q^d}{d^4(1-q^d)}, \qquad L_G(\tau) = \sum_{d\geq 1} \frac{q^d}{d^4(1+q^{2d})},$$
$$F(\tau) = L_F(\tau) - 2L_F(2\tau) = \sum_{d \text{ odd}} \frac{\chi(d)q^d}{d^4(1+q^d)}, \qquad G(\tau) = L_G(\tau) - \frac{L_G(2\tau)}{16} = \sum_{d \text{ odd}} \frac{q^d}{d^4(1+q^{2d})}. \tag{14.1}$$

All sums converge absolutely. A specialization of Bradley's Ramanujan-type formula [14, Corollary 4] gives

$$L_F(\tau) + \frac{\theta}{2} = \frac{i\tau^3}{2} L_G\left(-\frac{1}{4\tau}\right) + i\pi^4\left(\frac{5}{3072\tau} - \frac{\tau}{192} - \frac{\tau^3}{360}\right). \tag{14.2}$$

For clarity about normalization, in that corollary take the integer parameter 2, put its positive parameters equal to $\alpha = \pi\Im\tau$, $\beta = \pi/\Im\tau$, and use $E_0 = 1, E_2 = -1, E_4 = 5$, $B_0 = 1, B_2 = 1/6, B_4 = -1/30$. This gives (14.2) on the positive imaginary axis; analytic continuation gives the upper half-plane identity. Subtract twice the same formula at $2u$ from the formula at u, then put $u = -1/(8\tau)$. The result is

$$\tau^3 F\left(-\frac{1}{8\tau}\right) - \frac{i}{64} G(\tau) = \frac{\theta}{2}\tau^3 - i\pi^4\left(\frac{\tau^2}{512} + \frac{1}{12288}\right). \tag{14.3}$$

Thus the special value enters a transformation law with an explicit polynomial correction. This is the Eichler-integral viewpoint: four derivatives $q\,d/dq$ of these series are holomorphic modular forms of weight five. Their Fourier expansions and transformations are instances of the Eisenstein-series construction in Berndt–Straub [15, Section 4 and Example 6.4].

Choose coefficients

$$a_1 = 16, \qquad a_2 = -10, \qquad a_4 = 1.$$

They satisfy $\sum a_r r = \sum a_r r^3 = 0$ and $\sum a_r = 7$. Consequently, with

$$A = 16F - 10F(2\tau) + F(4\tau), \qquad H = A + \frac{\sqrt{2}}{2}\big(4G - 5G(2\tau) + G(4\tau)\big), \tag{14.4}$$

all π^4 terms cancel and

$$H\left(-\frac{1}{32\tau}\right) - \frac{7}{2}\theta = \frac{i}{32^{3/2}\tau^3}\left(H(\tau) - \frac{7}{2}\theta\right). \tag{14.5}$$

Here is a direct way to check the coefficients, without guessing a modular combination. Solve (14.3) for $F(r\tau)$. Its polynomial terms are proportional to $r\tau$ and $r^3\tau^3$, and its G term is $-8i(r\tau)^3 G(-1/(8r\tau))$. Under $\tau \mapsto -1/(32\tau)$ the indices $r = 1, 2, 4$ reverse to $4/r$; their cubic weights give exactly $(\sqrt{2}/2)(4, -5, 1)$. The two displayed zero sums remove the polynomial terms in both directions. At the fixed point $\tau = i/\sqrt{32}$ the multiplier in (14.5) is -1, so $H(i/\sqrt{32}) = 7\theta/2$.

14.2 Deleting prime denominators while retaining an exact error

Let S be a finite set of odd primes, $P = \prod_{p\in S} p$, and $\alpha_S = \prod_{p\in S}(1 - \chi(p)/p^4)$. Empty products are one. For $V_d f(\tau) = f(d\tau)$ set

$$A_S = \prod_{p\in S}(1 - \chi(p)p^{-4}V_p)A = \sum_{r\in\{1,2,4\}} \sum_{d\mid P} a_r \mu(d)\chi(d) d^{-4} F(rd\tau), \qquad c_S = \frac{7}{2}\alpha_S. \tag{14.6}$$

The symbol $\mu(d)$ here is (-1) to the number of prime factors of the squarefree divisor d. The elementary identity

$$(1-\chi(p)p^{-4}V_p)F=\sum_{\substack{d\ \mathrm{odd}\\ p\nmid d}}\frac{\chi(d)q^d}{d^4(1+q^d)}$$

shows that every coefficient of A_S is p-integral for $p\in S$. It removes an outer divisor in the Lambert sum; it does not remove every Fourier coefficient whose index is divisible by p. Dominated convergence gives $A_S(iy)\to c_S\theta$ as $y\downarrow 0$.

Apply the solved form of (14.3) term by term in (14.6). The same two zero sums eliminate the polynomial terms for every d, giving the exact remainder

$$A_S(\tau)-c_S\theta=-8i\tau^3\sum_{r,d\mid P}\frac{a_r r^3\mu(d)\chi(d)}{d}G\left(-\frac{1}{8rd\tau}\right), \tag{14.7}$$

where the sum over r is still $1,2,4$. This identity yields a uniform sign and size estimate.

Proposition 14.1. *Put $\sigma_P=\sum_{d\mid P}d^{-1}$, $x=\pi/(16Py)$ and $z=e^{-x}$. If $x\ge\log(20P\sigma_P)$, then*

$$A_S(iy)-c_S\theta=-\frac{512\mu(P)\chi(P)y^3z}{P}(1+\varepsilon),\qquad |\varepsilon|\le 2z^2+5P\sigma_P z<\frac{1}{2}. \tag{14.8}$$

In particular the error is nonzero, with the displayed sign, and its absolute value lies strictly between $256y^3z/P$ and $768y^3z/P$.

Proof. Regard G temporarily as a power series in a real argument z. For $0<z\le 1/2$ its defining sum gives $|G(z)-z|\le 2z^3$ and $0<G(z)\le 2z$: the first summand differs from z by at most z^3, while the remaining odd summands are bounded by $z^3/[3^4(1-z^2)]$. In (14.7) the unique largest dilation is $rd=4P$, with coefficient $64\mu(P)\chi(P)/P$ and argument z. Every other argument is $z^{4P/(rd)}$ with integer exponent at least 2. Since $\sum_r|a_r|r^3=160$, their total absolute contribution inside the sum is at most $320\sigma_P z^2$. Divide by $64z/P$ and include the leading term's relative error $2z^2$. Finally $z\le(20P\sigma_P)^{-1}\le 1/20$ makes the bound less than $1/2$. □

The required approach scale is explicit: $y\le\pi/[16P\log(20P\sigma_P)]$. These are analytic evaluations at $q=e^{-2\pi y}$. They are not rational approximants merely because their error is small. A further construction must turn the identity into rational coefficients without losing its decay.

14.3 A fixed-filter obstruction to the ordinary contour estimate

We give the analytic limitation rather than infer it from a plot of singularities. A holomorphic modular form of positive integer weight, periodic with period one and holomorphic at the cusps of a finite-index subgroup, cannot extend meromorphically through a finite rational cusp unless it is zero. Indeed, choose $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)$ with $c\neq 0$ taking infinity to the cusp. After increasing the cusp width if necessary, $f|_k\gamma=a_0+O(e^{-\eta\Im z})$ there. Meromorphic continuation through a/c would make this a convergent Laurent series in $1/z$. Its difference from a_0 is exponentially flat on the imaginary axis, so every Laurent coefficient is zero. It follows that $f(\tau)=a_0(a-c\tau)^{-k}$, which contradicts periodicity if $a_0\neq 0$ and nonzeroness if $a_0=0$.

For fixed S, $(q\,d/dq)^4A_S$ is such a modular form after passing to a common subgroup for its finitely many dilations. Its coefficient of q is 16, so it is nonzero. If A_S continued through a root of unity, its fourth derivative would too. Every root of unity is therefore a singularity; their density makes the unit circle a *natural boundary*, across which no analytic continuation is possible.

There are two useful consequences. First, for $h = 2A_S$, the cumulative germ $(X - h(q))/(1-q)$ is nonrational for every complex X. The Hankel criterion used in Section 12 therefore gives infinitely many nonzero endpoint determinants for each fixed filter. Second, the compact set removed in the ordinary reciprocal-plane absolute-contour estimate must contain the whole unit circle, whose logarithmic capacity is one. The estimate used in Section 12 cannot supply a negative quadratic decay constant here. This is a limitation of that estimate. It is neither a lower bound for the actual determinants nor a uniform statement when S grows with the matrix size.

14.4 An explicit coefficient realization and its prime cost

For readers wishing to reproduce the coefficient experiment, the level-32 choice was the following formal change of variable and prefactor:

$$t(q) = q \prod_{d\in\{1,2,4,8,16,32\}} \prod_{j\ge 1} (1-q^{dj})^{e_d}, \quad (e_1,e_2,e_4,e_8,e_{16},e_{32}) = (2,-3,1,1,-3,2),$$

$$J(q) = \vartheta_3(2\tau)^2 + 2\sqrt{2}\,\vartheta_3(16\tau)^2, \qquad \vartheta_3(u) = \sum_{m\in\mathbb{Z}} e^{\pi i m^2 u},$$

$$Y(q) = q\,\frac{d}{dq}\log t(q), \qquad \mathcal{A}(t(q)) = \frac{J(q)^5}{Y(q)}, \qquad \mathcal{B}(t(q)) = \mathcal{A}(t(q))H(\tau).$$

These definitions alone suffice to compute every coefficient exactly: $t = q + O(q^2)$ has an integral compositional inverse, $Y \in 1 + q\mathbb{Z}[[q]]$, and all operations take place in $\mathbb{Q}(\sqrt{2})[[q]]$. Write $\mathcal{A} = \sum(a_{n,0} + \sqrt{2}a_{n,1})t^n$ and $\mathcal{B} = \sum(b_{n,0} + \sqrt{2}b_{n,1})t^n$. Then $a_{n,j} \in \mathbb{Z}$, and $2\,\mathrm{lcm}(1,\dots,n)^4 b_{n,j}$ is integral. The latter follows directly from the Lambert coefficients in (14.1); integral composition and multiplication preserve that denominator bound.

The fourth powers are real costs of these coefficient sequences, not just artifacts of this sufficient bound. For an odd prime p, all coefficients before degree p are p-integral, while

$$p^4[q^p]H \equiv 16\chi(p) + 2\sqrt{2} \pmod{p}.$$

Since $\mathcal{A}(0) = (1+2\sqrt{2})^5 = 401 + 298\sqrt{2}$, and composition with $q = t + O(t^2)$ does not change this first nonintegral coefficient,

$$\begin{aligned} p^4 b_{p,0} &\equiv 6416\chi(p) + 1192 \pmod{p}, \\ p^4 b_{p,1} &\equiv 4768\chi(p) + 802 \pmod{p}. \end{aligned} \tag{14.9}$$

Thus $v_p(b_{p,j}) = -4$ apart from the finitely many primes dividing one of the four displayed nonzero integers. The least denominator clearing a whole prefix through n consequently has logarithm $4n + o(n)$, by the prime number theorem and the matching upper bound. This does not say that every individual coefficient has that denominator, or determine the primitive content of a determinant built from them.

What remains. The transformation and cusp estimate are available exactly, and the coefficient construction is reproducible. A continuation of this route must control primitive coefficient combinations, or supply a different arithmetic interpretation of the transformed values. Neither selected prime deletion nor the nonzero cusp error supplies this missing step. The more specialized coordinate and rational-system tests remain in records M03–M04; their finite comparisons do not exclude other modular choices.

15 Coupled period reductions and rationality-dependent sources

The constructions in this section illustrate two different ways of moving beyond a single positive matrix. Coupling produces a small space of special values, which one can then eliminate by linear algebra. Alternatively, assuming θ rational can create endpoint zeros that were unavailable before the assumption. Both operations are exact; neither automatically preserves the property needed for an irrationality proof.

15.1 An explicit five-period family

[record P03] A coupled integral led to the rational series defined below. We use the series as the definition here, so that reconstructing its coefficients does not require the original geometric or spherical-integral derivation. For an integer $n \geq 0$, put $\lambda = 4n + 1$ and

$$A_j(k) = \frac{(4n)!}{n!} \frac{(-1)^j \binom{3n}{j}}{\prod_{q=n+j+1}^{4n} (k+q) \prod_{q=4n+2}^{4n+j+1} (k+q)}, \qquad 0 \leq j \leq 3n. \tag{15.1}$$

An empty product is one. Define

$$L_j = 2(k+3n+1+j), \quad M_l = 2(k+2n+1+l), \quad T_{jl} = 2(k+2n+1+j+l), \quad V_{jl} = 2(k+1+j+l),$$

and

$$\begin{aligned} I_n(k) &= 2 \sum_{j,l=0}^{3n} A_j(k) A_l(k) \left(\frac{1}{L_j T_{jl}} + \frac{1}{L_j M_l} + \frac{1}{M_l V_{jl}} \right), \\ C_n(k) &= \frac{\prod_{r=1}^{4n} (k+2r)}{2^{4n}(4n)!}, \qquad R_n(k) = \frac{4(n!)^4 C_n(k) I_n(k)}{(4n)!(k+\lambda)^2}, \\ J_n &= \sum_{\ell \geq 0} (-1)^\ell R_n(2\ell). \end{aligned} \tag{15.2}$$

Every uncombined summand of $R_n(k)$ is $O_n(k^{-2n-4})$: each A_j has $3n$ denominator factors, the radial fractions supply two more, the angular denominator another two, and C_n has degree $4n$. Thus the series defining J_n converges absolutely. The formulas are finite rational operations followed by an elementary convergent sum; they define the family without fitting a recurrence.

Proposition 15.1. *For each $n \geq 0$, the formulas above give an explicitly computable rational decomposition*

$$J_n = r_n + b_n \beta(2) + a_n \theta + \ell_n \log 2 + z_n \zeta(3). \tag{15.3}$$

No linear independence of these five numbers is assumed.

Proof. All poles of R_n have the form $-q$, $1 \leq q \leq 8n+1$, with order at most four. Away from $q = \lambda$, the two A factors supply at most two orders and the radial fraction at most two. At $q = \lambda$ the A factors are regular, so the angular double pole and the radial fractions still give at most four. Properness was established above. Consequently

$$R_n(k) = \sum_{q=1}^{8n+1} \sum_{s=1}^{4} \frac{c_{q,s}}{(k+q)^s}, \qquad c_{q,s} \in \mathbb{Q}.$$

There is a useful reflection symmetry. Under $k' = -2\lambda - k$, $j' = 3n - j$, formula (15.1) gives $A_j(k') = A_{j'}(k)$: the denominator factors reverse, and their signs cancel the change in $(-1)^j$. The radial factors transform as $L \leftrightarrow -M$ and $T \leftrightarrow -V$; swapping j, l then preserves their sum. Both C_n and $(k+\lambda)^2$ are invariant. Hence

$$R_n(-2\lambda - k) = R_n(k), \qquad c_{2\lambda-q,s} = (-1)^s c_{q,s}.$$

For $q = 2h+1$ and $q = 2h+2$, respectively, the required alternating sums are

$$\sum_{\ell\ge 0} \frac{(-1)^\ell}{(2\ell+2h+1)^s} = (-1)^h \left(\beta(s) - \sum_{j=0}^{h-1} \frac{(-1)^j}{(2j+1)^s} \right),$$

$$\sum_{\ell\ge 0} \frac{(-1)^\ell}{(2\ell+2h+2)^s} = 2^{-s}(-1)^h \left(\eta(s) - \sum_{j=1}^{h} \frac{(-1)^{j-1}}{j^s} \right),$$

where $\eta(1) = \log 2$ and $\eta(3) = 3\zeta(3)/4$. Across reflected odd poles the tail signs agree, canceling odd beta orders; across reflected even poles they are opposite, canceling even eta orders. The central pole is odd and has only even orders. Only $\beta(2), \beta(4), \eta(1), \eta(3)$ and rational tails remain. The individual order-one sums converge conditionally; there are only finitely many of them, so taking their limits is legitimate. □

This proof is also an exact algorithm. At a pole of maximal order four, expand $(k+q)^4 R_n(k)$ through degree three in $k+q$ to recover the four coefficients, then use the two finite-tail formulas. For example,

$$J_0 = 6\theta, \qquad J_1 = \frac{301}{384} - \frac{503}{64}\beta(2) + \frac{243}{32}\theta - \frac{73}{24}\log 2 + \frac{27}{32}\zeta(3).$$

The second displayed coefficient vector has primitive integer form $(301, -3018, 2916, -1168, 324)$ in the order $(1, \beta(2), \theta, \log 2, \zeta(3))$.

Elimination and what it does not preserve. Choose four indices $n_0, \dots, n_3$ and form the 3×4 rational matrix whose columns are $(b_{n_j}, \ell_{n_j}, z_{n_j})^T$. Let w_j be its signed maximal minors, with signs chosen so that $\sum_j w_j(b_{n_j}, \ell_{n_j}, z_{n_j}) = 0$. Then

$$\sum_{j=0}^{3} w_j J_{n_j} = \left(\sum_j w_j a_{n_j} \right) \theta + \sum_j w_j r_{n_j}. \tag{15.4}$$

If the matrix has rank less than three, use an explicitly computed rational null vector instead; the maximal minors would all vanish. Thus the auxiliary constants can be eliminated without assuming they are independent. But positive individual integrals can cancel in this signed sum, and the minors can have large denominators. To use (15.4) one still needs nonvanishing and a bound after dividing out the full common content of its two coefficients. The four recorded finite cofactor experiments do not supply those all-index estimates. The fixed period-space theorem is the reusable conclusion here; it is not a proof that the final linear form is small.

15.2 Endpoint zeros obtained by assuming rationality

[record E07] Niven-type proofs suggest a different starting point: use a hypothetical rational value to construct a positive function with many endpoint zeros. Let

$$\mathcal{F}(x) = \sum_{m\ge 1} \frac{\chi_4(m) x^m}{m^4}, \qquad E = x\frac{d}{dx}, \qquad E^4\mathcal{F} = \frac{x}{1+x^2}.$$

The integral representation

$$\mathcal{F}(x) = \frac{x}{6}\int_0^1 \frac{(-\log t)^3}{1+x^2t^2}\,\mathrm{d}t$$

shows that $\mathcal{F}'(x) > 0$ on $[0,1]$: its differentiated integrand contains $(1-x^2t^2)/(1+x^2t^2)^2$, positive almost everywhere. Assume $\theta = u/v$ with positive integers u, v, and define

$$A_n(x) = \frac{[v\mathcal{F}(x)(u - v\mathcal{F}(x))]^n}{n!}, \qquad I_n = \int_0^1 \frac{A_n(x)}{1+x^2}\,\mathrm{d}x. \tag{15.5}$$

Then $0 < I_n \le (u^2/4)^n/n! \to 0$. The source has a zero of order n at each endpoint. An integrality theorem for an unbounded sequence of these I_n would therefore give a contradiction. The issue is that the endpoint zeros do not provide the requisite integrality theorem.

Proposition 15.2. *For any fixed $n \ge 1$ and nonzero integers u, v, every odd prime $p > n$ with $p \nmid uv$ satisfies*

$$v_p(A_n^{(p+n-1)}(0)) = -3.$$

Under the rationality assumption the first nonzero Euler derivative at the other endpoint is

$$E^n A_n(1) = (-uv)^n \left(\frac{\pi^3}{32}\right)^n.$$

Proof. Expand

$$A_n = \frac{1}{n!}\sum_{k=0}^{n}(-1)^k\binom{n}{k}u^{n-k}v^{n+k}\mathcal{F}^{n+k}.$$

At degree $N = p+n-1 < 2p$, a p-denominator can only come from the coefficient $\chi_4(p)/p^4$ in one factor. In $\mathcal{F}^{n+k}$, using that coefficient requires degree at least $p+n+k-1$. Only $k=0$ contributes, with n choices for that factor and the linear coefficient in every other factor. Its contribution is $u^n v^n \chi_4(p)/((n-1)!p^4)$, of valuation -4; all other terms are p-integral. Multiplying by $N!$, whose valuation is one, proves the claim. At $x=1$, use the coordinate $t = \log x$. The linear term of $v\mathcal{F}(e^t)(u - v\mathcal{F}(e^t))$ is $-uv(E\mathcal{F})(1)t$. Since $(E\mathcal{F})(1) = \beta(3) = \pi^3/32$, the coefficient of t^n gives the second formula. □

Thus not all origin derivatives can be integral, and rationality of the endpoint does not rationalize its derivative. This excludes a blanket endpoint-integrality argument, while leaving open constructions that use only selected derivatives and control their denominators.

15.3 Changing variables and the differential obstruction

In the coordinate $z = \mathcal{F}(x)$, the source becomes a polynomial, but the natural corresponding integral is explicitly

$$\int_0^1 A_n(x)\mathcal{F}'(x)\,\mathrm{d}x = \frac{u^{2n+1}n!}{v(2n+1)!}. \tag{15.6}$$

Indeed substitute $w = v\mathcal{F}(x)$ and use $\int_0^u w^n(u-w)^n\,\mathrm{d}w = u^{2n+1}(n!)^2/(2n+1)!$. This is a positive rational number. For $n = (p-1)/2$, $p \nmid uv$, its valuation at the odd prime p is -1. Clearing its denominator and removing common numerator content leaves the constant polynomial 1, rather than shrinking integers.

One cannot instead keep the original Euler derivative and treat $\mathcal{F}$ as its only additional rational-function coordinate. In fact

$$E\mathcal{F} \notin \mathbb{C}(x, \mathcal{F}). \tag{15.7}$$

Here is a direct functional proof. The third Euler derivative is $\arctan x$; continuation once around $x = i$ changes it by π. Integrating with E, and taking the lower changes to vanish at i, gives

$$\Delta\mathcal{F} = \frac{\pi}{6}\log^3(x/i), \qquad \Delta(E\mathcal{F}) = \frac{\pi}{2}\log^2(x/i).$$

If $E\mathcal{F} = R(x, \mathcal{F})$ with $R \in \mathbb{C}(x, z)$, repeated continuation at a generic nearby x would give $R(x, \mathcal{F} + m\Delta\mathcal{F}) = E\mathcal{F} + m\Delta(E\mathcal{F})$ for every integer m. A rational function of z agreeing with an affine function on this infinite set must itself be affine. Its slope would be $3/\log(x/i)$, which is not rational in x. This proves (15.7). The larger field containing all four Euler primitives does close under differentiation. It would require a new endpoint-elimination and arithmetic argument, which this calculation does not exclude.

A useful control example. The failure is not repaired by imposing rational Taylor coefficients, positive rational forcing and a nonpolynomial primitive. The function

$$\mathcal{H}(x) = x + \frac{\mathrm{Li}_4(x/2) - \mathrm{Li}_4(x^2/2)}{32}$$

has $\mathcal{H}(0) = 0$, $\mathcal{H}(1) = 1$, and

$$\frac{E^4\mathcal{H}}{x} = 1 + \frac{1}{32}\left(\frac{1}{2-x} - \frac{16x}{2-x^2}\right) \geq \frac{33}{64}, \qquad \mathcal{H}'(x) \geq \frac{251}{256} \quad (0 \leq x \leq 1).$$

For the derivative estimate use $z \leq \mathrm{Li}_3(z) \leq z + z^2/[8(1-z)]$ and differentiate the displayed formula; the two polylogarithmic terms divided by x are bounded below by $1/2$ and above by $9/8$, respectively. All Taylor coefficients are rational and the forcing has no pole on the interval. Nevertheless its endpoint is the rational number 1. Thus those structural properties alone cannot supply the missing irrationality implication. The target's particular pole geometry would have to enter an additional argument.

16 Quarter-shifted paired hypergeometric identities

[record Q04][record Q05] Leibtag et al. [8, Section 6.2] represent the same approximation coefficients in two ways: a terminating hypergeometric sum and minors of a dual coefficient matrix. Different presentations can expose different prime divisors. The experiment here deforms that pair of presentations to the quarter arguments $1/4$ and $3/4$, then subtracts to isolate θ. Shifting only the final harmonic sums does not preserve the identity. The kernels, their coefficients and the normalizing scalar all have to change together.

This is the most computationally substantial case study. We give the finite definitions, the exact remainder formulas, and the mechanism of the paired identities. The accompanying directory `checks/paired/` contains the rational matrix data and a standalone exact replay of the ten polynomial certificates. Those data are part of this case study, not a reference to an unspecified research log. The mathematical argument explains what the certificates prove and why their boundary terms vanish.

16.1 Forward coefficients and an exact remainder

Let $\alpha = -3/4$ or $-1/4$, and put $a = 1 + \alpha$. For $n \geq 0$ set

$$B_n^{\alpha}(k) = \prod_{j \in \{12,13,14\}} \binom{k + jn + \alpha}{10n} \binom{15n}{n+k} \binom{15n}{k}. \tag{16.1}$$

For differentiation this means the full Gamma interpolation, with $\binom{x}{m} = x(x-1)\cdots(x-m+1)/m!$ for integer $m \geq 0$ and $\binom{15n}{z} = \Gamma(15n+1)/[\Gamma(z+1)\Gamma(15n-z+1)]$. In particular, $B_n^\alpha(k)$ can be zero at an integer while its derivative is nonzero. Put

$$H_m^{(s)}(a) = \sum_{j=0}^{m-1} (j+a)^{-s},$$

and define the rational numbers

$$\begin{aligned} Q_n^a &= \sum_{k=0}^{14n} B_n^\alpha(k), \\ P_{s,n}^a &= \sum_{k=-n}^{15n} \left(B_n^\alpha(k) H_{4n+k}^{(s)}(a) - \frac{(B_n^\alpha)'(k)}{s-1} H_{4n+k}^{(s-1)}(a) \right), \quad s = 3, 4. \end{aligned} \tag{16.2}$$

The summation range in the second line includes both simple-zero tails. For reproduction without Gamma differentiation, the derivative in the interior $0 \leq k \leq 14n$ is $B_n^\alpha(k)$ times

$$\sum_{j=12}^{14} \sum_{i=0}^{10n-1} \frac{1}{k+jn+\alpha-i} + H_{14n-k}^{(1)}(1) - H_{n+k}^{(1)}(1) + H_{15n-k}^{(1)}(1) - H_k^{(1)}(1).$$

For $14n < k \leq 15n$, replace the last two binomials in (16.1) by $\binom{15n}{k}(-1)^k/[(n+k)\binom{n+k-1}{15n}]$; for $k = -i$, $1 \leq i \leq n$, replace them by $\binom{15n}{n-i}(-1)^{i-1}/[(15n+i)\binom{15n+i-1}{i-1}]$. These are exactly their simple-zero derivatives.

Write $(z)_m = z(z+1)\cdots(z+m-1)$ and define

$$\begin{aligned} \mathcal{H}_n(s) &= \frac{(-1)^n (15n)!^2}{(10n)!^3} \frac{\prod_{j=12}^{14} (s-jn-\alpha)_{10n}}{(s)_{14n+1}(s-n)_{16n+1}}, \\ T_n(z) &= \mathcal{H}_n(z+4n). \end{aligned} \tag{16.3}$$

Gamma reflection gives $B_n^\alpha(x) = (\sin(\pi x)/\pi)^2 T_n(-x-4n)$. Taking the double and simple Laurent coefficients therefore gives

$$T_n(z) = \sum_{k=-n}^{15n} \left(\frac{B_n^\alpha(k)}{(z+4n+k)^2} - \frac{(B_n^\alpha)'(k)}{z+4n+k} \right). \tag{16.4}$$

Both sides are proper rational functions with the same principal parts; the difference is a polynomial vanishing at infinity. Also $\sum_k (B_n^\alpha)'(k) = 0$ because $T_n = O(z^{-2})$. Differentiate and sum (16.4) at $z = m+\alpha$, $m \geq 1$. The elementary identity $\sum_{m\geq 1}(m+\alpha+h)^{-s} = \zeta(s,a) - H_h^{(s)}(a)$ then proves

$$\begin{aligned} Q_n^a \zeta(3,a) - P_{3,n}^a &= -\frac{1}{2} \sum_{m\geq 1} T_n'(m+\alpha), \\ Q_n^a \zeta(4,a) - P_{4,n}^a &= \frac{1}{6} \sum_{m\geq 1} T_n''(m+\alpha). \end{aligned} \tag{16.5}$$

All differentiated tails converge absolutely. Their first $9n$ and $8n$ sampled terms, respectively, vanish by the multiplicities of the numerator roots. These formulas already define useful exact starting points independently of the paired representation.

16.2 Dual coefficient rows and the paired identities

Define a different rational function

$$\mathcal{R}_n(s) = \frac{(-1)^n(1+2n-s)_{14n}(1+n-s)_{16n}}{(s+\alpha)_{11n+1}(s+n+\alpha)_{11n+1}(s+2n+\alpha)_{8n+2}}. \tag{16.6}$$

For $k = 0, 1, 2$, take the partial fractions

$$(-s)^k\mathcal{R}_n(s) = \sum_{j\geq 0}\sum_{i=1}^{3}\frac{c_{j,i}^{(k)}}{(s+\alpha+j)^i}.$$

Absent poles and orders have coefficient zero. Define three rational row vectors, whose coordinates are indexed by k, by

$$A_k = 3\sum_j c_{j,3}^{(k)}, \qquad B_k = 2\sum_j c_{j,2}^{(k)}, \qquad D_k = -\sum_{j,i} i\, c_{j,i}^{(k)} H_{2n+j}^{(i+1)}(a). \tag{16.7}$$

The simple coefficients sum to zero, since the rational function is $O(s^{-2})$ or smaller. Thus differentiation and summation yield the vector identity

$$G_n = A_n\zeta(4,a) + B_n\zeta(3,a) + D_n = \left(-\sum_{m>2n}\frac{d}{ds}[(-s)^k\mathcal{R}_n(s)]\Big|_{s=m}\right)_{k=0}^{2}. \tag{16.8}$$

This proves the formal absence of $\zeta(2,a)$ without a numerical linear-independence assumption. At $n = 0$ the rows are

$$A_0 = 3(1,\alpha,\alpha^2), \quad B_0 = -2(1,1+\alpha,\alpha^2+2\alpha), \quad D_0 = (a^{-2},a^{-1},1).$$

Put $u_n = (1, 10n+a, (10n+a)^2)$ and

$$\Lambda_n = \frac{(8n)!(9n)!^2(11n)!^2(12n)!}{6(14n)!(16n)!}\frac{(a)_{11n}(a)_{12n}^2(a)_{13n}^2(a)_{14n}}{(a)_{16n}(a)_{17n}^2(a)_{18n}^2(a)_{19n}}. \tag{16.9}$$

The complete paired formulas are

$$Q_n^a = \Lambda_n \det\begin{bmatrix}B_n\\A_n\\u_n\end{bmatrix}, \qquad P_{3,n}^a = -\Lambda_n \det\begin{bmatrix}D_n\\A_n\\u_n\end{bmatrix}, \qquad P_{4,n}^a = \Lambda_n \det\begin{bmatrix}D_n\\B_n\\u_n\end{bmatrix}. \tag{16.10}$$

The next subsection details the finite certificate proof of these all-index identities. It is longer than merely checking several n, because the derivative tails and the separate coefficient rows need their own recurrence justification.

16.3 How to verify the all-index identities

The data files specify rational matrices $M(n)$ of size 4, $Q(n)$ of size 3, a 4×3 matrix $K(n)$, and rank-three companion matrices $C(n), C_Q(n)$. They are exact rational functions in n, not fitted floating-point recurrences. A rational entry is stored as two lists of coefficients in ascending powers of n, for its numerator and denominator. The replay performs the following checks.

First, with $e_0 = (1,0,0,0)^T$ and $E_n = (8n+1)(9n+1)(10n+1)(11n+a)$, it verifies the rational identities

$$M(n)K(n+1)Q(n)^T = c(n)K(n), \qquad K(n)u_n^T = -E_n e_0. \tag{16.11}$$

The rank-three reduction is explicit. Form the Krylov matrix

$$U(n) = [e_0,\ M(n)e_0,\ M(n)M(n+1)e_0].$$

Then $M(n)U(n+1) = U(n)C(n)$; the analogous 3×3 identity holds for Q, C_Q. All denominators, all required determinants, and the leading recurrence coefficients are nonzero at every integer $n \geq 0$. This is certified by a polynomial sign test: after a fixed shift the coefficients have one sign and the constant is nonzero, with the finitely preceding integers checked exactly.

For context, K comes from an Euler Lagrange identity. With $\partial = z\, d/dz$, use

$$\mathcal{L} = z \prod_{j=1}^{5} (\partial - a_j + 1) - \prod_{j=1}^{5} (\partial - b_j),$$

where

$$(a_j) = (-16n, -17n, 1 + 11n + \alpha, 1 + 12n + \alpha, 2 + 10n + \alpha), \qquad (b_j) = (-2n, -n, \alpha, n + \alpha, 2n + \alpha).$$

If $\mathcal{L} = \sum_j l_j(z)\partial^j$, then

$$\partial \sum_{j \geq 1} \sum_{h=0}^{j-1} (-1)^h \partial^h (l_j f) \partial^{j-1-h} g = f\mathcal{L}g - g\mathcal{L}^* f.$$

Apply the contiguous change $f = z^{2n}(\partial + 12n + a)F/(12n + a)$ and evaluate at $z = 1$. The fourth F derivative reduces by its hypergeometric equation; the extra g-jet column vanishes. The remaining bilinear matrix is K. The certificate proof only needs (16.11), so the supplied rational data can be checked without importing a computer algebra package for hypergeometric transformations.

Second, the scalar recurrences must describe the actual sums. For each quarter, `problems.py` constructs five explicit polynomial problems over $\mathbb{Q}[n][k]$ of the form

$$\mathcal{A}(n,k)P(n,k+1) - \mathcal{B}(n,k)P(n,k) = \mathcal{F}(n,k). \tag{16.12}$$

They are the forward scalar recurrence, the dual scalar recurrence, and the three coordinate transports of the dual row. The right side is obtained by putting the relevant shifted hypergeometric summands over their common denominator. Every linear factor, shift and factorial normalization is retained explicitly in that file. The solver uses *exact descending coefficients*, then independently substitutes the full resulting polynomial in (16.12). For these problems $\deg_k \mathcal{A} = \deg_k \mathcal{B} = 5$, their leading coefficients are one and their next coefficients differ by $-d$, where d is the prescribed degree of P. The coefficient of k^{i+4} has pivot $i - d$. All lower coefficients are consequently determined by the leading one; the four remaining equations fix that one by polynomial division. A nonzero residual rejects the certificate.

The degrees $(\deg_n P, \deg_k P)$ are

α	forward	dual scalar	row 0	row 1	row 2
$-3/4$	$(553, 114)$	$(448, 107)$	$(125, 35)$	$(126, 35)$	$(127, 35)$
$-1/4$	$(553, 114)$	$(446, 107)$	$(124, 35)$	$(125, 35)$	$(126, 35)$.

These large degrees explain the use of machine-readable coefficients. The complete objects are generated and checked by `checks/paired/verify.py`; no recurrence is accepted because it matches a finite run of examples.

Third, transfer from a polynomial identity to a sum requires its boundary terms. For the forward sum the factorial continuation terminates at $k = 0$ and $k = 14n + 43$ for the four shifted summands. For the derivative tails use the *uncentered* kernel $\mathcal{H}_n$ in (16.3), sampled at $s = m + \alpha$, $m > 4n$. For $n \geq 2$, the four shifted tails can all start at $m = 12n$: omitted first and second derivatives lie in their common triple-zero interval. The lower certificate has a zero of order three there; its denominator is nonzero by the exact shifted sign test. At infinity it is constant plus $O(s^{-1})$, so its first and second derivatives vanish. Absolute locally uniform tail convergence permits differentiation. The two exceptional recurrence indices $n = 0, 1$ are checked exactly for both numerators.

For the dual rows the three certificates become rational identities

$$(-s)^j \mathcal{R}_{n+1}(s) - \sum_{i=0}^{2} Q(n)_{ij} (-s)^i \mathcal{R}_n(s) = V_j(n, s+1) - V_j(n, s).$$

At $n \geq 1$ all finite poles of V_j lie in $-\alpha + \mathbb{Z}$; the apparent positive integer poles cancel against numerator zeros. Translation preserves the total Laurent coefficient of each order, so A_n and B_n separately transport by $Q(n)$. For the differentiated sums in (16.8), both tails can start at $m = 16n$. Omitted terms lie in the common double-zero interval, the lower certificate derivative vanishes, and its derivative is $O(s^{-2})$ at infinity. This proves transport of G_n; subtracting the separate A, B transports proves transport of D. The exceptional $n = 0$ step checks all nine rational row coordinates. No assertion of numerical period independence enters this step.

Finally let $\Phi_M(n) = M(0) \cdots M(n-1)$, and similarly for Q. The forward readout rows before multiplication by $C_n^\alpha = B_n^\alpha(0)$ are

$$\rho_q = (1, 0, 0, 0), \quad \rho_3 = (0, 0, -1/2, (2\alpha + 1)/4), \quad \rho_4 = (0, 0, 0, -1/6).$$

The preceding recurrences and their first three values identify $C_n^\alpha \rho_j \Phi_M(n) e_0$ with the forward quantities. At $n = 0$, multiplication by $K(0)$ gives respectively $-(a/6)(B_0 \times A_0)$, $(a/6)(D_0 \times A_0)$ and $-(a/6)(D_0 \times B_0)$. Telescope (16.11) and use $(vQ) \times (wQ) = \det(Q)(v \times w)Q^{-T}$. This gives (16.10) with scalar

$$\frac{C_n^\alpha a \prod_{j<n} c(j)}{6E_n \det \Phi_Q(n)}.$$

Its initial value is $1/6$; the one-step ratio agrees identically with (16.9), completing the proof. All the stated matrix, polynomial, boundary-degree and initial-value checks are included in the standalone replay.

16.4 Character projection, rates and denominator normalization

The elementary identity $\zeta(4, 1/4) - \zeta(4, 3/4) = 256\theta$ gives, with $a = 1/4, b = 3/4$,

$$W_n = Q_n^a Q_n^b \theta - \frac{P_{4,n}^a Q_n^b - P_{4,n}^b Q_n^a}{256}. \tag{16.13}$$

Thus all auxiliary constants disappear. The coefficient of the target is a product of two different denominators, which matters in estimating its size.

For reference, we state precisely the analytic input to the arithmetic comparison below. With

$$\sigma = 65.45153656\ldots, \qquad g = 112.80753924\ldots,$$

the fixed-quarter construction has

$$\frac{\log Q_n^a}{n} \longrightarrow \sigma, \qquad \left| \zeta(s, a) - \frac{P_{s,n}^a}{Q_n^a} \right| \leq e^{-(g - o(1))n}, \quad s = 3, 4. \tag{16.14}$$

Here the analytic recurrence argument is the dual-convergence result of Leibtag et al. [8, Theorem 2.6, Corollary 2.7 and Appendix B.1], with the shifted matrices just certified. The required checks can be made explicitly as follows. Under $\operatorname{diag}(1, n+1, (n+1)^2)$ the two $Q(n)$ limits coincide. Their eigenvalues are a real $\lambda_0 = 7.2027\ldots\ 10^{-24}$ and a conjugate pair of modulus $e^{59.5199\ldots}$; $g = \log(|\lambda_1|/\lambda_0)$. The vector u_n tends, after scaling, to $(1, 10, 100)$ and has nonzero projection on the required eigenspace. The exact limiting matrix and its cyclic-vector determinant are checked in the accompanying `checks/paired/rates.py` replay.

To identify the actual state as the small solution, one does not assume the conclusion of a numerical recurrence fit. On $2n \le s \le 17n$, the products in (16.6), and their first derivatives, are bounded by a polynomial in n times $e^{c_0 n}$, where

$$c_0 = 14\log 15 + 16\log 16 - 11\log 2 - 11\log 3 - 8\log 4 = 51.4744\ldots < \log|\lambda_1|.$$

For example bound the numerator factors by $15n$ and $16n$, and the three denominator blocks below by $2n+\alpha$, $3n+\alpha$, $4n+\alpha$. Fixed shifts change these exponential bounds only by bounded or polynomial factors. Differentiating products adds only polynomial factors, including at numerator zeros. For $s \ge 17n$, all numerator factors have magnitude at most s, all denominator factors are at least $s - 3/4$, and differentiation gives an integrable polynomial tail $O(ns^{k-5})$, $k \le 2$. The state therefore has growth strictly below the conjugate pair. It is nonzero: its initial coefficient matrix is invertible and its constant coordinate is 1. The cited separation argument identifies it with the recessive mode. The initial readouts and nonzero limiting projection then give the relative-error estimate in (16.14).

The coefficient growth follows independently from the positive sum (16.2). Stirling's formula and the Laplace maximum for $k/n \in [0, 14]$ give $\sigma = \max \phi(x)$, where, with $L(t) = t\log t$ and $L(0) = 0$,

$$\phi(x) = \sum_{j=12}^{14} \big(L(x+j) - L(10) - L(x+j-10)\big) + 2L(15) - L(x+1) - L(14-x) - L(x) - L(15-x).$$

Its second derivative is strictly negative on $(0, 14)$, and its first derivative tends to opposite infinities at the endpoints. Its derivative therefore vanishes at the unique root in $(0, 14)$ of

$$\prod_{j\in\{0,1,2,3,4\}} (x+j) - \prod_{j=12}^{14} (x+j)(14-x)(15-x) = 0.$$

Fixed quarter shifts do not affect this exponential rate. Consequently (16.13) has upper error exponent $2\sigma - g = 18.0955\ldots$ before primitive normalization. This is an upper bound, not an assertion that the projected error actually grows or has that exact exponent.

A convenient uniform integer denominator is also explicit. Let O_m be the odd part of $\operatorname{lcm}(1, \ldots, m)$ and set, for $n \ge 1$,

$$D_n = 6\, 2^{60n + 3v_2((10n)!) + \lfloor \log_2(16n) \rfloor} O_{76n}^4. \tag{16.15}$$

It clears $Q_n^a, P_{3,n}^a, P_{4,n}^a$ at both quarters, so $256 D_n^2$ clears (16.13). To verify this, each degree-m quarter-binomial is integral at odd primes by counting multiples in an arithmetic progression of step four; its exact dyadic cost is $2m + v_2(m!)$. The derivative identity

$$\frac{d}{dx}\binom{x}{m} = \sum_{j=1}^{m} \frac{(-1)^{j-1}}{j} \binom{x}{m-j}$$

and the simple-zero tail formulas add one factor $\operatorname{lcm}(1, \ldots, 16n)$. The harmonic sums have index at most $19n$ and odd denominator dividing O_{76n}^s. Combining the one derivative layer with at most three harmonic layers, and clearing $1/2, 1/3$, proves (16.15). This bound is sufficient, and need not equal the primitive multiplier.

16.5 What the transferred arithmetic estimates can certify

It is possible to quantify a limitation without claiming a theorem about all actual common factors. The following calculation concerns only the termwise forward valuations and leading-Laurent-coefficient bounds in the dual rows. For $p > 3$, $p^2 > 120n$, put $x = n/p$. The two quarter arguments give phases $t = 1/4, 3/4$, interchanged when the residue class of p changes. Define the floor functions

$$\begin{aligned} f_t(x,r) &= \sum_{j=12}^{14} \big(\lfloor (r+j)x - t\rfloor - \lfloor (r+j-10)x - t\rfloor - \lfloor 10x\rfloor\big) \\ &\quad + 2\lfloor 15x\rfloor - \lfloor (1+r)x\rfloor - \lfloor (14-r)x\rfloor - \lfloor rx\rfloor - \lfloor (15-r)x\rfloor, \\ \nu_t(x,r) &= \lfloor (16+r)x - t\rfloor - \lfloor (2+r)x - t\rfloor + \lfloor (17+r)x - t\rfloor - \lfloor (1+r)x - t\rfloor \\ &\quad - \sum_{j\in\{r,11-r,r-1,12-r,r-2,10-r\}} \lfloor jx\rfloor, \\ \varphi_t(x) &= \sum_j e_j \lfloor jx\rfloor + \sum_j f_j \lfloor jx - t\rfloor, \end{aligned}$$

where the two coefficient lists are

$$(e_8, e_9, e_{11}, e_{12}, e_{14}, e_{16}) = (1, 2, 2, 1, -1, -1),$$

$$(f_{11}, f_{12}, f_{13}, f_{14}, f_{16}, f_{17}, f_{18}, f_{19}) = (1, 2, 2, 1, -1, -2, -2, -1).$$

They come respectively from a forward summand at $k = rn$, a triple pole at $j = rn$, and the scalar (16.9). We sample only integer r: $0, \dots, 14$ for the forward sum and $3, \dots, 9$ for the triple poles. For the specified estimates an optimistic ceiling on the quarter's lower valuation bound for Q_n^a is

$$U_t(x) = \max\left\{ \min_{r\in\{0,\dots,14\}} f_t(x,r), \quad \varphi_t(x) + 2 \min_{r\in\{3,\dots,9\}} \nu_t(x,r) - 1 \right\}. \tag{16.16}$$

Indeed, restricting a minimum to selected terms makes it larger. The two dual rows contributing to Q have lower bounds at most the selected leading valuation and that valuation minus one, respectively. Taking the better of the two presentations gives the maximum. Costs from P_4 have been discarded altogether. Every relaxation makes this a more generous ceiling for the specified estimates.

The functions are periodic with period one and vanish near zero. All breakpoints on $[0, 1]$ belong to

$$\left\{ \frac{4m + l}{4d} : 1 \le d \le 28,\ l \in \{0, 1, 3\},\ 0 \le 4m + l \le 4d \right\}.$$

The union has 854 open cells. One rational midpoint gives the exact constant value on each cell. Integrating through 40 uses only the rational weights $1/a - 1/b$; the remaining tails are at most $9/40$ and $11/40$, the respective essential maxima divided by 40. The standalone calculation `checks/paired/arithmetic.py` gives

$$\int_0^\infty \frac{U_{1/4}(x) + U_{3/4}(x)}{x^2}\, \mathrm{d}x < 88.208. \tag{16.17}$$

To pass from limiting floors to finite prime sums, first restrict x to a bounded set away from the rational walls. The $O(1/p)$ offsets then preserve all floors; the prime number theorem gives the integral. Small neighborhoods of the finitely many walls contribute arbitrarily little because the one-level exponents are uniformly bounded. A bound of 32 per quarter suffices: forward selected

valuations are at most 5, triple-pole leading valuations at most 8, and the balanced scalar valuation at most 12. The tail $p < n/X$ is thus at most $64n/X + o(n)$; primes below $\sqrt{120n}$ contribute $o(n)$, since each has $O(\log n)$ bounded layers. Let n, then X, tend to infinity. The sum of the two phases is independent of p mod 4, so no prime number theorem in progressions is required for this passage.

For these same estimates the dyadic cost of each quarter's Q is $90n + O(\log n)$: the three quarter-binomials cost $60n + 3v_2((10n)!)$, and ordinary binomial carries contribute only $O(\log n)$. The selected dual bound has the same leading cost. Even after forgiving every P_4 cost, success with (16.14) would require odd-prime savings exceeding

$$2\sigma + 180\log 2 - g = 142.8620\ldots,$$

well above (16.17). This establishes the claimed limitation of the transferred valuation estimates. It does not bound the actual gcd of the sums or minors from above, and it does not exclude faster decay after the character projection.

What remains. The paired identities are available, the auxiliary constants cancel exactly, and both coefficient presentations can be reconstructed. What is missing is eventual or suitably frequent nonvanishing of W_n, together with shrinking values after its *full* primitive normalization. The five recorded indices $n = 1, 2, 3, 4, 6$ give nonzero but large primitive errors; those computations are finite evidence. Cancellation inside the sums, a cheaper character projection, or a stronger projected-error estimate would go beyond the specific arithmetic comparison just proved.

17 Compatible multipliers and slow logarithmic dilations

Two further constructions are useful to retain explicitly. Both try to save denominators before elimination: one matches the endpoint values of different multipliers, and the other arranges cancellations between logarithmic arguments moving at different speeds.

17.1 Three pole multipliers with the same endpoint

[record L04] Use the completed identity (7.1), with

$$(z_1, z_3, z_5) = (i/2, (1+i)/4, (1-i)/8),$$

$$(c_1, c_3, c_5) = (225/136, -25/34, 9/68),$$

$$(\alpha_1, \alpha_3, \alpha_5) = (1, 3/2, 5/2).$$

Thus $\theta = \sum_j c_j(\Im \operatorname{Li}_4(z_j) + \alpha_j \log 2 \Im \operatorname{Li}_3(z_j) + (\alpha_j^2/3)\log^2 2 \Im \operatorname{Li}_2(z_j))$. Let

$$P_1 = t^2 + 4, \quad P_3 = t^2 - 4t + 8, \quad P_5 = t^2 - 8t + 32, \qquad (m_1, m_3, m_5) = (2n, 2n, n).$$

The key elementary compatibility is $P_j(1)^{m_j} = 25^n$ for all three indices. Their zeros are the two reciprocal conjugate arguments. Writing $[\,]_{\le N}$ for Taylor truncation, set

$$\begin{aligned} A_{j,s} &= [P_j(t)^{m_j}\mathcal{I}_s(z_j t)]_{\le 2m_j+2n}\Big|_{t=1}, \\ B_r &= [(2-t)^{4n}(-\log(1-t/2))^r]_{\le 8n}\Big|_{t=1}, \quad r = 1, 2, \\ R_n &= \sum_j c_j \left(A_{j,4} + \alpha_j B_1 A_{j,3} + \frac{\alpha_j^2}{3} B_2 A_{j,2} \right), \qquad \Lambda_n = 25^n\theta - R_n. \end{aligned}$$

Here $\mathcal{I}$ is the analytic conjugate-difference germ used in Section 12. All coefficients are rational. In particular B_2 approximates the square of the logarithm directly; it is not B_1^2.

An exact tail formula makes the analytic success transparent. For $z = z_j$, $m = m_j$, $\eta = z/\bar{z}$, put $E_{j,s} = 25^n \Im \operatorname{Li}_s(z) - A_{j,s}$. Expanding the geometric tail and using $k^{-s} = \int_0^1 x^{k-1}(-\log x)^{s-1}\,\mathrm{d}x/(s-1)!$ gives

$$E_{j,s} = \Im\left\{ \frac{z^{2n+1}}{(s-1)!} \int_0^1 \frac{x^{2n}[(1-x)(1-\eta x)]^m(-\log x)^{s-1}}{1-zx}\,\mathrm{d}x \right\}. \tag{17.1}$$

Indeed $P_j(1/(zx)) = (1-x)(1-\eta x)/(z^2x^2)$, which cancels $2m$ powers from the tail starting at degree $2m+2n+1$. Absolute convergence for $|z|<1$ justifies the interchange.

Let $l = \log 2$ and $\delta_r = l^r - B_r$. For $h > 4n$ the logarithmic Taylor coefficients are

$$[t^h](2-t)^{4n}(-\log(1-t/2)) = b_h = \frac{2^{4n-h}}{h\binom{h-1}{4n}}, \qquad [t^h](2-t)^{4n}(-\log(1-t/2))^2 = 2b_h(H_{h-4n-1} - H_{4n}).$$

These follow by differentiating $2^{4n}(1-t/2)^{4n-a}$ in a at zero, once and twice. For $h > 8n$, all terms are nonnegative. The integral formula for the first tail and the ratio $b_{h+1}/b_h = (h-4n)/(2(h+1)) < 1/2$ yield

$$0 < \delta_1 \le 4096^{-n}, \qquad 0 \le \delta_2 \le \frac{4\delta_1}{4n+1}.$$

The second inequality follows by bounding the harmonic difference by $(h-8n-1)/(4n+1)$ and summing a geometric first moment. Since $A_{j,s} = O(25^n)$, replacing l, l^2 by B_1, B_2 costs only $O((25/4096)^n)$ in Λ_n.

For $j = 3$ and 5, formula (17.1) gives respectively $O(64^{-n})$ and $O(144^{-n})$. The elementary bounds needed are $x^2(1-x)^2(1+x^2)/8 \le 1/64$ and $x^2(1-x)\sqrt{1+x^2}/32 \le 1/144$ on $[0,1]$. For $j = 1$ the completed tail is exactly

$$(-1)^n 2^{-2n-1} \int_0^1 [x^2(1-x^2)^2]^n \frac{(-\log x)^3/6 + l(-\log x)^2/2 + l^2(-\log x)/3}{1+x^2/4}\,\mathrm{d}x.$$

The bracket has its unique maximum $4/27$ at $x = 1/\sqrt{3}$; the second derivative of its logarithm there is -18. Taylor expansion at the maximum and exponential domination off a neighborhood give

$$\Lambda_n \sim (-1)^n K \frac{27^{-n}}{\sqrt{n}}, \qquad K = \frac{225\sqrt{\pi}}{884}\left(\frac{u^3}{6} + \frac{lu^2}{2} + \frac{l^2u}{3}\right) > 0, \quad u = \frac{\log 3}{2}. \tag{17.2}$$

The other errors are exponentially smaller. Thus eventual nonvanishing and analytic decay are established for these rational forms.

Writing $L_m = \operatorname{lcm}(1,\dots,m)$, a sufficient multiplier is

$$D_n = 17\,2^{9n+4} L_{6n}^2 L_{8n}^2. \tag{17.3}$$

For verification, the coefficient of t^r in $P_j^{m_j}$ has dyadic valuation at least $\lceil \alpha_j(2m_j - r)\rceil$; convolution preserves this bound. Combining it with the rational coefficients of $\mathcal{I}_s(z_j t)$ gives $2^{2\alpha_j n} L_{2m_j+2n}^s A_{j,s} \in \mathbb{Z}$. Similarly $2^{4n} L_{8n}^r B_r \in \mathbb{Z}$. The fixed rational constants are cleared by $17 \cdot 16$, and $L_{6n}^2 L_{8n}^2$ is a multiple of the required products L_{6n}^4 and $L_{6n}^3 L_{8n}$. This proves (17.3). Its logarithmic rate is $28 + 9\log 2$, far above the decay rate $\log 27$. The comparison says that this sufficient bound does not prove irrationality. It is not a lower bound for the primitive multiplier. The twelve recorded primitive forms failed, but useful all-index primitive arithmetic remains the missing estimate.

17.2 A logarithmic filter whose coefficient cost decreases

[record A01][record A02][record A03][record A06] The slow-dilation family can be defined without any special-value arithmetic. Let $h(0) = h'(0) = h''(0) = 0$ and $h'''(x) = x^2 \operatorname{sech}'(x)/8$. For a fixed odd $k \geq 3$ put

$$u = \log \frac{1+t}{1-t}, \qquad v = \log \frac{1+t^k}{1-t^k}, \qquad r_k(t) = 2h(u) + 4h(v) - 4h(u+v) - 2h(2v) + 2h(u+2v).$$

It has rational Taylor coefficients and begins $r_k(t) = -4t^{2k+4} + O(t^{2k+6})$. The endpoint identity $r_k(1^-) = -6\theta$ follows directly from

$$h(x) = T(x) - 3\theta + \tfrac{3}{2}\beta(3)x - \tfrac{1}{4}\beta(2)x^2, \qquad T(x) = \frac{1}{8}\int_x^\infty y(y-x)(2y-x)\operatorname{sech} y \, \mathrm{d}y.$$

Both sides have the prescribed third derivative and initial values; the three needed moments of $\operatorname{sech} y$ are obtained from its exponential series. In the displayed combination of h values, the linear and quadratic terms cancel, the constants sum to -6θ, and all T terms vanish as $t \uparrow 1$. Thus $h_k = -r_k$ is an explicit rational-coefficient germ with endpoint 6θ, to which the bordered determinant (12.2) applies with target $6X$.

There is an elementary coefficient-denominator improvement. If $x = au + bv$ is any of the five arguments above, put $A(t) = (1+t)^a(1+t^k)^b$. Then dx has integral power-series coefficients and

$$\operatorname{sech} x = \frac{A(t)A(-t)}{(A(t)^2 + A(-t)^2)/2} \in \mathbb{Z}[[t]],$$

since the denominator is an integer polynomial with constant one. With $I_j = \int_0^x s^j \operatorname{sech} s \, \mathrm{d}s$, integration gives

$$h(x) = \frac{3xI_2 - x^2 I_1 - 2I_3}{8}.$$

This is a rational combination of length-four iterated integrals with regular integral letters dx and $\operatorname{sech} x \, dx$. A coefficient of degree at most N has integration indices $1 \leq n_1 < \cdots < n_4 \leq N$. Its denominator therefore divides

$$W_4(N) = \prod_{j=1}^{4} L_{\lfloor N/j \rfloor}, \qquad v_p(W_4(N)) = \sum_{a \geq 1} \min(4, \lfloor N/p^a \rfloor).$$

Products obey the same bound by the integer shuffle identity. Hence $8W_4(N)$ clears the coefficient prefix. For $p > N/k$, expansion in v reduces the necessary word length to two: explicitly

$$r_k = 2\sum_{j \geq 2} \frac{(2^j - 2)v^j}{j!}(h^{(j)}(u) - h^{(j)}(0)), \qquad h''(u) = \frac{u^2 \operatorname{sech} u}{8} - \frac{1}{4}\int_0^u s \operatorname{sech} s \, \mathrm{d}s.$$

Through degree N only $j < p$ occurs, the coefficients of v are p-integral, and further derivatives of h involve at most two logarithms. Thus the exponent $v_p(W_2(N))$ suffices at these primes. This supplies a uniform, reconstructible coefficient bound; the sharper Frobenius bounds and fixed-k lower rates are recorded in A01.

The unresolved question is what survives after taking a determinant and its full primitive content. Positivity of a function does not prove positivity of its endpoint determinant. The logarithmic filter also develops nonreal branch points, so it cannot simply be identified with a positive real-supported Cauchy transform in the same variable. The fixed-k coefficient bounds do not answer either issue or give uniform estimates for k growing with the determinant order.

18 The wider investigation, by method

The case studies above give explicit constructions and proofs for the principal unfinished routes as well as the selected obstructions. This shorter catalog retains related variants, exact rediscoveries and bounded experiments. An entry described only here retains its archived scope; it is not promoted to a theorem proved in this supplement. The linked labels lead to the source register. They let a reader pursue an exploratory variant without making the research logs prerequisites for the case studies.

18.1 Rational Euler and two-endpoint Hermite sources

These constructions cancel Taylor coefficients at one or two points. The aim is to force a small remainder while keeping its coefficients arithmetically manageable. Solving the interpolation equations is only the first step; their endpoint values still need nonvanishing and primitive-size estimates.

Records H01–H03 extend (3.1) to two-endpoint sources. Section 13 proves the four-by-five reduction, including singular cases and the constant coefficient. The circular Hermite specialization has an archived eventual rank result and a signed error asymptotic proportional to $(-1)^n n^{5/2} \log^2 n\,((\sqrt{2}-1)/2)^n$. Its actual primitive height remains uncontrolled. Factorial rescaling of the entire pair does not change that question.

One arithmetic ingredient, H02, can be stated independently of that source. Products of r distinct integers in $[1, N]$, $N \geq r$, are all divisors of

$$W_r(N) = \prod_{j=1}^{r} \operatorname{lcm}(1, \ldots, \lfloor N/j \rfloor).$$

At a prime p the maximum possible product valuation is $\sum_{k\geq 1} \min(r, \lfloor N/p^k \rfloor)$; this is exactly $v_p(W_r(N))$. The maximum is attained by choosing the r integers with highest valuations, since the sets of multiples of p^k are nested. For the strict integration indices of the archived source this gives a sufficient coefficient denominator. In particular $\log W_4(N) \sim (1 + 1/2 + 1/3 + 1/4)N = 25N/12$ by the prime number theorem. It is not an endpoint primitive-value bound.

18.2 Continued fractions, derivatives and asymptotic summation

These methods improve a series tail by a rational approximation or an asymptotic correction. Differentiation can change the target constant, but also changes the denominator cost. It is essential to specify which endpoint and truncation parameters are allowed to grow together.

S04 and S05 are Theorems 4.1 and 5.1; their different quantifiers matter. Euler–Boole covers all endpoint and truncation schedules, whereas the continued-fraction theorem fixes the endpoint and restricts arithmetic to even depth. E05 concerns rational gauges (invertible rational changes of dependent variables) of a particular two-component tail system: the archived argument excludes its desired polynomial unimodular transfer. It says nothing about factorial gauges, higher-rank systems or unrelated recurrences. The binomial-shift reduction at the end of Section 5 also prevents counting an old scalar source as a new continued-fraction construction.

18.3 Modular and Eichler sources

Modular transformations relate expansions of a function at different points. The hope is to combine arithmetic information from those expansions with an analytic estimate improved by the

symmetry. The experiments below test cancellation of unwanted constants, removal of selected prime denominators, and the singularities that remain. The Eichler construction integrates a modular form; its transformation law introduces a polynomial containing auxiliary constants, which must be eliminated before isolating θ.

Section 14 develops the period cancellation of M01, a prime-deleted cusp source, and its fixed-filter natural boundary. M02 also constructs quadratic prime filters compatible with the Fricke transformation, obtaining selected savings in both endpoint germs. Neither transformation identities nor local savings alone establish favorable primitive arithmetic for growing filters. M03 records uncancelled interior poles in proposed higher-level coordinates and prefactors. A later annihilator removes ordinary and shifted pole families but leaves six fixed-point poles. These are limitations of those coordinates and raw coefficient estimates.

M04 compares three specified rational-system templates of ranks $9, 17, 18$. Its optimized displayed logarithmic derivative bound is below 1.3, against a required threshold above 4 in that test. This closes the prescribed comparison, not all modular coordinates. The exact template and source choices are retained in the record; they are essential hypotheses, not an incidental implementation detail.

18.4 Polylogarithm ladders and functional equations

Functional identities replace one special value by combinations at other arguments. Small arguments improve series convergence, and shared arguments can permit joint cancellation. The remaining task is to control all coefficients and auxiliary logarithms in the resulting form.

L01 is the completed Broadhurst identity used in Section 7; its attribution is part of the statement. L02 studies equal-speed scalar Taylor truncations of one rational realization. Using the c_m, z_m from Section 7, set $\ell(t) = -\log(1 - t/2)$ and, for real t near zero,

$$H(t) = \sum_{m\in\{2,3,5\}} c_m \Im\left(\mathrm{Li}_4(z_m t) + \frac{m\ell(t)}{2}\,\mathrm{Li}_3(z_m t) + \frac{m^2\ell(t)^2}{12}\,\mathrm{Li}_2(z_m t)\right).$$

Write $H(t) = \sum_{j\ge1} h_j t^j$; its coefficients are rational and $H(1) = \theta$. Let q_N be the reduced denominator of $\sum_{j=1}^{N} h_j$. The archived exact dyadic exponents are

$$v_2(q_{4m}) = 10m + 4, \quad v_2(q_{4m+1}) = 10m + 5, \quad v_2(q_{4m+2}) = 10m + 11, \quad v_2(q_{4m+3}) = 10m + 12,$$

with the first formula restricted to $m \ge 1$. Combined with the signed error estimate, these give a primitive obstruction for that scalar truncation, not for other uses of the ladder.

The common-base construction L03 and its precise missing odd-prime estimate are developed in Section 12. The compatible pole multipliers L04 and their eventual signed error asymptotic are developed in Section 17. Neither is excluded by the scalar Taylor-truncation result above. L05 specializes Gangl's weight-four functional equation [6] to a 21-argument algebraic presentation. The archive checks the symbol identity and tracks completion terms. A naive trace after separate inversions fails; no theorem of numerical period independence follows from a symbol computation.

18.5 Slow dilations, prime deletion and sparse Padé

These attempts modify the generating series to reduce its arithmetic cost: spread out its exponents, remove contributions from selected primes, or retain a sparse set of terms. Such changes also alter the singularities and the available approximation estimates, so an arithmetic saving alone does not settle the construction.

Section 17 defines the slow-dilation germ and proves its regular-word coefficient bound. The sharper archived result A01, for fixed odd k, finds coefficient denominator rates between $3/2+1/(k+2)$ and $3/2+1/k+1/(2k+4)$ in the normalization defined in its record. These are fixed-k coefficient statements, not primitive determinant rates or estimates uniform in a jointly growing k. A02 relates a bordered determinant's content to maximal-minor gcds and modular Padé defects. Here H is the archived rational moment matrix, b its bordering vector, and D clears their entries. With $T = DH$, $u = Db$, $d = \det T$, $m = \operatorname{adj}(T)u$, and $Q = u^T \operatorname{adj}(T)u$, the specialized gcds obey $\Delta \mid g \mid 6D\Delta^2$, where $g = \gcd(6Dd, Q)$ and $\Delta = \gcd(d, m_i)$. An example contains prime 23 absent from the coefficient denominator. The determinant identities are classical; the bounds do not settle all asymptotic primitive content.

A03 concerns failed fixed-cut absolute-contour estimates for $k = 3, 5$. It is not a lower bound on actual determinant size. A06 explains why a proposed same-variable positivity repair fails: the slow filter has genuine nonreal branch points satisfying $z^k = -(\sqrt{2}-1)^2$, inconsistent with the proposed compact real-supported Cauchy transform after the analyzed algebraic repairs: such a transform must be holomorphic off its real support. Different changes of variable are outside that statement.

Section 13 proves the selected-prime saving of A04 and the sparse determinant collapse of A05. The latter's further primitive-growth theorem is restricted to an oversieved regime: its cutoff prime is at least the fourth power of its size parameter. The natural cutoff was not excluded.

18.6 Positive determinants and interleaved frames

Changing the row and column nodes of a determinant may improve the balance between its analytic size and its prime denominators. The key distinction is between changing a basis within the same polynomial pencil, which preserves its primitive polynomial up to sign, and choosing a different set of rows, columns or moments, which can change the problem.

C01–C08 are the companion paper's positive cubic-tail kernel, uniform Cauchy comparison, consecutive-principal obstruction, difference-Gram construction and exact large-prime primitive arithmetic. Its two primitive lower exponents are $2 - 2\log 2$ and $6 - \log 4$ on the n^2 scale. S08 is a separate archived obstruction for an overshifted dense grid with $m \geq 768r^4$; it does not address proportional shifts. The unrestricted source cone is covered in Section 2.

Section 10 develops the nonprincipal experiment P06, including its local prime layers and the distinction between integer clearing and rational primitive normalization. Section 9 develops the separate moment-differentiation construction P07. Neither remaining arithmetic question is settled by the companion paper's principal-family theorems.

18.7 Hypergeometric and quarter-pole constructions

Factorial ratios and hypergeometric identities can provide rational coefficients together with exact remainder formulas. Shifting or differentiating their parameters changes the special values that occur. The tests here seek a combination containing only θ and a rational term, then examine the denominator powers and common factors.

Q01 differentiates an integer-pole tail to isolate θ, with signed error proportional to

$$(-1)^{n+1} n^{-5/2} \log^2 n (27/256)^n.$$

Its displayed clearing multiplier retains fifth odd-prime layers; the resulting unfavorable bound is not a theorem about full primitive content. Q02 tests three nonsymmetric half-parameter jets and a 120-element Thomae orbit at one fixture. Its six primitive eliminations are finite evidence, not a classification of transformations.

Q03 supplies an all-index moved-base Whipple identity, a positive terminating coefficient expression, dyadic valuations and eventual signed nonvanishing. Its proved multipliers obey a divisibility relation, and integer clearing of the selected cofactor fails even before odd-prime costs. Unbounded odd common content remains outside the result. The symmetric candidate was exactly an earlier archived family.

Section 16 gives the complete quarter-shifted paired construction Q04, the proof mechanism and standalone algebraic certificates. It also specifies and proves the transferred-estimate limitation Q05. The earlier numerator mismatch was repaired; it is not a reason to reject the corrected identities. The unresolved questions concern projected nonvanishing and primitive size.

18.8 Coupled integrals, geometry, and exact rediscoveries

Coupling several integrals or introducing extra parameters can cancel unwanted constants while retaining a positive expression. One must then check both the resulting arithmetic and whether the apparent new construction is actually a reparametrization of an earlier one.

P01 and O01 navigate the earlier lower-period cancellations, coupled holonomy, lattice, radial and orthogonal constructions. They are umbrella records, not single consolidated theorems. Some fitted recurrences were conjectural before later certification; those stages remain distinct.

P02 identifies the positive dimensional family exactly as $J_n = (n!)^4 4^{-n} L_{n+1,n+1}$ in the archived balanced-ladder notation. Here J_n is the proposed coupled integral and $L_{b,h}$ the earlier family with two integer parameters; their full integrands are given in the record. The displayed factor is a rational scalar, so primitive normalization removes it. This is the precise sense in which the proposed family is a rediscovery. Section 15.1 gives the full rational-series construction and five-period reduction of P03, followed by an explicit cofactor elimination. Section 11 develops the squared and unsquared quarter-pole pencils P04–P05, including the latter's conditional nonvanishing proof. Their actual primitive asymptotics remain unresolved.

S07 is a quarter-Gamma fourth-cumulant extraction from the weighted projective data $(1,3,4,4,4,4,4)$. Its scalar sequence uses finite Hurwitz fourth-power sums, with weights $1,3,4,4,4,4,4$ and total 24, at the two quarter arguments. The fourth logarithmic coefficient is formed with all lower products retained. Its archived arithmetic proof gives reduced-denominator logarithm at least $24n + o(n)$, whereas the error is of order n^{-5}. That scalar obstruction does not certify the larger geometric interpretation or exclude the separate nonconstant deformation. No geometric theorem is needed for the finite-sum obstruction.

E09 records other exact collapses: signed divisibility/LCM Gram determinants become linear approximants by rank one, and complete Boolean bases reproduce an Euler product. Polynomial Dirichlet convolution retains the fourth denominator layer but also pays the hypothetical rational denominator to the polynomial degree. Untested bases are not excluded.

18.9 Euler functionals and proposed analytic bridges

Here the main issue is whether a proposed connection between two kinds of information is valid: for example, whether a p-adic functional estimate controls the relevant coefficients, or whether an analytic property supplies the arithmetic conclusion needed for irrationality. A counterexample to such a connection closes only the stated implication.

E01 computes a norm on polynomials taking integral values on $\mathbb{Z}_p$, rather than on polynomials with integral monomial coefficients. Define the Euler functional by $\mathcal{E}(x^j) = E_j(0)$, where $\sum_{j\geq 0} E_j(0) z^j/j! = 2/(e^z+1)$, and set $\tau(P) = -\mathcal{E}(P''')/12$. For $p \geq 5$, let $m_p(h)$ be the least degree for which an integer-valued polynomial P satisfies $v_p(\tau(P)) = -h$. The exact onset formula is

$m_p(3q+r) = (3-r)p^q + rp^{q+1}$ for integers $q \geq 0$, $0 \leq r < 3$; the first three positive layers start at $p+2, 2p+1, 3p$. This says nothing by itself about rational poles or primitive determinant content.

E02 applies Euler distribution to the whole polynomial-plus-pole functional, with the affine target change $Y = \chi_4(p)p^4X + C_p$. The positive csch functional was already present in Carreira [4, Section 11]. Neither the distribution identity nor a formal target change equates real and p-adic limits. E03 supplies countermodels to particular proposed analytic implications, including an E-function Stokes inference, a compact-support inference from rational moments, and a spectral-raising bridge. Each countermodel has its stated hypotheses; none excludes the entire surrounding subject.

18.10 Cyclotomic parity and rationality-dependent constructions

The remaining tests use algebraic conjugation or begin by supposing θ rational. In the first case a trace can vanish by symmetry; in the second, an integrality claim or a differential construction must actually distinguish the hypothetical rational value. Neither property follows merely from writing down the corresponding formal expression.

The direct even-power signature extension E08 lies in a cyclotomic component negated by an involution fixing $\mathbb{Q}$, so its prescribed rational traces vanish. This parity statement does not apply to the nonlinear norm of Section 6; that family needed its own all-conjugate argument.

Section 15.2 constructs the rationality-dependent Niven source E07 and proves the surviving origin denominator, endpoint derivative and differential-field obstructions. Its scope is the specified integrality and coordinate proposals.

E06 subtracts the prescribed polar terms of a rationality-dependent q-series. At $q = (p-1)/p$, its p^2 truncation has exact denominator p^{p^2} and growing numerator. This is a result for that rational sampling and truncation, not for all q-series. E04 is the different monodromy theorem proved in Section 8. In both cases the hypothesis of rationality supplies no unspecified extra geometric or p-adic relation.

19 Computations, corrections, and review status

The repository separates three layers: the mathematical exposition, the inventory and method catalog, and verbatim research snapshots. The latter are historical evidence. A tentative conclusion inside an old note is not a current theorem if its later review corrected it. The register covers all 57 grouped entries and all twelve method groups; this is coverage of the selected archive, not a claim that every idea ever proposed for $\beta(4)$ has been investigated.

The companion paper's core exact and Arb checks have their own runner. For this supplement, a separate runner executes preserved scripts in a temporary directory, records fresh results, and checks the frozen source hashes. It includes the rational Sturm and tail certificate used in Theorem 7.1. Additional finite algebra checks cover the rewritten scalar, Euler–Boole, continuant and monodromy formulas. The precise cases, counts, dependencies and results are in `checks/supplement-reproduction.json`. The expanded case studies have an additional independent implementation of finite exact formulas in `checks/case_studies.py`, including partial fractions, prime reductions, endpoint reconstruction and determinant identities. Its report is `checks/case-studies-result.json`. Here "independent implementation" means a separate calculation by the assembling model, not an independent model or human audit.

The paired-identity directory is self-contained: its rational matrices, polynomial problem definitions and exact solver reconstruct all ten certificates, rather than depending on the original research environment. It records exact matrix transport, regularity, exceptional initial cases and the closed normalizing scalar. Separate scripts reproduce the floor integration and enclose the numerical rates

with Arb. The written proof in Section 16.3 explains the transfer from algebraic certificates to finite sums and differentiated tails. Computation does not replace those analytic boundary arguments. No large parameter search is part of this writing pass. Archived tests outside that runner retain their original status and are not described as rerun.

Three corrections are particularly consequential. The quarter forward/dual identity mismatch was repaired in the later deformation record. The ordinary first-variation examples require removal of the *full* common numerator gcd; their original clearing-only table is superseded. Finally, the interleaved-frame negative Smith layers and integer-clearing obstruction do not bound every rational primitive rescaling. The final method record preserves these distinctions.

All work was prepared with Codex. The main paper, this supplement and accompanying code have passed an independent model review. The historical record also contains model cross-checks and calculations. The independent model review is not a human mathematical audit or proof-assistant verification. Finite exact calculations check identities and specified instances; they do not prove asymptotic statements by extrapolation. The Sturm and tail calculation is explicitly distinguished because it certifies finite inequalities used in one proof.

The source ledger records versions, local PDFs, hashes, actual reading scope and unchecked dependencies. The specific obstruction statements are candidates for external scrutiny, not priority claims inferred from a failed literature search. Classical positivity, interpolation, orthogonal-polynomial, continued-fraction and content-removal mechanisms remain classical even when applied in a new experiment.

A Claim and source register

Each entry gives the archived classification, the scope of this writing pass, and one primary navigation source. Record labels elsewhere link here; the source links open files in the local checkout. All displayed paths are relative to the repository root. The complete proof and calculation path lists, limitations and literature links are in `planning/theorem-inventory.json`. The frozen archive classification does not mean every claim has been independently re-audited.

C01 – A positive cubic-tail kernel isolating beta(4) positive construction. Companion-paper proof; not repeated here.

evidence/research/dilations-and-positive-kernels.txt

C02 – Uniform two-sided comparison with a Cauchy determinant analytic theorem. Companion-paper proof; not repeated here.

evidence/research/collaboration/round-01/construction/findings.txt

C03 – Primitive growth for every consecutive principal offset primitive value obstruction. Companion-paper proof; not repeated here.

evidence/research/collaboration/round-01/construction/findings.txt

C04 – Difference-variable Gram moments with full target rank positive construction. Companion-paper proof; not repeated here.

evidence/research/collaboration/round-06/fresh/findings.txt

C05 – Matching real upper and lower rates for the difference Gram analytic theorem. Companion-paper proof; not repeated here.

evidence/research/collaboration/round-06/fresh/findings.txt

C06 – Exact large-prime primitive exponent for the difference Gram arithmetic theorem. Companion-paper proof; not repeated here.

evidence/research/collaboration/round-06/fresh/findings.txt

C07 – Primitive growth for the difference-variable Gram family primitive value obstruction. Companion-paper proof; not repeated here.

evidence/research/collaboration/round-06/fresh/findings.txt

C08 – Primitive invariance under full rational basis changes standard fact. Companion-paper proof; not repeated here.

evidence/research/collaboration/round-01/construction/findings.txt

S01 – The complete cone of scalar positive polynomial forms structural theorem. Unrestricted cone and degree law proved here; fixed-degree boundary formulas remain in the archive.

evidence/research/collaboration/round-01/fresh/findings.txt

S02 – Finite positive-cone extremizers do not exclude interior forms finite evidence. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-01/fresh/findings.txt

S03 – Repeated quadratic positive Mellin sources primitive value obstruction. Repeated-quadratic obstruction proved here.

evidence/research/collaboration/round-03/fresh/findings.txt

S04 – Uniform failure of the Euler-Boole truncation family primitive value obstruction. Uniform Euler–Boole obstruction proved here.

evidence/research/collaboration/round-05/fresh/findings.txt

S05 – Third-derivative continued fractions at fixed endpoints primitive value obstruction. Fixed-endpoint even-depth obstruction proved here.

evidence/research/collaboration/round-07/fresh/findings.txt

S06 – Cyclotomic minimal-polynomial obstruction primitive value obstruction. Prime-level sine-normalized obstruction proved here.

evidence/research/collaboration/round-11/norms/findings.txt

S07 – Quarter-Gamma cumulant sequence primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-09/orbifold/findings.txt

S08 – Overshifted dense grids primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-04/coordinator/loewner_shift.txt

S09 – A positive-source obstruction for the common-base Mellin kernel primitive value obstruction. Rational-kernel inequality proved here with exact certificates; beta identification uses the declared Broadhurst input.

evidence/research/collaboration/round-03/fresh/mellin_positive_obstruction.txt

A01 – Delayed prime layers in slowly varying dilations arithmetic theorem. Slow-dilation germ, endpoint and regular-word coefficient bound proved here; sharper Frobenius bounds and prefix lower rates remain archived.

evidence/research/collaboration/round-01/arithmetic/findings.txt

A02 – Bordered determinant content and modular defects arithmetic theorem. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-02/arithmetic/findings.txt

A03 – Fixed-cut capacity limitations estimate limitation. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-02/analysis/findings.txt

A04 – Prime deletion with subquadratic sufficient arithmetic cost partial construction. Exact Euler deletion, primitive selected-prime bound and subquadratic sufficient cost proved here; matching analytic bounds are not supplied.

evidence/research/collaboration/round-04/arithmetic/findings.txt

A05 – Sparse collapse and an oversieved primitive obstruction primitive value obstruction. Sparse Pade pair and determinant scalar collapse proved here; additional oversieved primitive obstruction remains archived, natural cutoff unresolved.

evidence/research/collaboration/round-04/construction/findings.txt

A06 – Nonreal branch points obstruct a particular positivity repair structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-01/construction/findings.txt

H01 – Rational Euler/Hermite source reduction positive construction. Euler and one-endpoint reductions plus the full two-endpoint four-by-five parametrization proved here, including singular cases.

evidence/research/collaboration/round-03/fresh/findings.txt

H02 – Strict integration denominator word arithmetic theorem. Strict-index denominator word and the factorial endpoint normalization proved here; source-specific sharpness remains archived.

evidence/research/collaboration/round-01/arithmetic/findings.txt

H03 – Circular Hermite sources: eventual nonvanishing and real error partial construction. The underlying endpoint construction is explicit here; the specialized eventual-rank and circular-error asymptotics remain archived.

evidence/research/collaboration/round-07/analysis/rank_asymptotic.txt

L01 – Completed Broadhurst weight-four ladder attributed identity. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-02/fresh/findings.txt

L02 – Scalar Taylor truncations of the ladder primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-02/fresh/findings.txt

L03 – Common-base ladder determinant partial construction. Common-base germ, nonvanishing, capacity bound, leading dyadic cost and signed-integral alternative developed here; the odd-prime inequality remains open.

evidence/research/collaboration/round-03/analysis/findings.txt

L04 – Matched simultaneous multipliers partial construction. Compatible pole multipliers, exact error formula, eventual signed asymptotic and sufficient clearing factor proved here; primitive arithmetic remains open.

evidence/research/collaboration/round-02/fresh/simultaneous.txt

L05 – Gangl specialization and completion/trace limitations attributed identity. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-06/identities/findings.txt

M01 – Eichler period cancellation and the level-32 source positive construction. Classical period identity specialized explicitly; period cancellation and coefficient prime cost proved here. Specialized Apery-limit singularity analysis remains archived.

evidence/research/modular-detour.txt

M02 – Fricke-compatible prime filters arithmetic theorem. F-only prime deletion, cusp-error sign/size and fixed-filter natural boundary proved here. Balanced quadratic/quartic filter variants remain archived.

evidence/research/collaboration/round-05/arithmetic/findings.txt

M03 – Interior poles in higher-level coordinate repairs structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-06/fresh/pole_annihilator_review.txt

M04 – A bounded level-32 rational-system comparison estimate limitation. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-13/findings.txt

Q01 – Differentiated integer-pole tail partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-04/fresh/findings.txt

Q02 – Nonsymmetric Whipple jets and one Thomae orbit finite evidence. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-10/symmetry/findings.txt

Q03 – Moved-base Whipple pairing estimate limitation. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-14/findings.txt

Q04 – Repaired paired forward/dual quarter identity positive construction. Full forward/dual definitions, remainder formulas, proof mechanism and standalone exact all-index certificates supplied; source recurrence theorem is explicitly cited for rates.

evidence/research/collaboration/round-15/coordinator/complete-deformation.txt

Q05 – Arithmetic ceiling for the proved paired-identity estimates estimate limitation. Specified transferred-valuation ceiling proved with exact floor integration; no upper bound on actual common content is asserted.

evidence/research/collaboration/round-15/coordinator/arithmetic-gate.txt

P01 – Lower-period cancellation in earlier positive/coupled integrals partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/structural-followup.txt

P02 – Dimensional ladder equals an archived balanced ladder equivalence. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-08/coordinator/dimensional-ladder-gate.txt

P03 – Nonuniform internal graph: fixed-period space and positive decay partial construction. Full rational-series family and five-period decomposition proved here, with explicit cofactor elimination; original geometric derivation and positive-integral bound remain archived.

evidence/research/collaboration/round-09/internal_graph/general-period-space.txt

P04 – Positive dense-quarter interpolation partial construction. Squared quarter-pole pencil, exact moment/pole evaluations, positivity, degree and analytic comparison proved here; primitive asymptotics remain open.

evidence/research/collaboration/round-10/coordinator/dense-quarter-gate.txt

P05 – Unsquared quarter poles: conditional nonvanishing partial construction. Unsquared quarter-pole pencil, coefficient recovery and auxiliary-prime conditional nonvanishing proved here; normalized size remains open.

evidence/research/collaboration/round-11/fresh/findings.txt

P06 – Interleaved nonprincipal frames: exact local layers, insufficient bound estimate limitation. Explicit interleaved family, positivity/energy, local Smith layers and integer-clearing limitation proved here; rational primitive content remains open.

evidence/research/collaboration/round-16/archive-audit.txt

P07 – Ordinary-moment first variation: positivity but open arithmetic partial construction. Analytic sign and rate proved here; primitive arithmetic remains open.

evidence/research/collaboration/round-18/findings.txt

E01 – Sharp Euler functional norm on integer-valued polynomials arithmetic theorem. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-10/adelic/findings.txt

E02 – Whole-functional Euler distribution attributed identity. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-08/discrete/findings.txt

E03 – Scoped countermodels for proposed analytic bridges structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/coordinated-round-08.txt

E04 – Monodromy obstruction to a linear endpoint regularization structural obstruction. Scoped monodromy statement proved here.

evidence/research/collaboration/round-12/geometry/findings.txt

E05 – Rational gauges of the ordinary two-state tail structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-12/fresh/tail_gauge_gate.txt

E06 – A rationality-dependent q-series truncation primitive value obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-12/arithmetic/findings.txt

E07 – Endpoint integrality fails for a particular Niven source structural obstruction. Rationality-dependent source, origin prime obstruction, endpoint derivative, coordinate integral and differential-field limitation proved here.

evidence/research/collaboration/round-11/rationality/findings.txt

E08 – Direct even-power signature transfer has zero rational trace structural obstruction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-10/fresh/findings.txt

E09 – Multiplicative and divisibility-Gram rediscoveries equivalence. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/collaboration/round-09/independent/multiplicative_gate.txt

O01 – Earlier coupled holonomy and lattice explorations partial construction. Catalogued with its archived scope; not promoted to a newly audited theorem.

evidence/research/coupled-holonomy.txt

References

[1] Research note prepared with Codex, *Arithmetic obstructions in positive determinant approximations to* $\beta(4)$, companion paper, 31 October 2026, PDF in the same repository.

[2] D. J. Broadhurst, *Polylogarithmic ladders, hypergeometric series and the ten millionth digits of* $\zeta(3)$ *and* $\zeta(5)$, arXiv:math/9803067v1 (1998).

[3] F. Calegari, V. Dimitrov and Y. Tang, *Arithmetic holonomy bounds and effective Diophantine approximation*, arXiv:2510.04156v1 (2025).

[4] M. C. S. Carreira, *Determinant constructions for the Dirichlet beta value at four*, pinned manuscript, commit `974a6e6c7b4b` (2026).

[5] NIST Digital Library of Mathematical Functions, Section 24.2, Section 24.17, Section 5.11, and Section 25.11, consulted 31 October–1 November 2026.

[6] H. Gangl, *Functional equations for higher logarithms*, arXiv:math/0207222v1 (2002).

[7] C. Krattenthaler, *A determinant identity for moments of orthogonal polynomials that implies Uvarov's formula for the orthogonal polynomials of rationally related densities*, arXiv:2103.03969v1 (2021).

[8] E. Leibtag, H. Barkan, R. Kalisch, S. Weinbaum, M. Shalyt and I. Kaminer, *New Directions in Irrationality*, arXiv:2610.27799v1 (2026).

[9] S. Litsey, *Irrationality measures for differences of trigamma values*, pinned manuscript, commit `aa2a42f4f383` (2026).

[10] J. Marché and G. Masbaum, *Signatures in TQFT: asymptotics and modularity*, author-hosted manuscript, version dated 21 February 2026.

[11] OpenAI, *Catalan's constant is irrational*, manuscript dated 24 September 2026, pinned manuscript, commit `adc7f1241b42`.

[12] M. Prévost, *Remainder Padé approximants for the Hurwitz zeta function*, arXiv:1709.05389v1 (2017).

[13] W. Zudilin, *A determinantal approach to irrationality*, arXiv:1507.05697v2 (2015; revised 2016).

[14] D. M. Bradley, *Series acceleration formulas for Dirichlet series with periodic coefficients*, Ramanujan J. **6** (2002), 331–346, arXiv:math/0505078v1.

[15] B. C. Berndt and A. Straub, *On a secant Dirichlet series and Eichler integrals of Eisenstein series*, arXiv:1406.2273v1 (2014).

[16] L. Lai, *The linear independence of* 1, $\zeta(3)$, *and* $\zeta(4)$, arXiv:2610.06838v2 (2026).

[17] Ozan Günyüz and Vyacheslav Zakharyuta, *On Pólya's theorem in several complex variables*, arXiv:1609.00218v1 (2016), The classical one-variable determinant inequality is stated as Theorem 2.1.

[18] A. Iribar López, *The Carlson–Pólya theorem on rational functions*, author-hosted note, Theorem 1 (Kronecker's Hankel criterion), consulted 1 November 2026.